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(58) **Field of Classification Search**

None
See application file for complete search history.

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(57) **ABSTRACT**

The invention relates to the production and use of Cas-
encoding sequences and vectors comprising these. Aspects
of the invention provide products, vectors, delivery vehicles,
uses and methods for producing Cas-encoding sequences in
bacterial or archaeal cells.

27 Claims, 6 Drawing Sheets

Specification includes a Sequence Listing.

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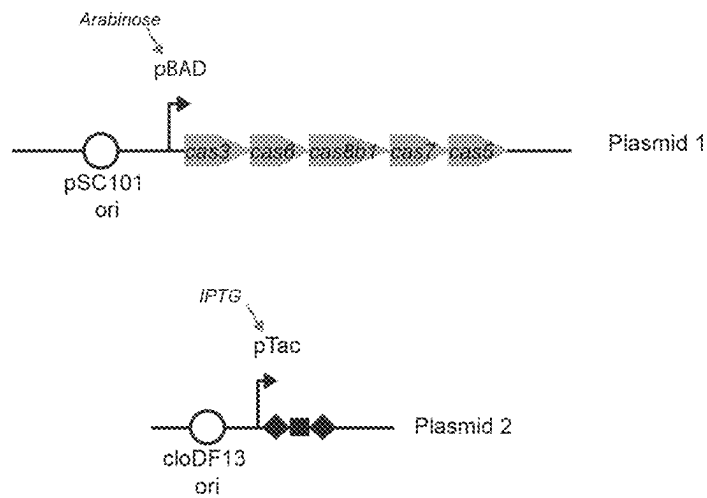


FIG. 1A

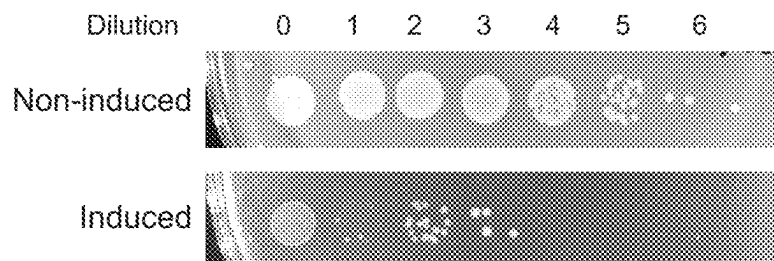


FIG. 1B

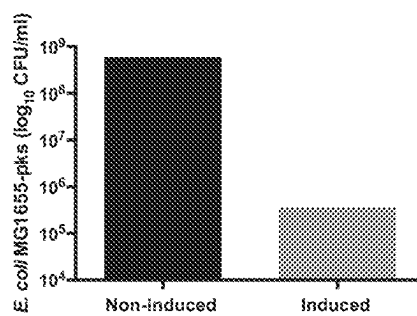


FIG. 1C

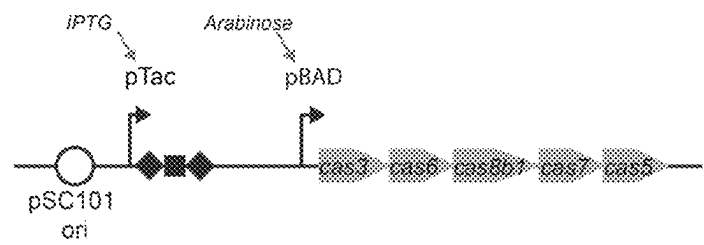


FIG. 2A

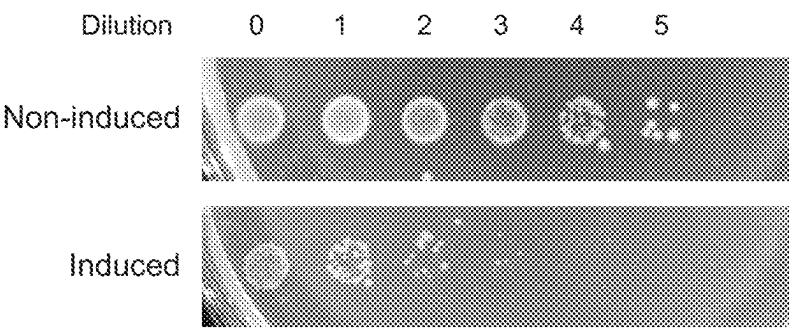


FIG. 2B

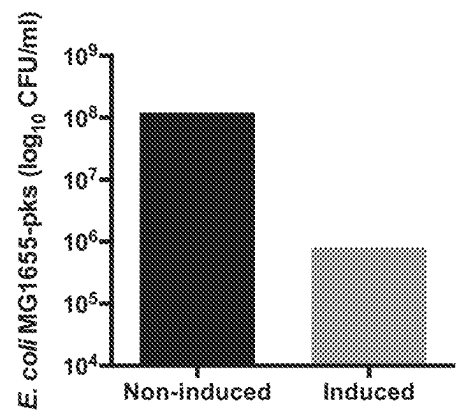


FIG. 2C

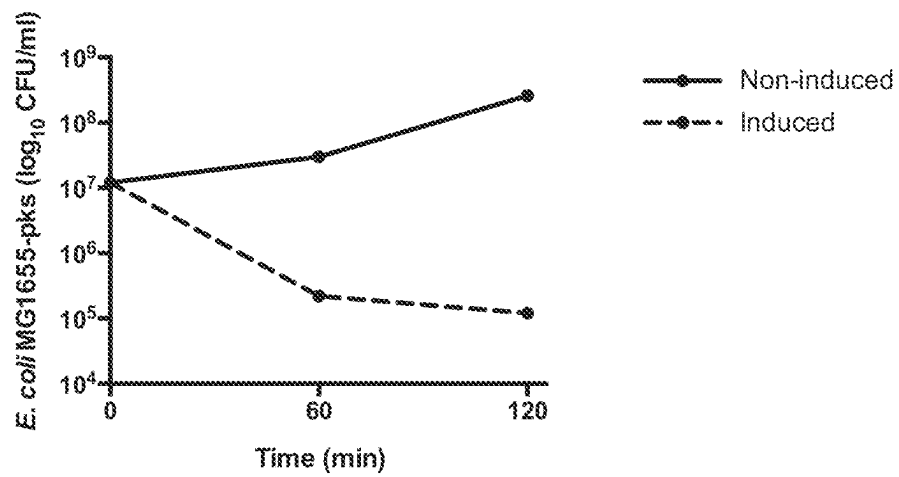


FIG. 3A

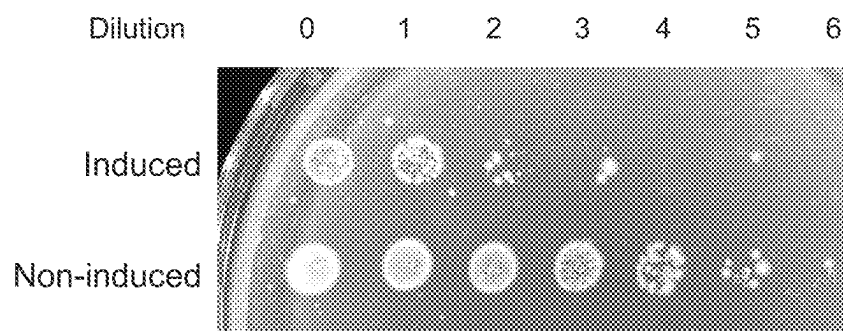


FIG. 3B

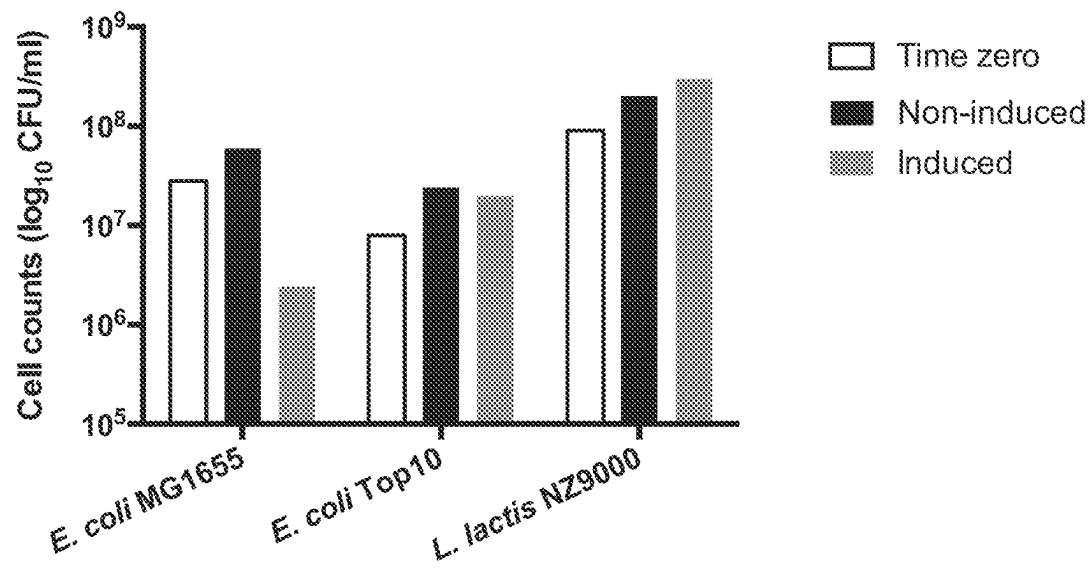


FIG. 4A

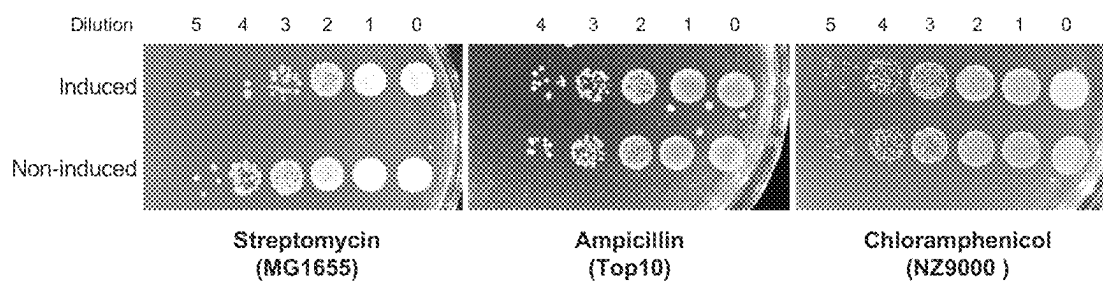


FIG. 4B

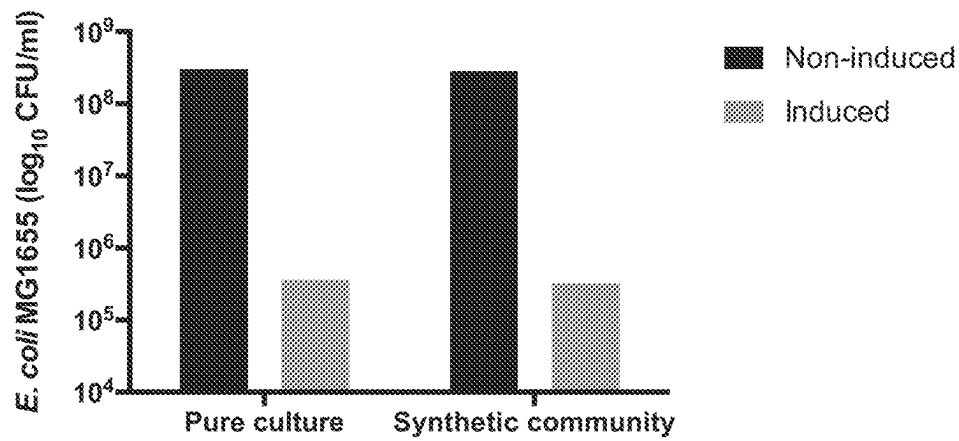


FIG. 5A

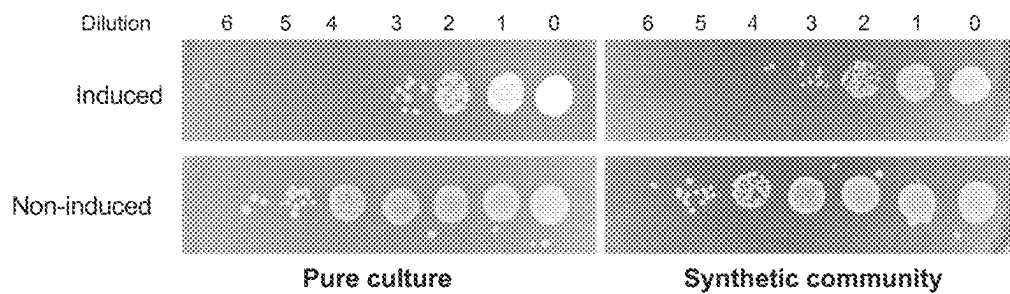


FIG. 5B

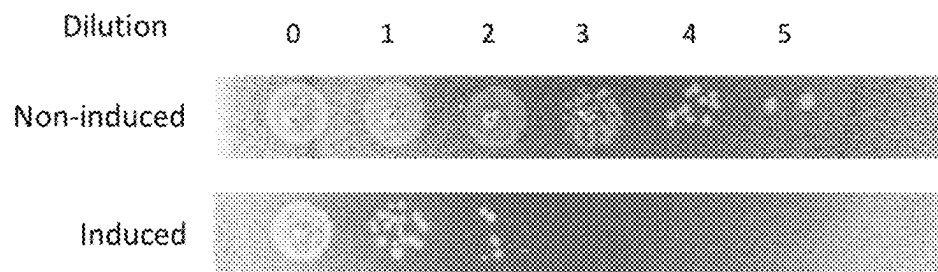


FIG. 6A

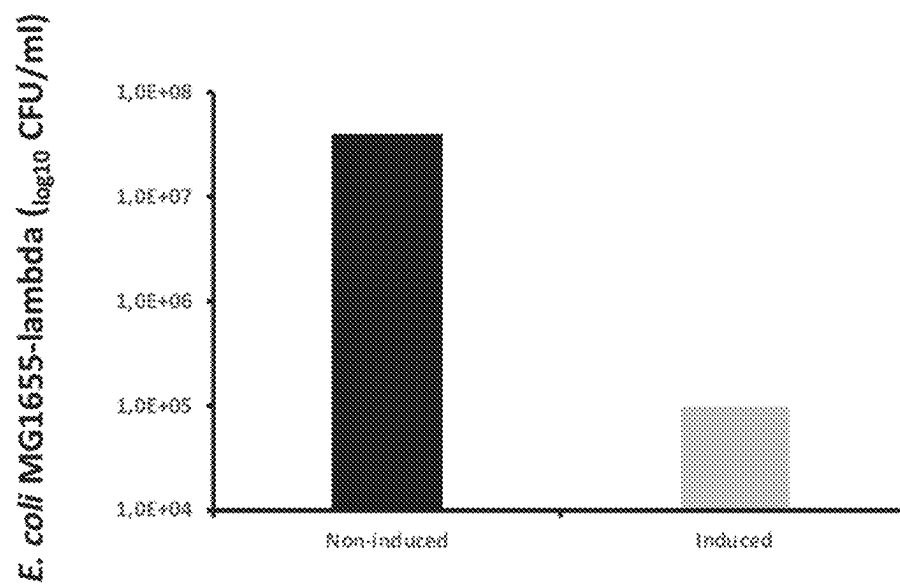


FIG. 6B

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SINGLE-VECTOR TYPE I VECTORS**CROSS-REFERENCE TO RELATED APPLICATIONS**

This application is a continuation of U.S. patent application Ser. No. 16/201,736, filed Nov. 27, 2018, which claims priority to Great Britain Patent Application No. 1816700.7, filed Oct. 14, 2018, and Great Britain Patent Application No. 1817509.1, filed Oct. 27, 2018, the contents of each of which are hereby incorporated herein by reference in their entirety.

REFERENCE TO AN ELECTRONIC SEQUENCE LISTING

The contents of the electronic sequence listing (786212000603seqlist.xml; Size: 38,386 bytes; and Date of Creation: Oct. 27, 2023) is herein incorporated by reference in its entirety.

TECHNICAL FIELD

The invention relates to the production and use of Cas-encoding sequences and vectors comprising these. Aspects of the invention provide products, vectors, delivery vehicles, uses and methods for producing Cas-encoding sequences in bacterial or archaeal cells.

BACKGROUND

The state of the art describes vectors and uses of these that employ CRISPR/Cas systems. For example, reference is made to WO2017/118598, US20180140698, US20170246221, US20180273940, US20160115488, US20180179547, US20170175142, US20160024510, US20150064138, US20170022499, US20160345578, US20180155729, US20180200342, WO2017112620, WO2018081502, PCT/EP2018/066954, PCT/EP2018/066980, PCT/EP2018/071454 and U.S. Ser. No. 15/985,658 and equivalent publications by the US Patent and Trademark Office (USPTO) or WIPO, the disclosures of which are incorporated herein by reference.

SUMMARY OF THE INVENTION

The invention provides the following configurations. In a First Configuration

A nucleic acid vector for introduction into a host cell, the vector comprising a first nucleotide sequence encoding a Type I Cas3 and a second nucleotide sequence encoding one or more Cascade proteins, wherein the first and second sequences are under the control of one or more promoters comprised by the vector for expression of the proteins in the cell.

In an example, the vector comprises an operon for expression in the cell of the Cas3 and Cascade proteins from a Cas module, the module comprising the nucleotide sequences encoding the Cas3 and Cascade proteins, and the operon comprising the Cas module under the control of a promoter for controlling the expression of both the Cas3 and Cascade proteins.

The invention also provides a delivery vehicle comprising the vector, as well as a pharmaceutical composition comprising the vector or vehicle and a pharmaceutically acceptable diluent, excipient or carrier.

The invention also provides a method of treating or reducing the risk of a disease or condition in a human or

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animal subject, the method comprising administering the vector, vehicle or composition to the subject, and introducing the vector into target host bacterial or archaeal cells in the subject (eg, in a gut microbiota, lung, eye or blood of the subject), wherein the Cas cuts (or otherwise modifies) one or more target sequences in the target cells and the cells are killed or growth or proliferation of the cells is reduced.

In a Second Configuration

A method of amplifying copies of a DNA encoding a functional Cas protein (optionally a Cas nuclease) in a bacterial or archaeal production strain of cells, the method comprising

(a) Providing production strain cells, each cell comprising a copy of said DNA, wherein each DNA comprises a nucleotide sequence encoding said Cas, wherein the nucleotide sequence is under the control of a promoter for controlling the expression of the Cas in the production strain cell, the DNA comprising an origin of replication that is operable in the cell for replication of the DNA;

(b) Culturing the cells to allow replication of the DNA, whereby the DNA is amplified; and

(c) Optionally isolating copies of the DNA,

Optionally wherein the promoter is an attenuated constitutive promoter.

In a Third Configuration

Use of an attenuated promoter in a DNA construct comprising a nucleotide sequence encoding a functional Cas protein (optionally a Cas nuclease) that is under the control of the promoter, in a method of amplifying copies of the DNA in a population of bacterial or archaeal production strain cells, the method comprising culturing the cells to allow replication of the DNA thereby amplifying the DNA in the cells, for enhancing the yield of amplified DNA produced by the production host cells.

In a Fourth Configuration

Use of an attenuated promoter in a DNA construct comprising a nucleotide sequence encoding a functional Cas protein (optionally a Cas nuclease) that is under the control of the promoter, in a method of amplifying copies of the DNA in a population of bacterial or archaeal production strain cells, the method comprising culturing the cells to allow replication of the DNA thereby amplifying the DNA in the cells, for reducing toxicity of the Cas in the production strain.

In a Fifth Configuration

Use of an attenuated promoter in a DNA construct comprising a nucleotide sequence encoding a functional Cas protein (optionally a Cas nuclease) that is under the control of the promoter, in a method of amplifying copies of the DNA in a population of bacterial or archaeal production strain cells, the method comprising culturing the cells to allow replication of the DNA thereby amplifying the DNA in the cells, for reducing mutation of the DNA (optionally the Cas-encoding sequence) in the production strain.

In a Sixth Configuration

Use of an attenuated promoter in a DNA construct comprising a nucleotide sequence encoding a functional Cas protein (optionally a Cas nuclease) that is under the control of the promoter, in a method of amplifying copies of the DNA in a population of bacterial or archaeal production strain cells, the method comprising culturing the cells to allow replication of the DNA thereby amplifying the DNA in the cells, for promoting production cell viability during the amplification of the DNA.

In a Seventh Configuration

Use of an attenuated promoter in a DNA construct comprising a nucleotide sequence encoding a functional Cas protein (optionally a Cas nuclease) that is under the control of the promoter, in a method of amplifying copies of the DNA in a population of bacterial or archaeal production strain cells, the method comprising culturing the cells to allow replication of the DNA thereby amplifying the DNA in the cells, for reducing the occurrence of Cas cutting of DNA.

In an Eighth Configuration

A method for enhancing the yield of amplified copies of a DNA construct in a population of bacterial or archaeal production strain cells, wherein the construct comprises a nucleotide sequence encoding a functional Cas protein (optionally a Cas nuclease) that is under the control of a promoter, the method comprising culturing the cells to allow replication of the DNA thereby amplifying the DNA in the cells, wherein the promoter is an attenuated promoter.

In a Ninth Configuration

A method for reducing toxicity of a functional Cas protein (optionally a Cas nuclease) in a population of bacterial or archaeal production strain cells in a process of amplifying copies of a DNA construct, wherein the construct comprises a nucleotide sequence encoding the Cas and the sequence is under the control of a promoter, the method comprising culturing the cells to allow replication of the DNA thereby amplifying the DNA in the cells, wherein the promoter is an attenuated promoter.

In a Tenth Configuration

A method for reducing mutation of a DNA construct encoding a functional Cas protein (optionally a Cas nuclease) in a population of bacterial or archaeal production strain cells in a process of amplifying copies of the construct, wherein the construct comprises a nucleotide sequence encoding the Cas and the sequence is under the control of a promoter, the method comprising culturing the cells to allow replication of the DNA thereby amplifying the DNA in the cells, wherein the promoter is an attenuated promoter.

In an Eleventh Configuration

A method for promoting production cell viability of a population of bacterial or archaeal production strain cells in a process of amplifying copies of a DNA construct comprised by the cells, wherein the construct comprises a nucleotide sequence encoding a functional Cas protein (optionally a Cas nuclease) and the sequence is under the control of a promoter, the method comprising culturing the cells to allow replication of the DNA thereby amplifying the DNA in the cells, wherein the promoter is an attenuated promoter.

In a Twelfth Configuration

A method for reducing the occurrence of Cas nuclease cutting of a DNA construct in a population of bacterial or archaeal production strain cells in a process of amplifying copies of the construct, wherein the construct comprises a nucleotide sequence encoding the Cas and the sequence is under the control of a promoter, the method comprising culturing the cells to allow replication of the DNA thereby amplifying the DNA in the cells, wherein the promoter is an attenuated promoter.

BRIEF DESCRIPTION OF THE DRAWINGS

FIGS. 1A-1C. Type I CRISPR-Cas system of *C. difficile* targeting *E. coli* MG1655. (FIG. 1A) Layout of the CRISPR Guided Vector™, CGV™. Plasmid 1: pSC101 on, pBAD promoter (induced by arabinose), cas3 and cascade genes.

Plasmid 2: pCloDF13 on, pTac promoter (induced by IPTG), CRISPR array. (FIG. 1B) Dilution series (10^1 - 10^6) of drop spots (5 μ l) of *E. coli* MG1655 harboring the CGV on LB agar plates with and without inducers. (FIG. 1C) CRISPR induction killed 99.9% of the population (grey bar). Growth in absence of induction is shown in black. CGV™ refers to a CRISPR Guided Vector™, which is a nucleic acid vector comprising nucleotide sequences encoding CRISPR/Cas components.

FIGS. 2A-2C. Type I CRISPR-Cas system of *C. difficile* targeting *E. coli* MG1655. (FIG. 2A) Layout of the CRISPR Guided Vector™, CGV™. pSC101 on, pTac promoter (induced by IPTG), CRISPR array, pBAD promoter (induced by arabinose), cas3 and cascade genes. (FIG. 2B) Dilution series (10^1 - 10^6) of drop spots (5 μ l) of *E. coli* MG1655 harboring the CGV on SM agar plates with and without inducers. (FIG. 2C) CRISPR induction killed 99% of the population (grey bar). Growth in absence of induction is shown in black. CGV™ refers to a CRISPR Guided Vector™, which is a nucleic acid vector comprising nucleotide sequences encoding CRISPR/Cas components.

FIGS. 3A-3B. Time-kill curves for *E. coli* MG1655 harboring the CGV. (FIG. 3A) CRISPR induction killed 99% of the population in 60 minutes (dashed line). Growth in absence of induction is shown in black lines. CRISPR/Cas was induced at time-point 0 and monitored until 120 minutes. (FIG. 3B) Dilution series (10^1 - 10^6) of drop spots (5 μ l) on SM agar plates of *E. coli* MG1655 after 60 minutes of induction.

FIGS. 4A-4B. Specific killing of *E. coli* MG1655 with type I-B CRISPR-Cas system of *C. difficile* in a synthetic microbial consortium. (FIG. 4A) Bacteria count of a synthetic population composed of three different strains. CRISPR was induced at time-point 0 and monitored for 60 minutes. Growth in absence of induction is shown in black. CRISPR induction prompted 1-log₁₀ reduction in viable cells of target strain *E. coli* MG1655, while leaving *E. coli* Top10 and *L. lactis* NZ9000 populations intact (dark grey bars). (FIG. 4B) Dilution series (10^1 - 10^6) of drop spots (5 μ l) of the bacterial community mixture after 60 minutes of induction. *E. coli* MG1655 grows selectively on BHI+ streptomycin, *E. coli* Top10 on ampicillin, and *L. lactis* NZ9000 on chloramphenicol.

FIGS. 5A-5B. Killing of *E. coli* MG1655 with type I-B CRISPR-Cas system of *C. difficile* in a synthetic microbial consortium compared to a pure culture of *E. coli* MG1655. (FIG. 5A) CRISPR induction generated 4-log₁₀ reductions in viable cells of target strain *E. coli* MG1655, both in the pure culture and in the community mixture (grey bars). Growth in absence of induction is shown in black. (FIG. 5B) Dilution series of a pure culture of *E. coli* MG1655 and the bacterial community mixture on streptomycin plates with and without inducers.

FIGS. 6A-6B. Type I CRISPR-Cas system of *E. coli* targeting *E. coli* MG1655. (FIG. 6A) Dilution series (10^1 - 10^6) of drop spots (5 μ l) of *E. coli* MG1655 harboring the CGV on SM agar plates with and without inducers. (FIG. 6B) CRISPR induction killed 99% of the population (grey bar). Growth in absence of induction is shown in black. CGV™ refers to a CRISPR Guided Vector™ which is a nucleic acid vector comprising nucleotide sequences encoding CRISPR/Cas components.

DETAILED DESCRIPTION

The invention relates to the production and use of Cas-encoding sequences and vectors comprising these. Aspects

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of the invention provide products, vectors, delivery vehicles, uses and methods for producing Cas-encoding sequences in bacterial or archaeal cells.

An aspect of the invention provides for the control of expression of Cas and optionally also Cascade proteins from single vectors, such as by regulated use of Cas modules in an operon and/or using attenuated promoters.

Concepts:

An aspect of the invention provides nucleic acid vectors that are useful for introducing into target host cells of any eukaryotic or prokaryotic species (eg, ex vivo or in vitro) for expressing Type I Cas and optionally other components of a Type I CRISPR/Cas system. Usefully, the vector may in some examples therefore provide a single-vector means for introducing a complete exogenous Type I CRISPR/Cas system into a target cell for modification (eg, cutting by Cas3) of DNA in the target cell. In an example, a chromosomal target sequence (ie, protospacer that is cognate with the Cas3) is modified. In another example, an episomal DNA sequence is modified, for example a plasmid sequence or a DNA that has been introduced into the cell. The latter may be useful in a recombinogenic method of the invention wherein exogenous DNA in the target cell is cut by the Cas3 and optionally this produces one or more recombinogenic ends for recombination of the cut DNA with a further DNA of interest, thereby producing a recombination product in the cell. For example, in such a recombinogenic method, the target cell is a recombinogenic *E. coli* cell, eg, comprising a red/ET system. In another example, the target cell is an undesired cell (eg, a cell of a species or strain that is pathogenic to humans or animals, such as a bacterial disease-causing species or strain) and the cutting by Cas3 kills the cell. This may be useful for treating or preventing an infection in a human or animal harbouring target cells. The provision of single-vector means that express minimally a Cas endonuclease (eg, Cas3), cognate accessory proteins (eg, Cascade proteins) and at least one CRISPR array (or nucleotide sequence encoding a guide RNA (eg, a single guide RNA)), wherein the Cas, accessory proteins and array (or nucleotide sequence) comprise a functional CRISPR/Cas system is more convenient and the inventors believe more efficient for introducing into a target cell and effecting CRISPR/Cas modification of a target sequence therein than the use of 2 or 3 or more separate vectors (eg, a vector encoding the Cas nuclease and a different vector encoding the accessory proteins, and possibly a further vector comprising the array (or gRNA-encoding nucleotide sequence) which all need to transform the target cell for the system to function). This may provide one or more benefits, therefore, such as simplifying delivery (and thus the design of delivery vehicles), simplifying construction of the vector and vehicle and/or providing for better cutting or killing efficiencies. Conveniently, an example of the invention therefore uses an operon for the coordinated expression in the target cells of the Cas and accessory proteins (and optional also the array or gRNA-encoding sequence(s)). Whilst not wishing to be bound by any particular theory, the introduction of a single vector (eg, using an operon) as per the invention may advantageously coordinate the expression of the Cas and accessory proteins (and optionally production of cRNAs or gRNAs) so that these are available to operate together without undue delay in the target cell. This may be important to tip the balance between, on the one hand the target cell using its endogenous anti-restriction, endogenous Cas or other endogenous mechanisms that seek out and degrade invading phage and DNA, and on the other hand efficient cell killing or deactivation of such mechanisms by the

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invading CRISPR components of the vector of the invention. In such an arms race, concerted and early operation of the CRISPR components in the cell are likely to be important to gain the upper hand and effect cell killing. The invention provides means to assist this.

By way of example, the invention thus provides the following Concepts:—

1. A nucleic acid vector for introduction into a host cell, the vector comprising a first nucleotide sequence encoding a Type I Cas3 and a second nucleotide sequence encoding one or more Cascade proteins, wherein the first and second sequences are under the control of one or more promoters comprised by the vector for expression of the proteins in the cell.

2. The vector of concept 1, wherein the vector comprises an operon for expression in the cell of the Cas3 and Cascade proteins from a Cas module, the module comprising the nucleotide sequences encoding the Cas3 and Cascade proteins, and the operon comprising the Cas module under the control of a promoter for controlling the expression of both the Cas3 and Cascade proteins.

3. The vector of concept 2, wherein

(a) the first sequence is between the promoter and the second sequence in the operon;

(b) the operon comprises no Cas-encoding nucleotide sequences between the promoter and the first nucleotide sequence; and/or

(c) the operon comprises (in 5' to 3' direction) the promoter, the first sequence and the second sequence.

4. The vector of any preceding concept, wherein each promoter is a constitutive promoter.

5. The vector of any one of concepts 1 to 3, wherein the promoter is repressible (optionally repressible by a tetracycline repressor or lac repressor).

6. The vector of any one of concepts 1 to 3, wherein the promoter is inducible.

7. The vector of any preceding concept, wherein the first sequence is under the control of a medium strength promoter.

8. The vector of any preceding concept, wherein the first sequence is under the control of a promoter that has an Anderson Score (AS) of $0.5 > AS > 0.1$.

9. The vector of any preceding concept, wherein the first sequence (and optionally the second sequence) is under the control of a promoter and translation initiation site (TIS) combination that is capable of producing expression of green fluorescent protein (GFP) from a first expression operating unit (EOU) in *E. coli* strain BW25113 cells with a fluorescence of from 0.5 to 4 times the fluorescence produced in *E. coli* strain BW25113 cells using a second EOU comprising a P10 promoter (SEQ ID NO: 1) combined with a BCD14 TIS (SEQ ID NO: 2), wherein the EOUs differ only in their promoter and TIS combinations, wherein each EOU comprises (in 5' to 3' direction) an upstream initiator, the respective promoter, the respective TIS, a nucleotide sequence encoding GFP, a 3' UTR, a transcription terminator and a downstream insulator.

10. The vector of concept 9, wherein fluorescence using the first EOU is 0.5 to 2 times the fluorescence using the second EOU.

11. The vector of any preceding concept, wherein the vector comprises an origin of replication that is operable in the host cell.

12. The vector of any preceding concept, wherein the vector comprises an origin of replication that is operable in a bacterial cell of a vector production strain, wherein the

Cas3 is not operable in the production strain cell to target and cut a chromosomal sequence thereof.

13. The vector of concept 12, wherein the first sequence is under the control of a promoter that is capable of controlling expression of the Cas3 at a level that is not toxic to the production strain cell.

14. The vector of any preceding concept, wherein the vector is a high copy number vector.

15. The vector of any preceding concept, wherein the first nucleotide sequence or operon is comprised by a mobile genetic element.

16. The vector of any preceding concept, wherein the vector is devoid of a Cas adaption module.

17. The vector of any preceding concept, wherein the vector is devoid of nucleotide sequence encoding one, more or all of a Cas1, Cas2, Cas4, Cas6, Cas7 and Cas8.

18. The vector of any preceding concept, wherein the vector comprises (optionally in 5' to 3' direction) nucleotide sequence encoding one, more or all of Cas11, Cas7 and Cas8a1.

19. The vector of concept 18, wherein the vector comprises nucleotide sequence encoding Cas3' and/or Cas3".

20. The vector of concept 19, wherein the nucleotide sequences encoding the Cas3' and/or Cas3" are between the promoter and the sequence(s) recited in concept 18.

21. The vector of any one of concepts 18 to 20, wherein the host cell comprises a Type IA CRISPR array that is cognate with the Cas3.

22. The vector of any one of concepts 18 to 20, wherein the host cell comprises an endogenous Type IB, C, U, D, E or F CRISPR/Cas system.

23. The vector of any one of concepts 1 to 17, wherein the vector comprises (optionally in 5' to 3' direction) nucleotide sequence encoding one, more or all of Cas8b1, Cas7 and Cas5.

24. The vector of concept 23, wherein the vector comprises a nucleotide sequence encoding Cas3 between the promoter and the sequence(s) recited in concept 23.

25. The vector of concept 23 or 24, wherein the host cell comprises a Type IB CRISPR array that is cognate with the Cas3.

26. The vector of concept 23 or 24, wherein the host cell comprises an endogenous Type IA, C, U, D, E or F CRISPR/Cas system.

27. The vector of any one of concepts 1 to 17, wherein the vector comprises (optionally in 5' to 3' direction) nucleotide sequence encoding one, more or all of Cas5, Cas8c and Cas7.

28. The vector of concept 27, wherein the vector comprises a nucleotide sequence encoding Cas3 between the promoter and the sequence(s) recited in concept 27.

29. The vector of concept 27 or 28, wherein the host cell comprises a Type IC CRISPR array that is cognate with the Cas3.

30. The vector of concept 27 or 28, wherein the host cell comprises an endogenous Type IA, B, U, D, E or F CRISPR/Cas system.

31. The vector of any one of concepts 1 to 17, wherein the vector comprises (optionally in 5' to 3' direction) nucleotide sequence encoding one, more or all of Cas8U2, Cas7, Cas5 and Cas6.

32. The vector of concept 31, wherein the vector comprises a nucleotide sequence encoding Cas3 between the promoter and the sequence(s) recited in concept 31.

33. The vector of concept 31 or 32, wherein the host cell comprises a Type IU CRISPR array that is cognate with the Cas3.

34. The vector of concept 31 or 32, wherein the host cell comprises an endogenous Type IA, B, C, D, E or F CRISPR/Cas system.

35. The vector of any one of concepts 1 to 17, wherein the vector comprises (optionally in 5' to 3' direction) nucleotide sequence encoding one, more or all of Cas10d, Cas7 and Cas5.

36. The vector of concept 35, wherein the vector comprises a nucleotide sequence encoding Cas3' and/or Cas3".

37. The vector of concept 36, wherein the nucleotide sequences encoding the Cas3' and/or Cas3" are between the promoter and the sequence(s) recited in concept 35.

38. The vector of any one of concepts 35 to 37, wherein the host cell comprises a Type ID CRISPR array that is cognate with the Cas3.

39. The vector of any one of concepts 35 to 37, wherein the host cell comprises an endogenous Type IA, B, C, U, E or F CRISPR/Cas system.

40. The vector of any one of concepts 1 to 17, wherein the vector comprises (optionally in 5' to 3' direction) nucleotide sequence encoding one, more or all of Cas8e, Cas11, Cas7, Cas5 and Cas6.

41. The vector of concept 40, wherein the vector comprises a nucleotide sequence encoding Cas3 between the promoter and the sequence(s) recited in concept 40.

42. The vector of concept 40 or 41, wherein the host cell comprises a Type IE CRISPR array that is cognate with the Cas3.

43. The vector of concept 40 or 41, wherein the host cell comprises an endogenous Type IA, B, C, D, U or F CRISPR/Cas system.

44. The vector of any one of concepts 1 to 17, wherein the vector comprises (optionally in 5' to 3' direction) nucleotide sequence encoding one, more or all of Cas8f, Cas5, Cas7 and Cas6f.

45. The vector of concept 44, wherein the vector comprises a nucleotide sequence encoding Cas3 between the promoter and the sequence(s) recited in concept 44, wherein the vector is devoid of nucleotide sequence encoding further Cas between the promoter and the sequence encoding the Cas3.

46. The vector of concept 44 or 45, wherein the host cell comprises a Type IF CRISPR array that is cognate with the Cas3.

47. The vector of concept 44 or 45, wherein the host cell comprises an endogenous Type IA, B, C, D, U or E CRISPR/Cas system.

48. The vector of any one of concepts 1 to 17, wherein the Cas and Cascade are

- (a) Type IA Cas and Cascade proteins;
- (b) Type IB Cas and Cascade proteins;
- (c) Type IC Cas and Cascade proteins;
- (d) Type ID Cas and Cascade proteins;
- (e) Type IE Cas and Cascade proteins;
- (f) Type IF Cas and Cascade proteins; or
- (g) Type IU Cas and Cascade proteins.

49. The vector of any preceding concept, wherein the Cas and Cascade are *E. coli* (optionally Type IE or IF) Cas and Cascade proteins.

50. The vector of concept 49, wherein the *E. coli* is ESBL-producing *E. coli* or *E. coli* ST131-O25b:H4.

51. The vector of any preceding concept, wherein the Cas and Cascade are

- (a) *Clostridium* (eg, *C. difficile*) Cas and Cascade proteins, optionally *C. difficile* resistant to one or more antibiotics selected from aminoglycosides, lincomycin, tet-

racyclines, erythromycin, clindamycin, penicillins, cephalosporins and fluoroquinolones;

- (b) *Pseudomonas aeruginosa* Cas and Cascade proteins, optionally *P. aeruginosa* resistant to one or more antibiotics selected from carbapenems, aminoglycosides, cefepime, ceftazidime, fluoroquinolones, piperacillin and tazobactam; or
- (c) *Klebsiella pneumoniae* (eg, carbapenem-resistant *Klebsiella pneumoniae* or Extended-Spectrum Beta-Lactamase (ESBL)-producing *K. pneumoniae*) Cas and Cascade proteins.

52. The vector of any preceding concept, wherein the Cas and Cascade are *E. coli*, *C. difficile*, *P. aeruginosa*, *K. pneumoniae*, *P. furiosus* or *B. halodurans* Cas and Cascade proteins.

53. The vector of any preceding concept, wherein the Cas3 is a Cas3 of a CRISPR/Cas locus of a first bacterial or archaeal species, wherein the distance between the Cas3-encoding sequence of the locus and its cognate promoter is further than the distance between the Cas3-encoding sequence and the respective promoter comprised by the vector.

54. The vector of any preceding concept, wherein the distance between the promoter and the Cas3-encoding sequence and/or Cascade protein-encoding sequence(s) is shorter than in a corresponding wild-type Type I locus.

55. The vector of any preceding concept, wherein the vector comprises (i) a CRISPR array for producing crRNAs in the host cell and/or (ii) one or more nucleotide sequences encoding one or more guide RNAs (gRNAs or single gRNAs), wherein the crRNAs or gRNAs are cognate to the Cas3 (and optionally cognate to the Cascade proteins).

56. The vector of concept 55 when dependent from concept 2, wherein the array or gRNA-encoding sequence(s) are comprised by the operon and under the control of the promoter.

57. The vector of concept 56, wherein the array or gRNA-encoding sequence(s) are under the control of a promoter that is different from the promoter that controls the expression of the Cas3.

58. The vector of concept 56 or 57, wherein one or more of the crRNAs or gRNAs comprises a spacer sequence that is capable of hybridising to a target nucleotide sequence of the host cell, wherein the target sequence is adjacent a PAM, the PAM being cognate to the Cas3.

59. The vector of concept 58, wherein the target sequence is a chromosomal sequence of the host cell.

60. The vector of concept 58 or 59, wherein the Cas3 is operable to cut the target sequence.

61. The vector of any preceding concept, wherein the vector is a plasmid or phagemid.

62. A delivery vehicle comprising the vector of any preceding concept, wherein the delivery vehicle is capable of delivering the vector into the host cell.

63. The vehicle of concept 62, wherein the delivery vehicle is a phage, non-replicative transduction particle, nanoparticle carrier, bacterium or liposome.

64. The vector or vehicle of any preceding concept, wherein the host cell is a bacterial or archaeal cell, optionally, the host cell is a *C. difficile*, *P. aeruginosa*, *K. pneumoniae* (eg, carbapenem-resistant *Klebsiella pneumoniae* or Extended-Spectrum Beta-Lactamase (ESBL)-producing *K. pneumoniae*), *E. coli* (eg, ESBL-producing *E. coli*, or *E. coli* ST131-O25b:H4), *H. pylori*, *S. pneumoniae* or *S. aureus* cell.

65. The vector or vehicle of any preceding concept for administration to a human or animal subject for treating or reducing the risk of a disease or condition in the subject.

66. The vector or vehicle of concept 65, wherein the disease or condition is an infection of the subject with host cells (eg, bacterial cells), or wherein the disease or condition is mediated by host cells (eg, bacterial cells).

67. A pharmaceutical composition comprising the vector or vehicle of any preceding concept and a pharmaceutically acceptable diluent, excipient or carrier.

68. A method of treating or reducing the risk of a disease or condition in a human or animal subject, the method comprising administering the vector, vehicle or composition of any preceding concept to the subject, and introducing the vector into target host bacterial or archaeal cells in the subject (eg, in a gut microbiota, lung, eye or blood of the subject), wherein the Cas cuts (or otherwise modifies) one or more target sequences in the target cells and the cells are killed or growth or proliferation of the cells is reduced.

69. The method of concept 68, wherein the target cells are cells of a disease pathogen species.

70. The method of concept 68 or 69, wherein the target cells are *C. difficile*, *P. aeruginosa*, *K. pneumoniae* (eg, carbapenem-resistant *Klebsiella pneumoniae* or Extended-Spectrum Beta-Lactamase (ESBL)-producing *K. pneumoniae*), *E. coli* (eg, ESBL-producing *E. coli*, or *E. coli* ST131-O25b:H4), *H. pylori*, *S. pneumoniae* or *S. aureus* cells.

Embodiments

An aspect of the invention provides improved ways of amplifying DNA constructs in bacterial and archaeal production strain cells. For example, the DNA may be a high copy number plasmid or phagemid comprising a constitutive promoter for controlling the expression of one or more Cas proteins when the DNA has been introduced into a target host bacterial or host cell. It is desirable, according to an aspect of the invention, to consider attenuating the promoter activity during amplification of the DNA in the production strain. This is useful, since the inventors have found that Cas expression in production strains may be toxic to production strain cells, thereby reducing the yield of amplified DNA. Toxicity may be due, for example, to off-target cutting of the production strain chromosomal DNA when the Cas is a nuclease (such as Cas9 or Cas3) and/or due to relatively high levels of expression of the Cas in the cells. Additionally or alternatively, undesirably the Cas expression or activity may impose a selective pressure that favours mutation and propagation of mutated DNA constructs (such as mutation in one more or all of a CRISPR/Cas operon, Cas-encoding gene, Cascade-encoding gene, CRISPR array and gRNA-encoding sequence of the DNA construct) in production cells, thereby reducing the yield of desired amplified constructs and imposing an undesired step of separating desired from mutated DNA constructs for further formulation into useful compositions. Such compositions may be pharmaceutical compositions, herbicides, pesticides, environmental remediation compositions etc. In one example, the promoter attenuation in production strains is achieved by using a medium strength (not high or low) promoter to control the Cas-encoding nucleotide sequence of the DNA constructs. A medium level of Cas expression may be tolerable in the production strains, and yet once the DNA is subsequently introduced into target host cells the Cas is expressed at sufficiently high levels to produce desired activity to modify (eg, cut) target sequences in target cells. In an alternative, the

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invention uses a repressible promoter, wherein the promoter is repressed in production strain, but not repressed in target host cells. For example, aspects of the invention use a tetracycline repressor (tetR) expressed in production strain cells that represses the promoter.

Thus, the yield can be enhanced by one or more of

- (a) reducing toxicity of the Cas in the production strain;
- (b) reducing mutation of the DNA (optionally the Cas-encoding sequence) in the production strain;
- (c) promoting production cell viability during the amplification of the DNA; and
- (d) reducing the occurrence of Cas cutting of DNA (optionally cutting of production host cell chromosomal DNA or said DNA construct).

To this end, the invention provides Embodiments as follows:—

1. A method of amplifying copies of a DNA encoding a functional Cas protein (optionally a Cas nuclease) in a bacterial or archaeal production strain of cells, the method comprising

- (a) Providing production strain cells, each cell comprising a copy of said DNA, wherein each DNA comprises a nucleotide sequence encoding said Cas, wherein the nucleotide sequence is under the control of a promoter for controlling the expression of the Cas in the production strain cell, the DNA comprising an origin of replication that is operable in the cell for replication of the DNA;
- (b) Culturing the cells to allow replication of the DNA, whereby the DNA is amplified; and
- (c) Optionally isolating copies of the DNA, wherein the promoter is an attenuated constitutive promoter.

In an example, promoter is a medium strength promoter. In another example, the promoter is repressed in the production strain cell. Hence, the promoter is an attenuated promoter in these examples.

2. Use of an attenuated promoter in a DNA construct comprising a nucleotide sequence encoding a functional Cas protein (optionally a Cas nuclease) that is under the control of the promoter, in a method of amplifying copies of the DNA in a population of bacterial or archaeal production strain cells, the method comprising culturing the cells to allow replication of the DNA thereby amplifying the DNA in the cells, for enhancing the yield of amplified DNA produced by the production host cells.

3. The use of paragraph 2, wherein the use is for enhancing said yield by

- (a) reducing toxicity of the Cas in the production strain;
- (b) reducing mutation of the DNA (optionally the Cas-encoding sequence) in the production strain;
- (c) promoting production cell viability during the amplification of the DNA; and/or
- (d) reducing the occurrence of Cas cutting of DNA (optionally cutting of production host cell chromosomal DNA or said DNA construct).

4. Use of an attenuated promoter in a DNA construct comprising a nucleotide sequence encoding a functional Cas protein (optionally a Cas nuclease) that is under the control of the promoter, in a method of amplifying copies of the DNA in a population of bacterial or archaeal production strain cells, the method comprising culturing the cells to allow replication of the DNA thereby amplifying the DNA in the cells, for reducing toxicity of the Cas in the production strain.

5. Use of an attenuated promoter in a DNA construct comprising a nucleotide sequence encoding a functional Cas

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protein (optionally a Cas nuclease) that is under the control of the promoter, in a method of amplifying copies of the DNA in a population of bacterial or archaeal production strain cells, the method comprising culturing the cells to allow replication of the DNA thereby amplifying the DNA in the cells, for reducing mutation of the DNA (optionally the Cas-encoding sequence) in the production strain.

6. Use of an attenuated promoter in a DNA construct comprising a nucleotide sequence encoding a functional Cas protein (optionally a Cas nuclease) that is under the control of the promoter, in a method of amplifying copies of the DNA in a population of bacterial or archaeal production strain cells, the method comprising culturing the cells to allow replication of the DNA thereby amplifying the DNA in the cells, for promoting production cell viability during the amplification of the DNA.

7. Use of an attenuated promoter in a DNA construct comprising a nucleotide sequence encoding a functional Cas protein (optionally a Cas nuclease) that is under the control of the promoter, in a method of amplifying copies of the DNA in a population of bacterial or archaeal production strain cells, the method comprising culturing the cells to allow replication of the DNA thereby amplifying the DNA in the cells, for reducing the occurrence of Cas cutting of DNA.

8. A method for enhancing the yield of amplified copies of a DNA construct in a population of bacterial or archaeal production strain cells, wherein the construct comprises a nucleotide sequence encoding a functional Cas protein (optionally a Cas nuclease) that is under the control of a promoter, the method comprising culturing the cells to allow replication of the DNA thereby amplifying the DNA in the cells, wherein the promoter is an attenuated promoter.

9. A method for reducing toxicity of a functional Cas protein (optionally a Cas nuclease) in a population of bacterial or archaeal production strain cells in a process of amplifying copies of a DNA construct, wherein the construct comprises a nucleotide sequence encoding the Cas and the sequence is under the control of a promoter, the method comprising culturing the cells to allow replication of the DNA thereby amplifying the DNA in the cells, wherein the promoter is an attenuated promoter.

10. A method for reducing mutation of a DNA construct encoding a functional Cas protein (optionally a Cas nuclease) in a population of bacterial or archaeal production strain cells in a process of amplifying copies of the construct, wherein the construct comprises a nucleotide sequence encoding the Cas and the sequence is under the control of a promoter, the method comprising culturing the cells to allow replication of the DNA thereby amplifying the DNA in the cells, wherein the promoter is an attenuated promoter.

11. A method for promoting production cell viability of a population of bacterial or archaeal production strain cells in a process of amplifying copies of a DNA construct comprised by the cells, wherein the construct comprises a nucleotide sequence encoding a functional Cas protein (optionally a Cas nuclease) and the sequence is under the control of a promoter, the method comprising culturing the cells to allow replication of the DNA thereby amplifying the DNA in the cells, wherein the promoter is an attenuated promoter.

12. A method for reducing the occurrence of Cas nuclease cutting of a DNA construct in a population of bacterial or archaeal production strain cells in a process of amplifying copies of the construct, wherein the construct comprises a nucleotide sequence encoding the Cas and the sequence is under the control of a promoter, the method comprising

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culturing the cells to allow replication of the DNA thereby amplifying the DNA in the cells, wherein the promoter is an attenuated promoter.

13. The use of paragraph 5 or 7, or the method of paragraph 10 or 12, wherein the mutation or cutting is mutation or cutting of host cell chromosomal DNA or the construct DNA.

14. The method or use of any one of paragraphs 2 to 13, wherein the promoter is a constitutive promoter.

15. The method or use of any preceding paragraph, wherein the promoter is repressed in the production strain cells (optionally repressed by a tetracycline repressor or a lac repressor).

16. The method or use of paragraph 15, wherein the promoter is $P_{LtetO-1}$, $P_{LlacO-1}$ or a repressible homologue thereof.

Other examples of suitable repressible promoters are Ptac (repressed by lad) and the Leftward promoter (pL) of phage lambda (which repressed by the λ cl repressor). In an example, the promoter comprises a repressible operator (eg, tetO or lacO) fused to a promoter sequence. The corresponding repressor is encoded by a nucleic acid in the production strain (eg, a chromosomally-integrated sequence or a sequence comprised by an episome) and the repressor is expressed during the DNA or vector amplification method of the invention, whereby the promoter controlling Cas expression is repressed. In delivery vehicles that are subsequently produced from isolated amplified DNA/vector, the vehicle is devoid of an expressible nucleotide sequence encoding the repressor, whereby the promoter is functional when the DNA/vector is introduced into a target host cell. For example, in the absence of the repressor the promoter is constitutively ON for expression of the Cas. The system is therefore primed to work once the DNA/vector is introduced into the host cells, and this effect can be enhanced further by using a high copy number DNA/vector comprising an origin of replication that is operable in the host cell. A high copy number vector or DNA is also desirable in the production strain cells for enhancing yield of the DNA/vector, and by use of an attenuated promoter as described herein (eg, medium strength promoter and/or repressed promoter in the production strain cells) one can minimize Cas toxicity whilst culturing to maximize amplification and thus yield of the DNA/vector.

17. The method or use of any preceding paragraph, wherein the promoter is a medium strength promoter.

18. The method or use of any preceding paragraph, wherein the promoter has an Anderson Score (AS) of $0.5 > AS > 0.1$.

19. The method or use of any preceding paragraph, wherein the nucleotide sequence encoding said Cas is under the control of a promoter and translation initiation site (TIS) combination that is capable of producing expression of green fluorescent protein (GFP) from a first expression operating unit (EOU) in *E. coli* strain BW25113 cells with a fluorescence of from 0.5 to 4 times the fluorescence produced in *E. coli* strain BW25113 cells using a second EOU comprising a P10 promoter (SEQ ID NO: 1) combined with a BCD14 TIS (SEQ ID NO: 2), wherein the EOUs differ only in their promoter and TIS combinations, wherein each EOU comprises (in 5' to 3' direction) an upstream initiator, the respective promoter, the respective TIS, a nucleotide sequence encoding GFP, a 3' UTR, a transcription terminator and a downstream insulator.

20. The method or use of paragraph 19, wherein fluorescence using the first EOU is 0.5 to 2 times the fluorescence using the second EOU.

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21. The method or use of any preceding paragraph, wherein the nuclease is Cas3 and optionally the DNA or cell encodes cognate Cascade proteins.

22. The method or use of any one of paragraphs 1 to 20, wherein the Cas is a Cas9.

23. The method or use of any preceding paragraph, wherein the production strain cells comprise a helper phage genome that is inducible to produce phage coat proteins in the cells, wherein the method further comprises inducing production of the phage proteins and causing packaging of the amplified DNA into phage particles or non-self-replicative transduction particles, and further isolating the phage or transduction particles and optionally formulating the particles into a pharmaceutical composition for administration to a human or animal subject for treating or reducing the risk of a disease or condition in the subject.

24. The method or use of paragraph 23, wherein the particles are capable of infecting target host cells in the subject and transducing the cells with the DNA, wherein the Cas and crRNAs (or guide RNAs, gRNAs) encoded by the DNA are expressed in the cells, the crRNAs or (gRNAs) being operable to guide the Cas to a target nucleotide sequence (optionally a chromosomal sequence) comprised by the cells, wherein the Cas cuts the target sequences in the cells, thereby killing host cells and treating or reducing the risk of the disease or condition.

25. The method or use of paragraph 24, wherein the host cells are bacterial or archaeal cells, optionally, the host cells are *C. difficile*, *P. aeruginosa*, *K. pneumoniae* (eg, carbapenem-resistant *Klebsiella pneumoniae* or Extended-Spectrum Beta-Lactamase (ESBL)-producing *K. pneumoniae*), *E. coli* (eg, ESBL-producing *E. coli*, or *E. coli* ST131-O25b:H4), *H. pylori*, *S. pneumoniae* or *S. aureus* cells.

26. The method or use of any preceding paragraph, wherein each DNA is comprised by a high copy number plasmid or phagemid.

27. The method or use of any preceding paragraph, wherein the DNA construct comprises one or more nucleotide sequences for producing crRNAs or gRNAs that are operable for Cas nuclease targeting in target host cells.

Paragraphs & Generally Applicable Features

The invention provides the following Paragraphs, which are supported by the Examples below. Any features of the Concepts are combinable with any features of the Embodiments. Any features of the Concepts are combinable with any features of the Embodiments. Any features of the Paragraphs are combinable with any features of the Embodiments.

Any cell herein (eg, a production strain cell or target host cell) may be a bacterial cell, archaeal cell, algal cell, fungal cell, protozoan cell, invertebrate cell, vertebrate cell, fish cell, bird cell, mammal cell, companion animal cell, dog cell, cat cell, horse cell, mouse cell, rat cell, rabbit cell, eukaryotic cell, prokaryotic cell, human cell, animal cell, rodent cell, insect cell or plant cell. Preferably, the cell is a bacterial cell. Alternatively, the cell is a human cell. Optionally, the production strain cell(s) and target host cell(s) are of the same phylum, order, family, genus, species or strain.

1. A nucleic acid vector for introduction into a host cell, the vector comprising a first nucleotide sequence encoding a Type I Cas3, wherein the sequence is under the control of a promoter comprised by the vector for expression of the Cas3 in the cell.

In an example, the vector is a DNA vector, eg, ssDNA vector or dsDNA vector.

2. The vector of paragraph 1, wherein the vector comprises a second nucleotide sequence encoding one or more

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Cascade proteins, wherein the first and second sequences are under the control of one or more promoters comprised by the vector for expression of the proteins in the cell.

3. The vector of paragraph 2, wherein the Cascade protein (s) are cognate with the Cas3.

In an example, the Cas3 is cognate with Cascade proteins encoded by the host cell and/or encoded by a second operon. Optionally, the second operon is comprised by the vector. Optionally, the second operon is comprised by a second vector that is capable of introducing the second operon into the host cell, whereby the Cas3 and Cascade proteins are expressed from the operons in the host cell and are operable with crRNA or gRNA to target the Cas to a host cell target sequence, wherein the Cas3 is capable of modifying the target sequence.

4. The vector of paragraph 2 or 3, wherein the vector comprises an operon for expression in the cell of the Cas3 and Cascade proteins from a Cas module, the module comprising the nucleotide sequences encoding the Cas3 and Cascade proteins, and the operon comprising the Cas module under the control of a promoter for controlling the expression of both the Cas3 and Cascade proteins.

The term "operon" is known to the skilled person such as relating to a functioning unit of DNA containing at least expressible 2 nucleotide sequences respectively encoding for an expression product (eg, a respective translatable mRNA), wherein the sequences are under common promoter control.

5. The vector of paragraph 4, wherein the first sequence is between the promoter and the second sequence in the operon.

6. The vector of paragraph 4 or 5, wherein the operon comprises no Cas-encoding nucleotide sequences between the promoter and the first nucleotide sequence.

Optionally, the Cas3 is a Cas3 encoded by a CRISPR/Cas locus of a first bacterial or archaeal species, wherein in the locus the Cas3-encoding sequence is 3' of Cascade protein-encoding sequences (ie, the latter are between the Cas3 and the 5'-most promoter of the locus).

Optionally, the Cas3 is a ygcB protein (eg, wherein the production strain cell and/or host target cell is an *E. coli*).

Optionally, the Cascade proteins comprise or consist of cas5 (casD, csy2)

cas6 (cas6f, cse3, casE)

cas7 (csc2, csy3, cse4, casC)

cas8 (casA, cas8a1, cas8b1, cas8c, cas10d, cas8e, cse1, cas8f, csy1).

Optionally herein the promoter and the Cas3-encoding sequence are spaced no more than 150, 100, 50, 40, 30, 20 or 10 bp apart, eg, from 30-45, or 30-40, or 39 or around 39 bp apart.

Optionally herein a ribosome binding site and the Cas3-encoding sequence are spaced no more than 20, 15, 14, 13, 12, 11, 10, 9, 8, 7, 6, 4 or 3 bp apart, eg, from 10-5, 6 or around 6 bp apart.

7. The vector of any one of paragraphs 4 to 6, wherein the operon comprises (in 5' to 3' direction) the promoter, the first sequence and the second sequence.

8. The vector of any preceding paragraph, wherein each promoter is a constitutive promoter.

9. The vector of any one of paragraphs 1 to 7, wherein the promoter is repressible (optionally repressible by a tetracycline repressor or lac repressor).

10. The vector of any one of paragraphs 1 to 7, wherein the promoter is inducible.

11. The vector of any preceding paragraph, wherein the first sequence is under the control of a weak promoter.

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12. The vector of any one of paragraphs 1 to 7, wherein the first sequence is under the control of a medium strength promoter.

13. The vector of any one of paragraphs 1 to 7, wherein the first sequence is under the control of a strong promoter.

In an example, the promoter is in combination with a Shine-Dalgarno sequence comprising the sequence 5'-aaagaggagaaa-3' (SEQ ID NO: 5) or a ribosome binding site homologue thereof.

14. The vector of any one of paragraphs 1 to 7, wherein the first sequence is under the control of a promoter that has an Anderson Score (AS) of $AS \geq 0.5$.

See Table 2 for more information on Anderson Scores in relation to promoters.

15. The vector of any one of paragraphs 1 to 7, wherein the first sequence is under the control of a promoter that has an Anderson Score (AS) of $0.5 > AS > 0.1$.

16. The vector of any one of paragraphs 1 to 7, wherein the first sequence is under the control of a promoter that has an Anderson Score (AS) of ≤ 0.1 .

17. The vector of any one of paragraphs 1 to 7, wherein the first sequence (and optionally the second sequence) is under the control of a promoter and translation initiation site (TIS) combination that is capable of producing expression of green fluorescent protein (GFP) from a first expression operating unit (EOU) in *E. coli* strain BW25113 cells with a fluorescence of from 0.5 to 4 times the fluorescence produced in *E. coli* strain BW25113 cells using a second EOU comprising a P10 promoter (SEQ ID NO: 1) combined with a BCD14 TIS (SEQ ID NO: 2), wherein the EOUs differ only in their promoter and TIS combinations, wherein each EOU comprises (in 5' to 3' direction) an upstream initiator, the respective promoter, the respective TIS, a nucleotide sequence encoding GFP, a 3' UTR, a transcription terminator and a downstream insulator.

18. The vector of paragraph 17, wherein fluorescence using the first EOU is 0.5 to 2 times the fluorescence using the second EOU.

For example, fluorescence using the first EOU is 0.5 to X times the fluorescence using the second EOU, wherein X is from 3.0 to 1.0, eg, 3, 2.5, 2, 1.5 or 1, wherein fluorescence is determined using excitation at 481 nm and emission at 507 nm. Optionally, *E. coli* cultures at OD600 of 0.3-0.5 in the exponential growth phase are used.

For example, the upstream insulator, the nucleotide sequence encoding GFP, 3' UTR, transcription terminator and downstream insulator of each EOU are as disclosed in Mutalik et al (2013). For example, the upstream insulator, the nucleotide sequence encoding GFP, 3' UTR, transcription terminator and downstream insulator of each EOU are corresponding sequences of SEQ ID NO: 4. For example, the *E. coli* is *E. coli* BW25113 is grown in MOPS EZ Rich Medium (Teknova) supplemented with 50 µg/ml kanamycin (kan) at 37° C., shaken at 900 r.p.m. For example, each EOUs is comprised by a medium copy plasmid, eg, a plasmid derived from pFAB217 comprising a p15A replication origin and a kan resistance gene.

19. The vector of any preceding paragraph, wherein the vector comprises an origin of replication that is operable in the host cell.

20. The vector of any preceding paragraph, wherein the vector comprises an origin of replication that is operable in a bacterial cell of a vector production strain, wherein the Cas3 is not operable in the production strain cell to target and cut a chromosomal sequence thereof.

An example of a production strain cell is an *E. coli* cell. A production strain cell is a cell that is used to amplify DNA

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encoding Cas (and optionally other components of a CRISPR/Cas system). Usefully, the strain may package the amplified DNA into transduction particles that are may be isolated to produce a composition that can be contacted with a population of target host cells (eg, bacterial, archaeal, prokaryotic, eukaryotic, human, animal, mammal, rodent, mouse, rat, rabbit, *Xenopus*, fish, bird, amphibian, insect, plant, amoeba or algae cells) wherein the DNA is introduced into the cells for expression of the Cas (and optional other CRISPR/Cas system components), wherein the Cas is guided to a protospacer target sequence in the host cells and modifies (eg, cuts) the sequence. In another example, the amplified DNA isolated from a population of production strain cells and is combined with a delivery vehicle (eg, a carrier bacterium, nanoparticle or liposome), wherein the delivery vehicle can be contacted with a population of target host cells (eg, bacterial, archaeal, prokaryotic, eukaryotic, human, animal, mammal, rodent, mouse, rat, rabbit, *Xenopus*, fish, bird, amphibian, insect, plant, amoeba or algae cells) wherein the DNA is introduced into the cells for expression of the Cas (and optional other CRISPR/Cas system components), wherein the Cas is guided to a protospacer target sequence in the host cells and modifies (eg, cuts) the sequence.

21. The vector of paragraph 20, wherein the first sequence is under the control of a promoter that is capable of controlling expression of the Cas3 at a level that is not toxic to the production strain cell.

In an example, substantially no production strain cells are killed when the Cas3-encoding sequence is amplified therein. In another example, no more than 40, 30, 20, 10, 5, 4, 3, 2, or 1% of production strain cells are killed when the Cas3-encoding sequence is amplified therein. For example this is in a 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 12 or 24 hour period of culturing the cells.

22. The vector of paragraph 20, wherein the first sequence is under the control of a promoter that controls expression of the Cas3 in the production strain cell such that the cell is capable of growth and propagation sufficient to produce at least 1000 copies of the vector.

For example this is in a 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 12 or 24 hour period of culturing the cells. For example, at least 10^4 , 10^5 , 10^6 , 10^7 , 10^8 , 10^9 , 10^{10} , 10^{11} , 10^{12} , 10^{13} , 10^{14} , 10^{15} , 10^{16} , 10^{17} or 10^{18} copies of the vector are produced per 10^3 , 10^4 , 10^5 , 10^6 , 10^7 , 10^8 , 10^9 , 10^{10} , 10^{11} , 10^{12} , 10^{13} , 10^{14} , 10^{15} , 10^{16} , 10^{17} production strain cells respectively.

23. The vector of any one of paragraphs 20 to 22, wherein the cell is capable of at least 2 or 3 logs of expansion when the vector is comprised therein.

For example, this is in a 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 12 or 24 hour period of culturing the cells.

24. The vector of any preceding paragraph, wherein the vector is a high copy number vector.

25. The vector of any preceding paragraph, wherein the first nucleotide sequence or operon is comprised by a mobile genetic element.

Suitable mobile genetic elements, eg, transposons, are disclosed in WO2016177682 and US20170246221, the disclosures of which are explicitly incorporated herein for possible use in the invention and for providing one or more features for the claims herein.

26. The vector of any preceding paragraph, wherein the vector is devoid of a Cas adaption module.

For example, the vector is devoid of nucleotide sequences encoding a Cas1, Cas2 and/or Cas4.

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27. The vector of any preceding paragraph, wherein the vector is devoid of nucleotide sequence encoding one, more or all of a Cas1, Cas2, Cas4, Cas6 (optionally Cas6f), Cas7 and Cas8 (optionally Cas8f).

28. The vector of any preceding paragraph, wherein the vector is devoid of a sequence encoding a Cas6 (optionally a Cas6f).

29. The vector of any one of paragraphs 1 to 27, wherein the module encodes a Cas6 (optionally a Cas6f).

30. The vector of any preceding paragraph, wherein the vector comprises (optionally in 5' to 3' direction) nucleotide sequence encoding one, more or all of Cas11, Cas7 and Cas8a1.

31. The vector of paragraph 30, wherein the vector comprises nucleotide sequence encoding Cas3' and/or Cas3" (optionally wherein the nucleotide sequences encoding the Cas3' and/or Cas3" are between the promoter and the sequence(s) recited in paragraph 30).

In one embodiment, the vector comprises nucleotide sequences (in 5' to 3' direction) that encode a Cas3 (eg, Cas3' and/or Cas3"), Cas11, Cas7 and Cas8a1. Optionally, a nucleotide sequence encoding Cas6 is between the Cas3 sequence(s) and the Cas11 sequence. Optionally, the vector comprises a Type IA CRISPR array or one or more nucleotide sequences encoding single guide RNA(s) (gRNA(s)), wherein the array and each gRNA comprises repeat sequence that is cognate with the Cas3. Thus, the array is operable in a host cell when the vector has been introduced into the cell for production of guide RNAs, wherein the guide RNAs are operable with the Cas and Cascade proteins to target and modify (eg, cut) a target nucleotide sequence in the host cell, optionally thereby killing the host cell. Similarly, the single guide RNAs encoded by the vector in one embodiment are operable with the Cas and Cascade proteins to target and modify (eg, cut) a target nucleotide sequence in the host cell, optionally thereby killing the host cell.

32. The vector of paragraph 30 or 31, wherein the host cell comprises a Type IA CRISPR array that is cognate with the Cas3.

33. The vector of paragraph 30 or 31, wherein the host cell comprises an endogenous Type IB, C, U, D, E or F CRISPR/Cas system.

34. The vector of any one of paragraphs 1 to 29, wherein the vector comprises (optionally in 5' to 3' direction) nucleotide sequence encoding one, more or all of Cas8b1, Cas7 and Cas5.

35. The vector of paragraph 34, wherein the vector comprises a nucleotide sequence encoding Cas3 between the promoter and the sequence(s) recited in paragraph 34.

In one embodiment, the vector comprises nucleotide sequences (in 5' to 3' direction) that encode a Cas3, Cas8b1, Cas7 and Cas5. Optionally, a nucleotide sequence encoding Cas6 is between the Cas3 sequence(s) and the Cas8b1 sequence. Optionally, the vector comprises a Type IB CRISPR array or one or more nucleotide sequences encoding single guide RNA(s) (gRNA(s)), wherein the array and each gRNA comprises repeat sequence that is cognate with the Cas3. Thus, the array is operable in a host cell when the vector has been introduced into the cell for production of guide RNAs, wherein the guide RNAs are operable with the Cas and Cascade proteins to target and modify (eg, cut) a target nucleotide sequence in the host cell, optionally thereby killing the host cell. Similarly, the single guide RNAs encoded by the vector in one embodiment are operable with the Cas and Cascade proteins to target and modify (eg, cut) a target nucleotide sequence in the host cell, optionally thereby killing the host cell.

36. The vector of paragraph 34 or 35, wherein the host cell comprises a Type IB CRISPR array that is cognate with the Cas3.

37. The vector of paragraph 34 or 35, wherein the host cell comprises an endogenous Type IA, C, U, D, E or F CRISPR/Cas system.

38. The vector of any one of paragraphs 1 to 29, wherein the vector comprises (optionally in 5' to 3' direction) nucleotide sequence encoding one, more or all of Cas5, Cas8c and Cas7.

39. The vector of paragraph 38, wherein the vector comprises a nucleotide sequence encoding Cas3 between the promoter and the sequence(s) recited in paragraph 38.

In one embodiment, the vector comprises nucleotide sequences (in 5' to 3' direction) that encode a Cas3, Cas5, Cas8c and Cas7. Optionally, a nucleotide sequence encoding Cas6 is between the Cas3 sequence(s) and the Cas5 sequence. Optionally, the vector comprises a Type IC CRISPR array or one or more nucleotide sequences encoding single guide RNA(s) (gRNA(s)), wherein the array and each gRNA comprises repeat sequence that is cognate with the Cas3. Thus, the array is operable in a host cell when the vector has been introduced into the cell for production of guide RNAs, wherein the guide RNAs are operable with the Cas and Cascade proteins to target and modify (eg, cut) a target nucleotide sequence in the host cell, optionally thereby killing the host cell. Similarly, the single guide RNAs encoded by the vector in one embodiment are operable with the Cas and Cascade proteins to target and modify (eg, cut) a target nucleotide sequence in the host cell, optionally thereby killing the host cell.

40. The vector of paragraph 38 or 39, wherein the host cell comprises a Type IC CRISPR array that is cognate with the Cas3.

41. The vector of paragraph 38 or 39, wherein the host cell comprises an endogenous Type IA, B, U, D, E or F CRISPR/Cas system.

42. The vector of any one of paragraphs 1 to 29, wherein the vector comprises (optionally in 5' to 3' direction) nucleotide sequence encoding one, more or all of Cas8U2, Cas7, Cas5 and Cas6.

43. The vector of paragraph 42, wherein the vector comprises a nucleotide sequence encoding Cas3 between the promoter and the sequence(s) recited in paragraph 42.

In one embodiment, the vector comprises nucleotide sequences (in 5' to 3' direction) that encode a Cas3, Cas8U2, Cas7, Cas5 and Cas6. Optionally, a nucleotide sequence encoding Cas6 is between the Cas3 sequence(s) and the Cas8U2 sequence. Optionally, the vector comprises a Type IU CRISPR array or one or more nucleotide sequences encoding single guide RNA(s) (gRNA(s)), wherein the array and each gRNA comprises repeat sequence that is cognate with the Cas3. Thus, the array is operable in a host cell when the vector has been introduced into the cell for production of guide RNAs, wherein the guide RNAs are operable with the Cas and Cascade proteins to target and modify (eg, cut) a target nucleotide sequence in the host cell, optionally thereby killing the host cell. Similarly, the single guide RNAs encoded by the vector in one embodiment are operable with the Cas and Cascade proteins to target and modify (eg, cut) a target nucleotide sequence in the host cell, optionally thereby killing the host cell.

44. The vector of paragraph 42 or 43, wherein the host cell comprises a Type IU CRISPR array that is cognate with the Cas3.

45. The vector of paragraph 42 or 43, wherein the host cell comprises an endogenous Type IA, B, C, D, E or F CRISPR/Cas system.

46. The vector of any one of paragraphs 1 to 29, wherein the vector comprises (optionally in 5' to 3' direction) nucleotide sequence encoding one, more or all of Cas10d, Cas7 and Cas5.

47. The vector of paragraph 46, wherein the vector comprises a nucleotide sequence encoding Cas3' and/or Cas3" (optionally wherein the nucleotide sequences encoding the Cas3' and/or Cas3" are between the promoter and the sequence(s) recited in paragraph 46).

In one embodiment, the vector comprises nucleotide sequences (in 5' to 3' direction) that encode a Cas3, Cas10d, Cas7 and Cas5. Optionally, a nucleotide sequence encoding Cas6 is between the Cas3 sequence(s) and the Cas10d sequence. Optionally, the vector comprises a Type ID CRISPR array or one or more nucleotide sequences encoding single guide RNA(s) (gRNA(s)), wherein the array and each gRNA comprises repeat sequence that is cognate with the Cas3. Thus, the array is operable in a host cell when the vector has been introduced into the cell for production of guide RNAs, wherein the guide RNAs are operable with the Cas and Cascade proteins to target and modify (eg, cut) a target nucleotide sequence in the host cell, optionally thereby killing the host cell. Similarly, the single guide RNAs encoded by the vector in one embodiment are operable with the Cas and Cascade proteins to target and modify (eg, cut) a target nucleotide sequence in the host cell, optionally thereby killing the host cell.

48. The vector of paragraph 46 or 47, wherein the host cell comprises a Type ID CRISPR array that is cognate with the Cas3.

49. The vector of paragraph 46 or 47, wherein the host cell comprises an endogenous Type IA, B, C, U, E or F CRISPR/Cas system.

50. The vector of any one of paragraphs 1 to 29, wherein the vector comprises (optionally in 5' to 3' direction) nucleotide sequence encoding one, more or all of Cas8e, Cas11, Cas7, Cas5 and Cas6.

51. The vector of paragraph 50, wherein the vector comprises a nucleotide sequence encoding Cas3 between the promoter and the sequence(s) recited in paragraph 50.

In one embodiment, the vector comprises nucleotide sequences (in 5' to 3' direction) that encode a Cas3, Cas8e, Cas11, Cas7, Cas5 and Cas6. Optionally, a nucleotide sequence encoding Cas6 is between the Cas3 sequence(s) and the Cas11 sequence. Optionally, the vector comprises a Type IE CRISPR array or one or more nucleotide sequences encoding single guide RNA(s) (gRNA(s)), wherein the array and each gRNA comprises repeat sequence that is cognate with the Cas3. Thus, the array is operable in a host cell when the vector has been introduced into the cell for production of guide RNAs, wherein the guide RNAs are operable with the Cas and Cascade proteins to target and modify (eg, cut) a target nucleotide sequence in the host cell, optionally thereby killing the host cell. Similarly, the single guide RNAs encoded by the vector in one embodiment are operable with the Cas and Cascade proteins to target and modify (eg, cut) a target nucleotide sequence in the host cell, optionally thereby killing the host cell.

52. The vector of paragraph 50 or 51, wherein the host cell comprises a Type IE CRISPR array that is cognate with the Cas3.

53. The vector of paragraph 50 or 51, wherein the host cell comprises an endogenous Type IA, B, C, D, U or F CRISPR/Cas system.

54. The vector of any one of paragraphs 1 to 29, wherein the vector comprises (optionally in 5' to 3' direction) nucleotide sequence encoding one, more or all of Cas8f, Cas5, Cas7 and Cas6f.

55. The vector of paragraph 54, wherein the vector comprises a nucleotide sequence encoding Cas3 between the promoter and the sequence(s) recited in paragraph 54, wherein the vector is devoid of nucleotide sequence encoding further Cas between the promoter and the sequence encoding the Cas3.

In one embodiment, the vector comprises nucleotide sequences (in 5' to 3' direction) that encode a Cas3, Cas8f, Cas5, Cas7 and Cas6f. Optionally, a nucleotide sequence encoding Cas6 is between the Cas3 sequence(s) and the Cas8f sequence. Optionally, the vector comprises a Type I CRISPR array or one or more nucleotide sequences encoding single guide RNA(s) (gRNA(s)), wherein the array and each gRNA comprises repeat sequence that is cognate with the Cas3. Thus, the array is operable in a host cell when the vector has been introduced into the cell for production of guide RNAs, wherein the guide RNAs are operable with the Cas and Cascade proteins to target and modify (eg, cut) a target nucleotide sequence in the host cell, optionally thereby killing the host cell. Similarly, the single guide RNAs encoded by the vector in one embodiment are operable with the Cas and Cascade proteins to target and modify (eg, cut) a target nucleotide sequence in the host cell, optionally thereby killing the host cell.

56. The vector of paragraph 54 or 55, wherein the host cell comprises a Type I CRISPR array that is cognate with the Cas3.

57. The vector of paragraph 54 or 55, wherein the host cell comprises an endogenous Type IA, B, C, D, U or E CRISPR/Cas system.

58. The vector of any one of paragraphs 1 to 29, wherein the Cas and Cascade are Type IA Cas and Cascade proteins.

59. The vector of any one of paragraphs 1 to 29, wherein the Cas and Cascade are Type IB Cas and Cascade proteins.

60. The vector of any one of paragraphs 1 to 29, wherein the Cas and Cascade are Type IC Cas and Cascade proteins.

61. The vector of any one of paragraphs 1 to 29, wherein the Cas and Cascade are Type ID Cas and Cascade proteins.

62. The vector of any one of paragraphs 1 to 29, wherein the Cas and Cascade are Type IE Cas and Cascade proteins.

63. The vector of any one of paragraphs 1 to 29, wherein the Cas and Cascade are Type IF Cas and Cascade proteins.

64. The vector of any one of paragraphs 1 to 29, wherein the Cas and Cascade are Type IU Cas and Cascade proteins.

65. The vector of any one of paragraphs 1 to 29, wherein the Cas and Cascade are *E. coli* (optionally Type IE or IF) Cas and Cascade proteins, optionally wherein the *E. coli* is ESBL-producing *E. coli* or *E. coli* ST131-O25b:H4.

66. The vector of any one of paragraphs 1 to 29, wherein the Cas and Cascade are *Clostridium* (eg, *C. difficile*) Cas and Cascade proteins, optionally *C. difficile* resistant to one or more antibiotics selected from aminoglycosides, lincomycin, tetracyclines, erythromycin, clindamycin, penicillins, cephalosporins and fluoroquinolones.

67. The vector of any one of paragraphs 1 to 29, wherein the Cas and Cascade are *Pseudomonas aeruginosa* Cas and Cascade proteins, optionally *P. aeruginosa* resistant to one or more antibiotics selected from carbapenems, aminoglycosides, cefepime, ceftazidime, fluoroquinolones, piperacillin and tazobactam.

68. The vector of any one of paragraphs 1 to 29, wherein the Cas and Cascade are *Klebsiella pneumoniae* (eg, carbapenem-resistant *Klebsiella pneumoniae* or Extended-

Spectrum Beta-Lactamase (ESBL)-producing *K. pneumoniae*) Cas and Cascade proteins.

69. The vector of any one of paragraphs 1 to 29, wherein the Cas and Cascade are *E. coli*, *C. difficile*, *P. aeruginosa*, *K. pneumoniae*, *P. furiosus* or *B. halodurans* Cas and Cascade proteins.

70. The vector of any preceding paragraph, wherein the Cas3 is a Cas3 of a CRISPR/Cas locus of a first bacterial or archaeal species, wherein the distance between the Cas3-encoding sequence of the locus and its cognate promoter is further than the distance between the Cas3-encoding sequence and the respective promoter comprised by the vector.

The cognate promoter here is the one that controls expression of Cas3 in the wild-type locus.

71. The vector of any preceding paragraph, wherein the distance between the promoter and the Cas3-encoding sequence and/or Cascade protein-encoding sequence(s) is shorter than in a corresponding wild-type Type I locus.

A corresponding locus is a wild-type locus of a bacterial or archaeal species or strain that comprises an endogenous CRISPR/Cas system encoding the Cas3 and/or Cascade proteins of the type that are also encoded by the vector. Thus, when the vector comprises an operon, the operon may comprise Cas3- and Cascade-encoding nucleotide sequences that are not in a natural configuration.

72. The vector of any preceding paragraph, wherein the vector comprises (i) a CRISPR array for producing crRNAs in the host cell and/or (ii) one or more nucleotide sequences encoding one or more single guide RNAs (gRNAs), wherein the crRNAs or gRNAs are cognate to the Cas3 (and optionally cognate to the Cascade proteins).

73. The vector of paragraph 72 when dependent from paragraph 4, wherein the array or gRNA-encoding sequence(s) are comprised by the operon and under the control of the promoter.

74. The vector of paragraph 72, wherein the array or gRNA-encoding sequence(s) are under the control of a promoter that is different from the promoter that controls the expression of the Cas3.

75. The vector of any one of paragraphs 72 to 74, wherein one or more of the crRNAs or gRNAs comprises a spacer sequence that is capable of hybridising to a target nucleotide sequence of the host cell, wherein the target sequence is adjacent a PAM, the PAM being cognate to the Cas3.

Thus, the spacer hybridises to the protospacer to guide the Cas3 to the protospacer. Optionally, the Cas3 cuts the protospacer, eg, using exo- and/or endonuclease activity of the Cas3. Optionally, the Cas3 removes a plurality (eg, at least 2, 3, 4, 5, 6, 7, 8, 9 or 10) nucleotides from the protospacer.

76. The vector of paragraph 75, wherein the target sequence is a chromosomal sequence of the host cell.

77. The vector of paragraph 75 or 76, wherein the Cas3 is operable to cut the target sequence.

78. The vector of any preceding paragraph, wherein the vector is a plasmid or phagemid.

79. A delivery vehicle comprising the vector of any preceding paragraph, wherein the delivery vehicle is capable of delivering the vector into the host cell.

80. The vehicle of paragraph 79, wherein the delivery vehicle is a phage, non-replicative transduction particle, nanoparticle carrier, bacterium or liposome.

The phage or particles comprise phage coat proteins encapsidating DNA, wherein the DNA comprises the vector. Suitable examples of phage and particles are disclosed in U.S. Ser. No. 15/985,658 (and its equivalent publication by

USPTO) the disclosures of which are incorporated herein by reference for possible use in the invention and for providing one or more features that may be included in the claims herein. Phage or particle is capable of infecting the cell, thereby introducing the vector into the cell.

81. The vector or vehicle of any preceding paragraph, wherein the host cell is a bacterial or archaeal cell, optionally, the host cell is a *C. difficile*, *P. aeruginosa*, *K. pneumoniae* (eg, carbapenem-resistant *Klebsiella pneumoniae* or Extended-Spectrum Beta-Lactamase (ESBL)-producing *K. pneumoniae*), *E. coli* (eg, ESBL-producing *E. coli*, or *E. coli* ST131-O25b:H4), *H. pylori*, *S. pneumoniae* or *S. aureus* cell.

82. The vector or vehicle of any preceding paragraph for administration to a human or animal subject for treating or reducing the risk of a disease or condition in the subject.

83. The vector or vehicle of paragraph 82, wherein the disease or condition is an infection of the subject with host cells (eg, bacterial cells), or wherein the disease or condition is mediated by host cells (eg, bacterial cells).

84. A pharmaceutical composition comprising the vector or vehicle of any preceding paragraph and a pharmaceutically acceptable diluent, excipient or carrier.

85. A method of amplifying copies of a DNA encoding a functional Cas protein (optionally a Cas nuclease) in a bacterial or archaeal production strain of cells, the method comprising

(a) Providing production strain cells, each cell comprising a copy of said DNA, wherein each DNA comprises a nucleotide sequence encoding said Cas, wherein the nucleotide sequence is under the control of a promoter for controlling the expression of the Cas in the production strain cell, the DNA comprising an origin of replication that is operable in the cell for replication of the DNA;

(b) Culturing the cells to allow replication of the DNA, whereby the DNA is amplified; and

(c) Optionally isolating copies of the DNA,

86. The method of paragraph 85, wherein the promoter is a constitutive promoter.

87. The method of paragraph 85, wherein the promoter is repressible (optionally repressible by a tetracycline repressor or a lac repressor).

88. The method of paragraph 85, wherein the promoter is inducible.

89. The method of any one of paragraphs 85 to 88, wherein the promoter is a medium strength promoter.

90. The method of any one of paragraphs 85 to 89, wherein the promoter has an Anderson Score (AS) of $0.5 > AS > 0.1$.

91. The method of any one of paragraphs 85 to 90, wherein the nucleotide sequence encoding said Cas is under the control of a promoter and translation initiation site (TIS) combination that is capable of producing expression of green fluorescent protein (GFP) from a first expression operating unit (EOU) in *E. coli* strain BW25113 cells with a fluorescence of from 0.5 to 4 times the fluorescence produced in *E. coli* strain BW25113 cells using a second EOU comprising a P10 promoter (SEQ ID NO: 1) combined with a BCD14 TIS (SEQ ID NO: 2), wherein the EOUs differ only in their promoter and TIS combinations, wherein each EOU comprises (in 5' to 3' direction) an upstream initiator, the respective promoter, the respective TIS, a nucleotide sequence encoding GFP, a 3' UTR, a transcription terminator and a downstream insulator.

92. The method of paragraph 91, wherein fluorescence using the first EOU is 0.5 to 2 times the fluorescence using the second EOU.

93. The method of any one of paragraphs 85 to 92, wherein the nuclease is Cas3 and optionally the DNA or cell encodes cognate Cascade proteins and/or one or more crRNAs that are operable for Cas nuclease targeting.

For example, the targeting is targeting of the Cas to a protospacer sequence comprised by a host cell chromosome or an episome thereof. In another example the targeting is in a recombineering method and the Cas is targeted to a protospacer sequence of a DNA that has been introduced into or amplified in the host cell. In an example of such recombineering, the host cell is an *E. coli* cell.

94. The method of any one of paragraphs 85 to 92, wherein the Cas is a Cas9.

95. The method of any one of paragraphs 85 to 92, wherein the Cas is a Type IIIA csm protein or a Type IIIB cmr protein.

96. The method of any one of paragraphs 85 to 92, wherein the Cas is a CsfI.

97. The method of any one of paragraphs 85 to 92, wherein the Cas is a CpfI.

98. The method of any one of paragraphs 85 to 92, wherein the Cas is a Cas13 (optionally Cas13a or Cas13b).

99. The method of any one of paragraphs 85 to 92, wherein the Cas is selected from a Cas3, Cas8a, Cas5, Cas8b, Cas8c, Cas10d, Cse1, Cse2, Csy1, Csy2, Csy3, GSU0054, Cas10, Csm2, Cmr5, Cas10, Csx11, Csx10, Csf1, Cas9, Csn2, Cas4, Cpf1, C2c1, C2c3, Cas13a, Cas13b and Cas13c.

100. The method of any one of paragraphs 85 to 99, wherein the production strain cells comprise a helper phage genome that is inducible to produce phage coat proteins in the cells, wherein the method further comprises inducing production of the phage proteins and causing packaging of the amplified DNA into phage particles or non-self-replicative transduction particles, and further isolating the phage or transduction particles and optionally formulating the particles into a pharmaceutical composition for administration to a human or animal subject for treating or reducing the risk of a disease or condition in the subject.

101. The method of paragraph 100, wherein the particles are capable of infecting target host cells in the subject and transducing the cells with the DNA, wherein the Cas and crRNAs (or gRNAs) encoded by the DNA are expressed in the cells, the crRNAs or (gRNAs) being operable to guide the Cas to a target nucleotide sequence (optionally a chromosomal sequence) comprised by the cells, wherein the Cas cuts the target sequences in the cells, thereby killing host cells and treating or reducing the risk of the disease or condition.

102. The method of paragraph 101, wherein the host cells are bacterial or archaeal cells, optionally, the host cells are *C. difficile*, *P. aeruginosa*, *K. pneumoniae* (eg, carbapenem-resistant *Klebsiella pneumoniae* or Extended-Spectrum Beta-Lactamase (ESBL)-producing *K. pneumoniae*), *E. coli* (eg, ESBL-producing *E. coli*, or *E. coli* ST131-O25b:H4), *H. pylori*, *S. pneumoniae* or *S. aureus* cells.

103. The method of any one of paragraphs 85 to 102, wherein each DNA is comprised by a high copy number vector, optionally a high copy number plasmid (an optionally the promoter is a constitutive promoter).

104. The method of any one of paragraphs 85 to 103, wherein each DNA is comprised by a vector or vehicle according to any one of paragraphs 1 to 83.

105. Use of an attenuated strength promoter in a DNA construct comprising a nucleotide sequence encoding a functional Cas protein (optionally a Cas nuclease) that is under the control of the promoter, in a method of amplifying copies of the DNA in a population of bacterial or archaeal production strain cells, the method comprising culturing the cells to allow replication of the DNA thereby amplifying the DNA in the cells, for enhancing the yield of amplified DNA produced by the production host cells.

Thus, said enhancing may be relative to the yield produced using a strong promoter, eg, a strong constitutive promoter (for example a promoter having an Anderson Score (AS) of $AS \geq 0.5$). In another example, the strong promoter is a promoter comprised by a promoter and translation initiation site (TIS) combination that is capable of producing expression of green fluorescent protein (GFP) from a first expression operating unit (EOU) in *E. coli* strain BW25113 cells with a fluorescence of >4 times the fluorescence produced in *E. coli* strain BW25113 cells using a second EOU comprising a P10 promoter (SEQ ID NO: 1) combined with a BCD14 TIS (SEQ ID NO: 2), wherein the EOUs differ only in their promoter and TIS combinations, wherein each EOU comprises (in 5' to 3' direction) an upstream initiator, the respective promoter, the respective TIS, a nucleotide sequence encoding GFP, a 3' UTR, a transcription terminator and a downstream insulator.

106. The use of paragraph 105, wherein the use is for enhancing said yield by

- (d) reducing toxicity of the Cas in the production strain;
- (e) reducing mutation of the DNA (optionally the Cas-encoding sequence) in the production strain;
- (f) promoting production cell viability during the amplification of the DNA; and/or
- (g) reducing the occurrence of Cas cutting of DNA (optionally cutting of production host cell chromosomal DNA or said DNA construct).

107. Use of an attenuated strength promoter in a DNA construct comprising a nucleotide sequence encoding a functional Cas protein (optionally a Cas nuclease) that is under the control of the promoter, in a method of amplifying copies of the DNA in a population of bacterial or archaeal production strain cells, the method comprising culturing the cells to allow replication of the DNA thereby amplifying the DNA in the cells, for reducing toxicity of the Cas in the production strain.

108. Use of an attenuated strength promoter in a DNA construct comprising a nucleotide sequence encoding a functional Cas protein (optionally a Cas nuclease) that is under the control of the promoter, in a method of amplifying copies of the DNA in a population of bacterial or archaeal production strain cells, the method comprising culturing the cells to allow replication of the DNA thereby amplifying the DNA in the cells, for reducing mutation of the DNA (optionally the Cas-encoding sequence) in the production strain.

109. Use of an attenuated strength promoter in a DNA construct comprising a nucleotide sequence encoding a functional Cas protein (optionally a Cas nuclease) that is under the control of the promoter, in a method of amplifying copies of the DNA in a population of bacterial or archaeal production strain cells, the method comprising culturing the cells to allow replication of the DNA thereby amplifying the DNA in the cells, for promoting production cell viability during the amplification of the DNA.

110. Use of an attenuated strength promoter in a DNA construct comprising a nucleotide sequence encoding a functional Cas protein (optionally a Cas nuclease) that is

under the control of the promoter, in a method of amplifying copies of the DNA in a population of bacterial or archaeal production strain cells, the method comprising culturing the cells to allow replication of the DNA thereby amplifying the DNA in the cells, for reducing the occurrence of Cas cutting of DNA.

111. A method for enhancing the yield of amplified copies of a DNA construct in a population of bacterial or archaeal production strain cells, wherein the construct comprises a nucleotide sequence encoding a functional Cas protein (optionally a Cas nuclease) that is under the control of a promoter, the method comprising culturing the cells to allow replication of the DNA thereby amplifying the DNA in the cells, wherein the promoter is an attenuated strength promoter.

112. A method for reducing toxicity of a functional Cas protein (optionally a Cas nuclease) in a population of bacterial or archaeal production strain cells in a process of amplifying copies of a DNA construct, wherein the construct comprises a nucleotide sequence encoding the Cas and the sequence is under the control of a promoter, the method comprising culturing the cells to allow replication of the DNA thereby amplifying the DNA in the cells, wherein the promoter is an attenuated strength promoter.

113. A method for reducing mutation of a DNA construct encoding a functional Cas protein (optionally a Cas nuclease) in a population of bacterial or archaeal production strain cells in a process of amplifying copies of the construct, wherein the construct comprises a nucleotide sequence encoding the Cas and the sequence is under the control of a promoter, the method comprising culturing the cells to allow replication of the DNA thereby amplifying the DNA in the cells, wherein the promoter is an attenuated strength promoter.

114. A method for promoting production cell viability of a population of bacterial or archaeal production strain cells in a process of amplifying copies of a DNA construct comprised by the cells, wherein the construct comprises a nucleotide sequence encoding a functional Cas protein (optionally a Cas nuclease) and the sequence is under the control of a promoter, the method comprising culturing the cells to allow replication of the DNA thereby amplifying the DNA in the cells, wherein the promoter is an attenuated strength promoter.

115. A method for reducing the occurrence of Cas nuclease cutting of a DNA construct in a population of bacterial or archaeal production strain cells in a process of amplifying copies of the construct, wherein the construct comprises a nucleotide sequence encoding the Cas and the sequence is under the control of a promoter, the method comprising culturing the cells to allow replication of the DNA thereby amplifying the DNA in the cells, wherein the promoter is an attenuated strength promoter.

116. The use of paragraph 108 or 110, or the method of paragraph 113 or 115, wherein the mutation or cutting is mutation or cutting of host cell chromosomal DNA or the construct DNA.

117. The use or method of any one of paragraphs 105 to 116, wherein the promoter is a constitutive promoter.

118. The use or method of any one of paragraphs 105 to 117, wherein the promoter is repressible (optionally repressible by a tetracycline repressor or a lac repressor).

In an example, the promoter is a constitutive promoter and optionally the DNA is comprised by a high copy number plasmid or phagemid.

119. The use or method of any one of paragraphs 105 to 118, wherein the promoter is $P_{LtetO-1}$, $P_{LlacO-1}$ or a repressible homologue thereof.

$P_{LlacO-1}$ is repressed by lac repressor (LacR). $P_{LtetO-1}$ is repressed by tet repressor (TetR).

120. The use or method of any one of paragraphs 105 to 119, wherein the promoter is a medium strength promoter.

121. The use or method of any one of paragraphs 105 to 120, wherein the promoter has an Anderson Score (AS) of $0.5 > AS > 0.1$.

122. The use or method of any one of paragraphs 105 to 121, wherein the nucleotide sequence encoding said Cas is under the control of a promoter and translation initiation site (TIS) combination that is capable of producing expression of green fluorescent protein (GFP) from a first expression operating unit (EOU) in *E. coli* strain BW25113 cells with a fluorescence of from 0.5 to 4 times the fluorescence produced in *E. coli* strain BW25113 cells using a second EOU comprising a P10 promoter (SEQ ID NO: 1) combined with a BCD14 TIS (SEQ ID NO: 2), wherein the EOUs differ only in their promoter and TIS combinations, wherein each EOU comprises (in 5' to 3' direction) an upstream initiator, the respective promoter, the respective TIS, a nucleotide sequence encoding GFP, a 3' UTR, a transcription terminator and a downstream insulator.

123. The use or method of paragraph 122, wherein fluorescence using the first EOU is 0.5 to 2 times the fluorescence using the second EOU.

124. The use or method of any one of paragraphs 105 to 123, wherein the nuclease is Cas3 and optionally the DNA construct encodes cognate Cascade proteins.

125. The use or method of any one of paragraphs 105 to 123, wherein the Cas is a Cas9.

126. The use or method of any one of paragraphs 105 to 123, wherein the Cas is a Type IIIA csm protein or a Type IIIB cmr protein.

127. The use or method of any one of paragraphs 105 to 123, wherein the Cas is a CsfI.

128. The use or method of any one of paragraphs 105 to 123, wherein the Cas is a CpfI.

129. The use or method of any one of paragraphs 105 to 123, wherein the Cas is a Cas13 (optionally Cas13a or Cas13b).

130. The use or method of any one of paragraphs 105 to 123, wherein the Cas is selected from a Cas3, Cas8a, Cas5, Cas8b, Cas8c, Cas10d, Cse1, Cse2, Csy1, Csy2, Csy3, GSU0054, Cas10, Csm2, Cmr5, Cas10, Csx11, Csx10, CsfI, Cas9, Csn2, Cas4, CpfI, C2c1, C2c3, Cas13a, Cas13b and Cas13c.

131. The use or method of any one of paragraphs 105 to 130, wherein the DNA construct comprises one or more nucleotide sequences for producing crRNAs or gRNAs that are operable for Cas nuclease targeting.

132. The use or method of any one of paragraphs 105 to 131, wherein the production strain cells comprise a helper phage genome that is inducible to produce phage coat proteins in the cells, wherein the method further comprises inducing production of the phage proteins and causing packaging of the amplified DNA into phage particles or non-self-replicative transduction particles, and further isolating the phage or transduction particles and optionally formulating the particles into a pharmaceutical composition for administration to a human or animal subject for treating or reducing the risk of a disease or condition in the subject.

133. The method of paragraph 132, wherein the particles are capable of infecting target host cells in the subject and transducing the cells with the DNA, wherein the Cas and

crRNAs (or gRNAs) encoded by the DNA are expressed in the cells, the crRNAs or (gRNAs) being operable to guide the Cas to a target nucleotide sequence (optionally a chromosomal sequence) comprised by the cells, wherein the Cas cuts the target sequences in the cells, thereby killing host cells and treating or reducing the risk of the disease or condition.

134. The method of paragraph 133, wherein the host cells are bacterial or archaeal cells, optionally, the host cells are *C. difficile*, *P. aeruginosa*, *K. pneumoniae* (eg, carbapenem-resistant *Klebsiella pneumoniae* or Extended-Spectrum Beta-Lactamase (ESBL)-producing *K. pneumoniae*), *E. coli* (eg, ESBL-producing *E. coli*, or *E. coli* ST131-O25b:H4), *H. pylori*, *S. pneumoniae* or *S. aureus* cells.

135. The use or method of any one of paragraphs 105 to 134, wherein each DNA is comprised by a high copy number vector, optionally a high copy number plasmid (an optionally the promoter is a constitutive promoter).

136. The use or method of any one of paragraphs 105 to 135, wherein each DNA is comprised by a vector according to any one of paragraphs 1 to 78 and 81 to 83.

Clauses

The invention provides, by way of example, the following clauses; the features of these are combinable with any other disclosure herein.

1. A nucleic acid vector for introduction into a host cell, the vector comprising a first nucleotide sequence encoding a Type I Cas3 and a second nucleotide sequence encoding one or more Cascade proteins, wherein the first and second sequences are under the control of one or more promoters comprised by the vector for expression of the proteins in the cell.

2. The vector of Clause 1, wherein the vector comprises an operon for expression in the cell of the Cas3 and Cascade proteins from a Cas module, the module comprising the nucleotide sequences encoding the Cas3 and Cascade proteins, and the operon comprising the Cas module under the control of a promoter for controlling the expression of both the Cas3 and Cascade proteins.

3. The vector of Clause 2, wherein

(a) the first sequence is between the promoter and the second sequence in the operon;

(b) the operon comprises no Cas-encoding nucleotide sequences between the promoter and the first nucleotide sequence; and/or

(c) the operon comprises (in 5' to 3' direction) the promoter, the first sequence and the second sequence.

4. The vector of any preceding Clause, wherein each promoter is a constitutive promoter.

5. The vector of any one of Clauses 1 to 3, wherein the promoter is repressible (optionally repressible by a tetracycline repressor or lac repressor).

6. The vector of any one of Clauses 1 to 3, wherein the promoter is inducible.

7. The vector of any preceding Clause, wherein the first sequence is under the control of a medium strength promoter.

8. The vector of any preceding Clause, wherein the first sequence is under the control of a promoter that has an Anderson Score (AS) of $0.5 > AS > 0.1$.

9. The vector of any preceding Clause, wherein the first sequence (and optionally the second sequence) is under the control of a promoter and translation initiation site (TIS) combination that is capable of producing expression of green fluorescent protein (GFP) from a first expression operating unit (EOU) in *E. coli* strain BW25113 cells with a fluorescence of from 0.5 to 4 times the fluorescence

produced in *E. coli* strain BW25113 cells using a second EOU comprising a P10 promoter (SEQ ID NO: 1) combined with a BCD14 TIS (SEQ ID NO: 2), wherein the EOUs differ only in their promoter and TIS combinations, wherein each EOU comprises (in 5' to 3' direction) an upstream initiator, the respective promoter, the respective TIS, a nucleotide sequence encoding GFP, a 3' UTR, a transcription terminator and a downstream insulator.

10. The vector of Clause 9, wherein fluorescence using the first EOU is 0.5 to 2 times the fluorescence using the second EOU.

11. The vector of any preceding Clause, wherein the vector comprises an origin of replication that is operable in the host cell.

12. The vector of any preceding Clause, wherein the vector comprises an origin of replication that is operable in a bacterial cell of a vector production strain, wherein the Cas3 is not operable in the production strain cell to target and cut a chromosomal sequence thereof.

13. The vector of Clause 12, wherein the first sequence is under the control of a promoter that is capable of controlling expression of the Cas3 at a level that is not toxic to the production strain cell.

14. The vector of any preceding Clause, wherein the vector is a high copy number vector.

15. The vector of any preceding Clause, wherein the first nucleotide sequence or operon is comprised by a mobile genetic element.

16. The vector of any preceding Clause, wherein the vector is devoid of a Cas adaption module.

17. The vector of any preceding Clause, wherein the vector is devoid of nucleotide sequence encoding one, more or all of a Cas1, Cas2, Cas4, Cas6, Cas7 and Cas8.

18. The vector of any preceding Clause, wherein the vector comprises (optionally in 5' to 3' direction) nucleotide sequence encoding one, more or all of Cas11, Cas7 and Cas8a1.

19. The vector of Clause 18, wherein the vector comprises nucleotide sequence encoding Cas3' and/or Cas3".

20. The vector of Clause 19, wherein the nucleotide sequences encoding the Cas3' and/or Cas3" are between the promoter and the sequence(s) recited in Clause 18.

21. The vector of any one of Clauses 18 to 20, wherein the host cell comprises a Type IA CRISPR array that is cognate with the Cas3.

22. The vector of any one of Clauses 18 to 20, wherein the host cell comprises an endogenous Type IB, C, U, D, E or F CRISPR/Cas system.

23. The vector of any one of Clauses 1 to 17, wherein the vector comprises (optionally in 5' to 3' direction) nucleotide sequence encoding one, more or all of Cas8b1, Cas7 and Cas5.

24. The vector of Clause 23, wherein the vector comprises a nucleotide sequence encoding Cas3 between the promoter and the sequence(s) recited in Clause 23.

25. The vector of Clause 23 or 24, wherein the host cell comprises a Type IB CRISPR array that is cognate with the Cas3.

26. The vector of Clause 23 or 24, wherein the host cell comprises an endogenous Type IA, C, U, D, E or F CRISPR/Cas system.

27. The vector of any one of Clauses 1 to 17, wherein the vector comprises (optionally in 5' to 3' direction) nucleotide sequence encoding one, more or all of Cas5, Cas8c and Cas7.

28. The vector of Clause 27, wherein the vector comprises a nucleotide sequence encoding Cas3 between the promoter and the sequence(s) recited in Clause 27.

29. The vector of Clause 27 or 28, wherein the host cell comprises a Type IC CRISPR array that is cognate with the Cas3.

30. The vector of Clause 27 or 28, wherein the host cell comprises an endogenous Type IA, B, U, D, E or F CRISPR/Cas system.

31. The vector of any one of Clauses 1 to 17, wherein the vector comprises (optionally in 5' to 3' direction) nucleotide sequence encoding one, more or all of Cas8U2, Cas7, Cas5 and Cas6.

32. The vector of Clause 31, wherein the vector comprises a nucleotide sequence encoding Cas3 between the promoter and the sequence(s) recited in Clause 31.

33. The vector of Clause 31 or 32, wherein the host cell comprises a Type IU CRISPR array that is cognate with the Cas3.

34. The vector of Clause 31 or 32, wherein the host cell comprises an endogenous Type IA, B, C, D, E or F CRISPR/Cas system.

35. The vector of any one of Clauses 1 to 17, wherein the vector comprises (optionally in 5' to 3' direction) nucleotide sequence encoding one, more or all of Cas10d, Cas7 and Cas5.

36. The vector of Clause 35, wherein the vector comprises a nucleotide sequence encoding Cas3' and/or Cas3".

37. The vector of Clause 36, wherein the nucleotide sequences encoding the Cas3' and/or Cas3" are between the promoter and the sequence(s) recited in Clause 35.

38. The vector of any one of Clauses 35 to 37, wherein the host cell comprises a Type ID CRISPR array that is cognate with the Cas3.

39. The vector of any one of Clauses 35 to 37, wherein the host cell comprises an endogenous Type IA, B, C, U, E or F CRISPR/Cas system.

40. The vector of any one of Clauses 1 to 17, wherein the vector comprises (optionally in 5' to 3' direction) nucleotide sequence encoding one, more or all of Cas8e, Cas11, Cas7, Cas5 and Cas6.

41. The vector of Clause 40, wherein the vector comprises a nucleotide sequence encoding Cas3 between the promoter and the sequence(s) recited in Clause 40.

42. The vector of Clause 40 or 41, wherein the host cell comprises a Type IE CRISPR array that is cognate with the Cas3.

43. The vector of Clause 40 or 41, wherein the host cell comprises an endogenous Type IA, B, C, D, U or F CRISPR/Cas system.

44. The vector of any one of Clauses 1 to 17, wherein the vector comprises (optionally in 5' to 3' direction) nucleotide sequence encoding one, more or all of Cas8f, Cas5, Cas7 and Cas6f.

45. The vector of Clause 44, wherein the vector comprises a nucleotide sequence encoding Cas3 between the promoter and the sequence(s) recited in Clause 44, wherein the vector is devoid of nucleotide sequence encoding further Cas between the promoter and the sequence encoding the Cas3.

46. The vector of Clause 44 or 45, wherein the host cell comprises a Type IF CRISPR array that is cognate with the Cas3.

47. The vector of Clause 44 or 45, wherein the host cell comprises an endogenous Type IA, B, C, D, U or E CRISPR/Cas system.

48. The vector of any one of Clauses 1 to 17, wherein the Cas and Cascade are

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- (a) Type IA Cas and Cascade proteins;
 - (b) Type IB Cas and Cascade proteins;
 - (c) Type IC Cas and Cascade proteins;
 - (d) Type ID Cas and Cascade proteins;
 - (e) Type IE Cas and Cascade proteins;
 - (f) Type IF Cas and Cascade proteins; or
 - (g) Type IU Cas and Cascade proteins.
49. The vector of any preceding Clause, wherein the Cas and Cascade are *E. coli* (optionally Type IE or IF) Cas and Cascade proteins.
50. The vector of Clause 49, wherein the *E. coli* is ESBL-producing *E. coli* or *E. coli* ST131-O25b:H4.
51. The vector of any preceding Clause, wherein the Cas and Cascade are
- (a) *Clostridium* (eg, *C. difficile*) Cas and Cascade proteins, optionally *C. difficile* resistant to one or more antibiotics selected from aminoglycosides, lincomycin, tetracyclines, erythromycin, clindamycin, penicillins, cephalosporins and fluoroquinolones;
 - (b) *Pseudomonas aeruginosa* Cas and Cascade proteins, optionally *P. aeruginosa* resistant to one or more antibiotics selected from carbapenems, aminoglycosides, cefepime, ceftazidime, fluoroquinolones, piperacillin and tazobactam; or
 - (c) *Klebsiella pneumoniae* (eg, carbapenem-resistant *Klebsiella pneumoniae* or Extended-Spectrum Beta-Lactamase (ESBL)-producing *K. pneumoniae*) Cas and Cascade proteins.
52. The vector of any preceding Clause, wherein the Cas and Cascade are *E. coli*, *C. difficile*, *P. aeruginosa*, *K. pneumoniae*, *P. furiosus* or *B. halodurans* Cas and Cascade proteins.
53. The vector of any preceding Clause, wherein the Cas3 is a Cas3 of a CRISPR/Cas locus of a first bacterial or archaeal species, wherein the distance between the Cas3-encoding sequence of the locus and its cognate promoter is further than the distance between the Cas3-encoding sequence and the respective promoter comprised by the vector.
54. The vector of any preceding Clause, wherein the distance between the promoter and the Cas3-encoding sequence and/or Cascade protein-encoding sequence(s) is shorter than in a corresponding wild-type Type I locus.
55. The vector of any preceding Clause, wherein the vector comprises (i) a CRISPR array for producing crRNAs in the host cell and/or (ii) one or more nucleotide sequences encoding one or more guide RNAs (gRNAs or single gRNAs), wherein the crRNAs or gRNAs are cognate to the Cas3 (and optionally cognate to the Cascade proteins).
56. The vector of Clause 55 when dependent from Clause 2, wherein the array or gRNA-encoding sequence(s) are comprised by the operon and under the control of the promoter.
57. The vector of Clause 56, wherein the array or gRNA-encoding sequence(s) are under the control of a promoter that is different from the promoter that controls the expression of the Cas3.
58. The vector of Clause 56 or 57, wherein one or more of the crRNAs or gRNAs comprises a spacer sequence that is capable of hybridising to a target nucleotide sequence of the host cell, wherein the target sequence is adjacent a PAM, the PAM being cognate to the Cas3.
59. The vector of Clause 58, wherein the target sequence is a chromosomal sequence of the host cell.
60. The vector of Clause 58 or 59, wherein the Cas3 is operable to cut the target sequence.

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61. The vector of any preceding Clause, wherein the vector is a plasmid or phagemid.
62. A delivery vehicle comprising the vector of any preceding Clause, wherein the delivery vehicle is capable of delivering the vector into the host cell.
63. The vehicle of Clause 62, wherein the delivery vehicle is a phage, non-replicative transduction particle, nanoparticle carrier, bacterium or liposome.
64. The vector or vehicle of any preceding Clause, wherein the host cell is a bacterial or archaeal cell, optionally, the host cell is a *C. difficile*, *P. aeruginosa*, *K. pneumoniae* (eg, carbapenem-resistant *Klebsiella pneumoniae* or Extended-Spectrum Beta-Lactamase (ESBL)-producing *K. pneumoniae*), *E. coli* (eg, ESBL-producing *E. coli*, or *E. coli* ST131-O25b:H4), *H. pylori*, *S. pneumoniae* or *S. aureus* cell.
65. The vector or vehicle of any preceding Clause for administration to a human or animal subject for treating or reducing the risk of a disease or condition in the subject.
66. The vector or vehicle of Clause 65, wherein the disease or condition is an infection of the subject with host cells (eg, bacterial cells), or wherein the disease or condition is mediated by host cells (eg, bacterial cells).
67. A pharmaceutical composition comprising the vector or vehicle of any preceding Clause and a pharmaceutically acceptable diluent, excipient or carrier.
- It will be understood that particular embodiments described herein are shown by way of illustration and not as limitations of the invention. The principal features of this invention can be employed in various embodiments without departing from the scope of the invention. Those skilled in the art will recognize, or be able to ascertain using no more than routine study, numerous equivalents to the specific procedures described herein. Such equivalents are considered to be within the scope of this invention and are covered by the claims. All publications and patent applications mentioned in the specification are indicative of the level of skill of those skilled in the art to which this invention pertains. All publications and patent applications and all US equivalent patent applications and patents are herein incorporated by reference to the same extent as if each individual publication or patent application was specifically and individually indicated to be incorporated by reference. Reference is made to WO2017/118598, US20180140698, US20170246221, US20180273940, US20160115488, US20180179547, US20170175142, US20160024510, US20150064138, US20170022499, US20160345578, US20180155729, US20180200342, WO2017112620, WO2018081502, PCT/EP2018/066954, PCT/EP2018/066980, PCT/EP2018/071454 and U.S. Ser. No. 15/985,658 and equivalent publications by the US Patent and Trademark Office (USPTO) or WIPO, the disclosures of which are incorporated herein by reference for providing disclosure that may be used in the present invention and/or to provide one or more features (eg, of a vector) that may be included in one or more claims herein.
- The use of the word “a” or “an” when used in conjunction with the term “comprising” in the claims and/or the specification may mean “one,” but it is also consistent with the meaning of “one or more,” “at least one,” and “one or more than one.” The use of the term “or” in the claims is used to mean “and/or” unless explicitly indicated to refer to alternatives only or the alternatives are mutually exclusive, although the disclosure supports a definition that refers to only alternatives and “and/or.” Throughout this application, the term “about” is used to indicate that a value includes the

inherent variation of error for the device, the method being employed to determine the value, or the variation that exists among the study subjects.

As used in this specification and claim(s), the words “comprising” (and any form of comprising, such as “comprise” and “comprises”), “having” (and any form of having, such as “have” and “has”), “including” (and any form of including, such as “includes” and “include”) or “containing” (and any form of containing, such as “contains” and “contain”) are inclusive or open-ended and do not exclude additional, unrecited elements or method steps.

The term “or combinations thereof” or similar as used herein refers to all permutations and combinations of the listed items preceding the term. For example, “A, B, C, or combinations thereof” is intended to include at least one of: A, B, C, AB, AC, BC, or ABC, and if order is important in a particular context, also BA, CA, CB, CBA, BCA, ACB, BAC, or CAB. Continuing with this example, expressly included are combinations that contain repeats of one or more item or term, such as BB, AAA, MB, BBC, AAABCCCC, CBBAAA, CABABB, and so forth. The skilled artisan will understand that typically there is no limit on the number of items or terms in any combination, unless otherwise apparent from the context.

Any part of this disclosure may be read in combination with any other part of the disclosure, unless otherwise apparent from the context.

All of the compositions and/or methods disclosed and claimed herein can be made and executed without undue experimentation in light of the present disclosure. While the compositions and methods of this invention have been described in terms of preferred embodiments, it will be apparent to those of skill in the art that variations may be applied to the compositions and/or methods and in the steps or in the sequence of steps of the method described herein without departing from the concept, spirit and scope of the invention. All such similar substitutes and modifications apparent to those skilled in the art are deemed to be within the spirit, scope and concept of the invention as defined by the appended claims.

The present invention is described in more detail in the following non-limiting Examples.

EXAMPLES

The examples illustrate fast and precision killing of *Escherichia coli* strains. As a model programmable nuclease system, we used a CRISPR guided vector (CGV™) to specifically target *Escherichia coli* MG1655.

Example 1. Single-Vector Cas3 & Cascade: Type I CRISPR-Cas System Targeting *E. coli*

A plasmid (which we call a CRISPR Guided Vector™, CGV™) was constructed comprising an operon with nucleotide sequences encoding a Type I Cas3 and Cascade proteins under the control of a common promoter. *C. difficile* Type IB Cas3 and Cascade was used. A cognate CRISPR array comprising *C. difficile* repeat sequences and spacer sequence for targeting an *E. coli* host cell chromosome was also introduced into target cells. An adaptation module containing Cas1, Cas2 and Cas4 was omitted in the vector (see FIG. 1A). In the wild-type *C. difficile* Type IB CRISPR/Cas locus, the cas3 gene is 3' of the Cascade genes (cas8b1, cas7 and cas5) and thus spaced away from the promoter upstream of the Cascade genes. When we tried this arrangement, we found killing of *E. coli* cells, but surprisingly when we

changed to a synthetic operon arrangement (in 5' to 3' orientation) of promoter, cas3, cas8b1, cas7 and cas5 we saw significantly higher killing of the target *E. coli* cells.

Results using this synthetic operon arrangement are shown in FIGS. 1A-1C. In FIG. 1B there is shown a dilution series (10^1 - 10^6) of drop spots (5 μ l) of target *E. coli* MG1655 cells harboring the CGV on LB agar plates with and without inducers. CRISPR/Cas induction surprisingly killed 99.9% of the population (FIG. 1C, grey bar). Growth in absence of induction is shown in black. CGV™ refers to a CRISPR Guided Vector™, which is a nucleic acid vector comprising nucleotide sequences encoding CRISPR/Cas components.

We also managed to achieve desirable targeted killing of *E. coli* cells using a similar set-up, except that *E. coli* Type IE Cas and Cascade were used, together with a cognate array targeting host cell *E. coli* chromosomal DNA (data not shown). In this case, a vector was used comprising (in 5' to 3' direction) a promoter controlling the expression of Cas3, Cas8e, Cas11, Cas7, Cas5 and Cas6 in an operon.

Materials and Methods

E. coli MG1655 was grown in lysogeny broth (LB) with shaking (250 rpm) at 37° C. When necessary, cultures were supplemented with tetracycline (10 μ g/mL), and spectinomycin (400 μ g/mL).

To construct a plasmid containing *C. difficile* CRISPR system under arabinose inducible pBAD promoter, cas3, cas6, cas8b, cas7 and cas5 genes from *C. difficile* were amplified and cloned in a low copy number plasmid (pSC101 on). cas3 was located in the beginning of the operon followed by cas6, cas8b, cas7 and cas5. The adaptation module (consisting of cas1, cas2, and cas4) was omitted in the vector (FIG. 1A). A second plasmid containing an IPTG inducible single-spacer array targeting a chromosomal intergenic region in *E. coli* MG1655 was constructed (FIG. 1A). The spacer was cloned under control of the IPTG-inducible P_{trc} promoter, in a CloDF13 on backbone. It contains 37 nucleotides from the genome of *E. coli* MG1655 (ctttgccgcgcgttcgtcagtaattctcgtcgcaa) (SEQ ID NO: 26). Additionally, the 3'-CCT protospacer adjacent motif (PAM) is located adjacent to the selected target sequence in the genome of *E. coli* MG1655 (FIG. 1A).

To perform killing assays, both plasmids were transformed into *E. coli* MG1655 by electroporation. Transformants were grown in liquid LB with antibiotics to mid-log phase, and the killing efficiency was determined by serial dilution and spot plating onto LB, and LB+inducers (0.5 mM IPTG and 1 arabinose). Viability was calculated by counting colony forming units (CFUs) on the plates and data were calculated as viable cell concentration (CFU/ml).

Example 2. Single-Vector Cas3-Cascade & Array: Type I CRISPR-Cas System Targeting *E. coli*

A plasmid (which we call a CRISPR Guided Vector™, CGV™, which is a nucleic acid vector comprising nucleotide sequences encoding CRISPR/Cas components) was constructed comprising an operon with nucleotide sequences encoding a Type I Cas3 and Cascade proteins under the control of a common promoter. *C. difficile* Type IB Cas3 and Cascade was used. Adaptation module containing Cas1, Cas2 and Cas4 was omitted in the vector. A cognate CRISPR array comprising *C. difficile* repeat sequences and spacer sequence for targeting an *E. coli* host cell chromosome was also cloned in the vector (see FIG. 2A). Similarly we also constructed a plasmid comprising of an operon with nucleotide sequences encoding *E. coli* Type IE Cas3 and Cascade proteins under control of a common promoter. The

E. coli adaption module containing Cas1 and Cas2 was omitted, in the vector. A cognate CRISPR array comprising *E. coli* repeat sequences and spacer sequence for targeting an *E. coli* host cell chromosome was also cloned in the vector.

The CGV containing the *C. difficile* CRISPR-Cas system was transformed into *E. coli* MG1655 which contains a pks sequence incorporated into the genome. Results using this synthetic operon arrangement are shown in FIGS. 2A-2C. In FIG. 2B there is shown a dilution series (10^1 - 10^5) of drop spots (5 μ l) of target *E. coli* MG1655 cells harboring the CGV on synthetic medium (SM) agar plates with and without inducers. CRISPR/Cas induction resulted in more than 2- $\log_{10}$ reductions in viable cells of *E. coli* MG1655 (FIG. 2C, grey bar). Growth in absence of induction is shown in black. CGV™ refers to a CRISPR Guided Vector™.

The survival of *E. coli* MG1655 upon induction was followed over time by plating the cultures in serial dilutions every 60 minutes, for 2 h (FIG. 3A). Killing curves revealed that CRISPR/Cas induction mediated rapid killing of *E. coli* MG1655, generating a two- $\log_{10}$ reduction in *E. coli* by the first 60 minutes. FIG. 3B shows a dilution series (10^1 - 10^6) of drop spots (5 μ l) of induced and non-induced cultures of target *E. coli* MG1655 on SM agar plates.

The CGV containing the *E. coli* CRISPR-Cas system was transformed into other *E. coli* MG1655 cells which contain a lambda sequence incorporated into the genome. Results using this synthetic operon arrangement are shown in FIGS. 6A-6B. In FIG. 6A there is shown a dilution series (10^1 - 10^5) of drop spots (5 μ l) of target *E. coli* MG1655 cells harboring the CGV on synthetic medium (SM) agar plates with and without inducers. CRISPR/Cas induction resulted in more than 2- $\log_{10}$ reductions in viable cells of *E. coli* MG1655 (FIG. 6B, grey bar). Growth in absence of induction is shown in black. In a repeat experiment (not shown) we saw a 3- $\log_{10}$ reductions in viable cells of *E. coli* MG1655 with CRISPR/Cas induction.

Materials and Methods

E. coli MG1655 was grown in synthetic medium (SM) with shaking (250 rpm) at 37° C. Cultures were supplemented with 10 μ g/mL tetracycline when required.

To construct a plasmid containing *C. difficile* CRISPR system under arabinose inducible pBAD promoter, cas3, cas6, cas8b, cas7 and cas5 genes from *C. difficile* were amplified and cloned in a low copy number plasmid (pSC101 ori). cas3 was located in the beginning of the operon followed by cas6, cas8b, cas7 and cas5. Additionally, an IPTG inducible single-spacer array targeting a chromosomal intergenic region in *E. coli* MG1655 was included in the vector under control of the IPTG-inducible P_{trc} promoter (FIG. 2A). It contains 37 nucleotides from the PKS gene (previously integrated into the genome of *E. coli* MG1655) (gtttggcgatggcgcggtgtgtgtgtgcttcgcgct) (SEQ ID NO: 27). Additionally, the 3'-CCT protospacer adjacent motif (PAM) is located adjacent to the selected target sequence in the genome of *E. coli* MG1655 (FIG. 2A).

To construct a plasmid containing *E. coli* CRISPR system under arabinose inducible pBAD promoter, cas3, cse1, cse2, cas7, cas5 and cas6 genes from *E. coli* were amplified and cloned in a low copy number plasmid (pSC101 ori). The operon comprised (in 5' to 3' direction) cas3 followed by cse1 cse2, cas7, cas5 and cas6. Additionally, an IPTG inducible single-spacer array targeting a chromosomal intergenic region in *E. coli* MG1655 was included in the vector under control of the IPTG-inducible P_{trc} promoter. It contained 32 nucleotides from the lambda sequence (previously integrated into the genome of *E. coli* MG1655) (tgggatgcc-

taccgcaagcagcttggtgcctgaa) (SEQ ID NO: 28) and found to efficiently target in Brouns et al., 2008 (Science. 2008 Aug. 15; 321(5891):960-4. doi: 10.1126/science.1159689; "Small CRISPR RNAs guide antiviral defense in prokaryotes"). Additionally, the 3'-ATG protospacer adjacent motif (PAM) is located adjacent to the selected target sequence in the genome of *E. coli* MG1655.

The CGVs were transformed into *E. coli* MG1655 by electroporation. Transformants were grown in liquid SM with antibiotics to mid-log phase, and the killing efficiency was determined by serial dilution and spot plating onto LB, and LB+inducers (0.5 mM IPTG and 1% arabinose). Viability was calculated by counting colony forming units (CFUs) on the plates and data were calculated as viable cell concentration (CFU/ml).

To perform killing curves, *E. coli* MG1655 harboring the CGV was grown in liquid SM with antibiotics to mid-log phase. The culture was divided into two tubes and either inducers (0.5 mM IPTG and 1 arabinose) or PBS were added. Survival of the strain was followed over time by plating the cultures in serial dilutions (10^1 - 10^6) of drop spots (5 μ l) every 60 minutes, for 2 h, on SM plates with antibiotics. Survival frequency was calculated by counting colony forming units (CFUs) on the plates and data were calculated as viable cell concentration (CFU/ml).

Example 3. Precision Killing of Target Strain *E. coli* MG1655 in a Microbiome

An artificial microbial consortium was constructed to study the efficiency of the CGV carrying the CRISPR-Cas system of *C. difficile*, to specifically target *E. coli* MG1655 in the presence of other microbes, mimicking the human microbiome.

The synthetic consortium consisted of three strains (two different species) with differential antibiotic resistance profiles: a streptomycin-resistant *E. coli* MG1655 (target strain), an ampicillin-resistant *E. coli* Top10, and a chloramphenicol-resistant *Lactococcus lactis* NZ9000. To create the consortium, bacterial cultures were grown separately in Brain Heart Infusion broth (BHI, optimal growth medium for *L. lactis*) to mid-log phase and mixed in fresh BHI broth with and without inducers. After 1 h induction at 30° C., the composition of the consortium was determined by counting viable colonies on selective plates. Induction of the CRISPR system in the mixed community, resulted in >10-fold killing of target *E. coli* MG1655, while leaving *E. coli* Top10 and *L. lactis* NZ9000 cell populations unharmed (FIG. 4A). In FIG. 4B there is shown a dilution series (10^1 - 10^5) of drop spots (5 μ l) of the synthetic consortium after 1 h induction on BHI agar plates.

Additionally, CRISPR killing of target strain *E. coli* MG1655 in the synthetic microbial consortium was compared to a pure culture (ie, target strain *E. coli* MG1655 that is not mixed with another strain or species). Unexpectedly, in both conditions, killing of 3 logs was achieved when plated on BHI agar plates with inducers (FIG. 5A). Thus, surprisingly the killing in the microbiome setting was as efficient as the killing in pure culture. In FIG. 5B there is shown a dilution series (10^1 - 10^5) of drop spots (5 μ l) of the synthetic consortium and *E. coli* MG1655 in pure culture on BHI agar plates with and without inducers.

Materials and Methods

E. coli MG1655, *E. coli* Top10, and *Lactococcus lactis* NZ9000 were grown in BHI broth with shaking (250 rpm)

at 30° C. Cultures were supplemented with 1000 µg/mL streptomycin, 100 µg/mL ampicillin, or 10 µg/mL chloramphenicol, respectively.

To create the consortium, bacterial cultures were grown in BHI with appropriate antibiotics to mid-log phase. Cultures were washed twice in PBS to remove the antibiotics and mixed in fresh BHI broth. The mixed culture was spotted onto BHI plates with streptomycin, ampicillin or chloramphenicol to quantify the initial concentration of *E. coli* MG1655, *E. coli* Top10 and *L. lactis* NZ9000, respectively. The mixed culture was divided into two tubes and either inducers (0.5 mM IPTG and 1% arabinose) or PBS were added. After 1 h induction at 30° C., the composition of the consortium was calculated by counting colony forming units (CFUs) on selective plates and data were calculated as viable cell concentration (CFU/ml).

Example 4. Use of Promoter Repression in Vector Amplification Strains

We engineered an *E. coli* Top10 production strain cell population comprising plasmid CGV DNA and an express-

ible sequence encoding a Tet repressor (TetR). The DNA comprised a Cas9-encoding nucleotide sequence under the control of a Tet promoter (pLtetO-1 promoter). The promoter is normally constitutively ON, but it was repressed by TetR in our cells. Thus, in this way we could successfully culture the cells and amplify the CGV without observing adverse toxicity due to Cas9 expression.

In an experiment in the absence of repression, we did not observe any colonies of production strain bacteria, and we surmise that this was due to Cas9 toxicity. We believe, in addition to providing a way of increasing CGV yield (eg, for subsequent packaging into phage or non-self-replicative transduction particles), our method using repression can minimize selection for mutations in the DNA that would otherwise be forced by higher Cas9 expression and cutting (eg, due to CGV cutting).

REFERENCES

Mutalik et al, Nat Methods. 2013 April; 10(4):354-60. doi: 10.1038/nmeth. 2404. Epub 2013 Mar. 10, "Precise and reliable gene expression via standard transcription and translation initiation elements".

TABLE 1

Optionally, the target host cells are cells of a genus or species selected from this Table and/or the production strain cells are cells of a genus or species selected from this Table			Example Bacteria
<i>Abiotrophia</i>	<i>Acidocella</i>	<i>Actinomyces</i>	<i>Alkalitlimicola</i>
<i>Abiotrophia defectiva</i>	<i>Acidocella aminolytica</i>	<i>Actinomyces boyds</i>	<i>Alkalitlimicola ehrlichii</i>
<i>Acaricomes</i>	<i>Acidocella facilis</i>	<i>Actinomyces deontocolens</i>	<i>Alkaliphilus</i>
<i>Acartomes phytoseutili</i>	<i>Acidomonas</i>	<i>Actinomyces europaeus</i>	<i>Alkaliphilus orelandii</i>
<i>Acetomaculum</i>	<i>Acidomonas methanolica</i>	<i>Actinomyces georgiae</i>	<i>Alkaliphilus transvaalensis</i>
<i>Acetomaculum ruminis</i>	<i>Acidothermus</i>	<i>Actinomyces gerencseriae</i>	<i>Allochromatium</i>
<i>Acetivibrio</i>	<i>Acidothermus cellulolyticus</i>	<i>Actinomyces</i>	<i>Allochromatium vinosum</i>
<i>Acetivibrio cellulolyticus</i>	<i>Acidovorax</i>	<i>Acidothiobacillus</i>	<i>Allocooccus</i>
<i>Acetivibrio cellulosignus</i>	<i>Acidovorax anthracis</i>	<i>Acidothiobacillus</i>	<i>Allocooccus otitis</i>
<i>Acetivibrio multivorans</i>	<i>Acidovorax caryi</i>	<i>Acidothiobacillus</i>	<i>Allocooccus</i>
<i>Acetomicrobium</i>	<i>Acidovorax catenae</i>	<i>Acidothiobacillus</i>	<i>Allocooccus</i>
<i>Acetomicrobium noterae</i>	<i>Acidovorax citrulli</i>	<i>Acidothiobacillus</i>	<i>Allocooccus</i>
<i>Acetobacter</i>	<i>Acidovorax defluvi</i>	<i>Acidothiobacillus</i>	<i>Allocooccus</i>
<i>Acetobacter aceti</i>	<i>Acidovorax facilis</i>	<i>Acidothiobacillus</i>	<i>Allocooccus</i>
<i>Acetobacter cerevisiae</i>	<i>Acidovorax konjaci</i>	<i>Acidothiobacillus</i>	<i>Allocooccus</i>
<i>Acetobacter cihonensis</i>	<i>Acidovorax temperans</i>	<i>Acidothiobacillus</i>	<i>Allocooccus</i>
<i>Acetobacter estuensis</i>	<i>Acidovorax valentianae</i>	<i>Acidothiobacillus</i>	<i>Allocooccus</i>
<i>Acetobacter fabrum</i>	<i>Acetobacter</i>	<i>Acidothiobacillus</i>	<i>Allocooccus</i>
<i>Acetobacter glaucus</i>	<i>Acetobacter baumannii</i>	<i>Acidothiobacillus</i>	<i>Allocooccus</i>
<i>Acetobacter indonesiensis</i>	<i>Acetobacter baylyi</i>	<i>Acidothiobacillus</i>	<i>Allocooccus</i>
<i>Acetobacter lovaniensis</i>	<i>Acetobacter boveii</i>	<i>Acidothiobacillus</i>	<i>Allocooccus</i>
<i>Acetobacter malorum</i>	<i>Acetobacter calcoaceticus</i>	<i>Acidothiobacillus</i>	<i>Allocooccus</i>
<i>Acetobacter nitrogenifigens</i>	<i>Acetobacter gertneri</i>	<i>Acidothiobacillus</i>	<i>Allocooccus</i>
<i>Acetobacter oeni</i>	<i>Acetobacter haemolyticus</i>	<i>Acidothiobacillus</i>	<i>Allocooccus</i>
<i>Acetobacter orientalis</i>	<i>Acetobacter johnsonii</i>	<i>Acidothiobacillus</i>	<i>Allocooccus</i>
<i>Acetobacter orleanensis</i>	<i>Acetobacter junii</i>	<i>Acidothiobacillus</i>	<i>Allocooccus</i>
<i>Acetobacter pasteurianus</i>	<i>Acetobacter lwoffi</i>	<i>Acidothiobacillus</i>	<i>Allocooccus</i>
<i>Acetobacter pomorum</i>	<i>Acetobacter parvus</i>	<i>Acidothiobacillus</i>	<i>Allocooccus</i>
<i>Acetobacter senegalensis</i>	<i>Acetobacter radiorisistens</i>	<i>Acidothiobacillus</i>	<i>Allocooccus</i>
<i>Acetobacter xylinus</i>	<i>Acetobacter schindleri</i>	<i>Acidothiobacillus</i>	<i>Allocooccus</i>
<i>Acetobacterium</i>	<i>Acetobacter soli</i>	<i>Acidothiobacillus</i>	<i>Allocooccus</i>
<i>Acetobacterium bakii</i>	<i>Acetobacter tjernbergiae</i>	<i>Acidothiobacillus</i>	<i>Allocooccus</i>
<i>Acetobacterium carbinolicum</i>	<i>Acetobacter townii</i>	<i>Acidothiobacillus</i>	<i>Allocooccus</i>
<i>Acetobacterium dehalogenans</i>	<i>Acetobacter ursingii</i>	<i>Acidothiobacillus</i>	<i>Allocooccus</i>
<i>Acetobacterium limetarium</i>	<i>Acetobacter venetianus</i>	<i>Acidothiobacillus</i>	<i>Allocooccus</i>
<i>Acetobacterium malicum</i>	<i>Acetocarpuspora</i>	<i>Acidothiobacillus</i>	<i>Allocooccus</i>
<i>Acetobacterium paludosum</i>	<i>Acetocarpuspora corrugata</i>	<i>Acidothiobacillus</i>	<i>Allocooccus</i>
<i>Acetobacterium tundiae</i>	<i>Acetocarpuspora</i>	<i>Acidothiobacillus</i>	<i>Allocooccus</i>
<i>Acetobacterium wieringae</i>	<i>Acetocarpuspora</i>	<i>Acidothiobacillus</i>	<i>Allocooccus</i>
<i>Acetobacterium woodii</i>	<i>Acetocarpuspora</i>	<i>Acidothiobacillus</i>	<i>Allocooccus</i>
<i>Acetofilamentum</i>	<i>Acetocarpuspora</i>	<i>Acidothiobacillus</i>	<i>Allocooccus</i>
<i>Acetofilamentum rigidum</i>	<i>Acetocarpuspora</i>	<i>Acidothiobacillus</i>	<i>Allocooccus</i>
<i>Acetohalobium</i>	<i>Acetobacter</i>	<i>Acidothiobacillus</i>	<i>Allocooccus</i>
<i>Acetohalobium arabaticum</i>	<i>Acetobacter sediminis</i>	<i>Acidothiobacillus</i>	<i>Allocooccus</i>
<i>Acetomicrobium</i>	<i>Acetobacterium</i>	<i>Acidothiobacillus</i>	<i>Allocooccus</i>
<i>Acetomicrobium faecale</i>	<i>Acetobacterium</i>	<i>Acidothiobacillus</i>	<i>Allocooccus</i>
<i>Acetomicrobium flavidum</i>	<i>Acetobacterium</i>	<i>Acidothiobacillus</i>	<i>Allocooccus</i>
<i>Acetonema</i>	<i>Acetobacterium</i>	<i>Acidothiobacillus</i>	<i>Allocooccus</i>

Optionally, the target host cells are cells of a genus or species selected from this Table and/or the production strain cells are cells of a genus or species selected from this Table

<i>Acetoneuma longum</i>	<i>hymeniacidinis</i>	<i>Actinotalea</i>	<i>Anabaena flos-aquae</i>	<i>Arthroacter</i>
<i>Acetothermus</i>	<i>Actinoallotrichus spitiensis</i>	<i>Actinotalea fermentans</i>	<i>Anabaena variabilis</i>	<i>psychrolactophilus</i>
<i>Acetothermus paucivorans</i>	<i>Actinobacillus</i>	<i>Aerococcus</i>	<i>Anaerococcus</i>	<i>Arthroacter ramosus</i>
<i>Acholeplasma</i>	<i>Actinobacillus capsulatus</i>	<i>Aerococcus sanguinicola</i>	<i>Anaerococcus burkinensis</i>	<i>Arthroacter sulfonivorans</i>
<i>Acholeplasma axanthum</i>	<i>Actinobacillus delphinicola</i>	<i>Aerococcus urinae</i>	<i>Anaerobaculum</i>	<i>Arthroacter sulfureus</i>
<i>Acholeplasma brassicae</i>	<i>Actinobacillus hominis</i>	<i>Aerococcus urinaeequi</i>	<i>Anaerobaculum mobile</i>	<i>Arthroacter uratoxydans</i>
<i>Acholeplasma cavintectatum</i>	<i>Actinobacillus indolicus</i>	<i>Aerococcus urinaehominis</i>	<i>Anaerobiospirillum</i>	<i>Arthroacter ureafaciens</i>
<i>Acholeplasma equifatale</i>	<i>Actinobacillus lignirestii</i>	<i>Aerococcus viridans</i>	<i>Anaerobiospirillum</i>	<i>Arthroacter viscosus</i>
<i>Acholeplasma granularum</i>	<i>Actinobacillus minor</i>	<i>Aeromicrobium</i>	<i>succiniciproducens</i>	<i>Arthroacter wolnensis</i>
<i>Acholeplasma hippikon</i>	<i>Actinobacillus muris</i>	<i>Aeromicrobium erythreum</i>	<i>Anaerobiospirillum thomasi</i>	<i>Asata</i>
<i>Acholeplasma laidlawii</i>	<i>Actinobacillus</i>	<i>Aeromonas</i>	<i>Anaerococcus</i>	<i>Asata bogorensis</i>
<i>Acholeplasma modicum</i>	<i>pleuropneumoniae</i>	<i>Aeromonas</i>	<i>Anaerococcus hydrogenalis</i>	<i>Asanoa</i>
<i>Acholeplasma morum</i>	<i>Actinobacillus porcinus</i>	<i>allosaccharophila</i>	<i>Anaerococcus lactolyticus</i>	<i>Asanoa ferruginea</i>
<i>Acholeplasma multilocale</i>	<i>Actinobacillus rossi</i>	<i>Aeromonas bestiarum</i>	<i>Anaerococcus prevotii</i>	<i>Asiticacaulis</i>
<i>Acholeplasma oculi</i>	<i>Actinobacillus sootiae</i>	<i>Aeromonas carvatae</i>	<i>Anaerococcus tetradius</i>	<i>Asiticacaulis hiprosthectum</i>
<i>Acholeplasma palmae</i>	<i>Actinobacillus seminis</i>	<i>Aeromonas encheleia</i>	<i>Anaerococcus vaginalis</i>	<i>Asiticacaulis excentricus</i>
<i>Acholeplasma parvum</i>	<i>Actinobacillus succinogenes</i>	<i>Aeromonas</i>	<i>Anaerofastis</i>	<i>Atopobacter</i>
<i>Acholeplasma plectae</i>	<i>Actinobacillus suis</i>	<i>enteropelogenes</i>	<i>Anaerofastis stercorihominis</i>	<i>Atopobacter phocae</i>
<i>Acholeplasma vituli</i>	<i>Actinobacillus urcae</i>	<i>Aeromonas eucrenophila</i>	<i>Anaeromusa</i>	<i>Atopobium</i>
<i>Achromobacter</i>	<i>Actinobaculum</i>	<i>Aeromonas ichthiosmia</i>	<i>Anaeromusa acidaminophila</i>	<i>Atopobium fossor</i>
<i>Achromobacter dentrificans</i>	<i>Actinobaculum massiliense</i>	<i>Aeromonas jondacii</i>	<i>Anaeromusa xylobacter</i>	<i>Atopobium minutum</i>
<i>Achromobacter insolitus</i>	<i>Actinobaculum schaalii</i>	<i>Aeromonas media</i>	<i>Anaeromyxobacter</i>	<i>Atopobium parvulum</i>
<i>Achromobacter piechaidii</i>	<i>Actinobaculum suis</i>	<i>Aeromonas popoffii</i>	<i>dehalogenans</i>	<i>Atopobium rhinae</i>
<i>Achromobacter rullandii</i>	<i>Actinomycetes urinale</i>	<i>Aeromonas sobria</i>	<i>Anaerorhabdus</i>	<i>Atopobium vaginae</i>
<i>Achromobacter spanius</i>	<i>Actinocatenispora</i>	<i>Aeromonas veronii</i>	<i>Anaerorhabdus furcosa</i>	<i>Aureobacterium</i>
<i>Acidaminobacter</i>	<i>Actinocatenispora ripis</i>	<i>Agrobacterium</i>	<i>Anaerosinus</i>	<i>Aureobacterium barkeri</i>
<i>Acidaminobacter</i>	<i>Actinocatenispora</i>	<i>Agrobacterium</i>	<i>Anaerosinus glycerinii</i>	<i>Aureobacterium</i>
<i>Acidaminobacter hydrogeniformans</i>	<i>thailandica</i>	<i>gelatinovorum</i>	<i>Anaerovirgula</i>	<i>Aureobacterium liquefaciens</i>
<i>Acidaminococcus</i>	<i>Actinocatenispora sera</i>	<i>Agrococcus</i>	<i>Anaerovirgula multivorans</i>	<i>Atribacterium</i>
<i>Acidaminococcus fermentans</i>	<i>Actinocorallia</i>	<i>Agrococcus citreus</i>	<i>Ancalomicrobium</i>	<i>Atribacterium avium</i>
<i>Acidaminococcus intestini</i>	<i>Actinocorallia aurantiaca</i>	<i>Agrococcus jenensis</i>	<i>Ancalomicrobium adetum</i>	<i>Atribacterium gallinarum</i>
<i>Acidicaldus</i>	<i>Actinocorallia aurea</i>	<i>Agromonas</i>	<i>Ancylobacter</i>	<i>Atribacterium paragalinarum</i>
<i>Acidicaldus organivorans</i>	<i>Actinocorallia cavernae</i>	<i>Agromonas oligotrophica</i>	<i>Ancylobacter aquaticus</i>	<i>Atribacterium volantium</i>
<i>Acidimicrobium</i>	<i>Actinocorallia glomerata</i>	<i>Agromyces</i>	<i>Aneurinibacillus</i>	<i>Azoarcus</i>
<i>Acidimicrobium ferrooxidans</i>	<i>Actinocorallia herbida</i>	<i>Agromyces fuscus</i>	<i>Aneurinibacillus aneurinilyticus</i>	<i>Azoarcus indigenus</i>
<i>Acidiphilium</i>	<i>Actinocorallia libanotica</i>	<i>Agromyces hippuratus</i>	<i>Aneurinibacillus migulanus</i>	<i>Azoarcus toluilyticus</i>
<i>Acidiphilium acidophilum</i>	<i>Actinocorallia longicatena</i>	<i>Agromyces luteolus</i>	<i>Aneurinibacillus</i>	<i>Azoarcus toluivorans</i>
<i>Acidiphilium angustum</i>	<i>Actinomadura</i>	<i>Agromyces medellanus</i>	<i>thermoaerophilus</i>	<i>Azohydromonas</i>
<i>Acidiphilium cryptum</i>	<i>Actinomadura alba</i>	<i>Agromyces ramosus</i>	<i>Angiococcus</i>	<i>Azohydromonas australica</i>
<i>Acidiphilium multivorum</i>	<i>Actinomadura atramentaria</i>	<i>Agromyces rhizosphaerae</i>	<i>Angitococcus disciformis</i>	<i>Azohydromonas lata</i>
<i>Acidiphilium organovorum</i>	<i>Actinomadura</i>	<i>Akkermansia</i>	<i>Angulomicrobium</i>	<i>Azomonas</i>
<i>Acidiphilium rubrum</i>	<i>bangladeshensis</i>	<i>Akkermansia muciniphila</i>	<i>Angulomicrobium tetradrale</i>	<i>Azomonas agilis</i>
<i>Acidsoma</i>	<i>Actinomadura caellatispora</i>	<i>Albidifex</i>	<i>Akkermansia</i>	<i>Azomonas insigis</i>
<i>Acidsoma sibiricum</i>	<i>Actinomadura tibensis</i>	<i>Albidifex ferriveducens</i>	<i>Akkermansia</i>	<i>Azomonas macrocytogenes</i>
<i>Acidsoma tundrae</i>	<i>Actinomadura chokortensis</i>	<i>Albidovulum</i>	<i>Akkermansia</i>	<i>Azorhizobium</i>
<i>Acidsphaera</i>	<i>Actinomadura citrea</i>	<i>Albidovulum inexpectatum</i>	<i>Akkermansia</i>	<i>Azorhizobium caulinodans</i>
<i>Acidsphaera rubrifaciens</i>	<i>Actinomadura coerulea</i>	<i>Alcaligenes</i>	<i>Akkermansia</i>	<i>Azorhizobium</i>
<i>Acidithiobacillus</i>	<i>Actinomadura echinospira</i>	<i>Alcaligenes dentrificans</i>	<i>Akkermansia</i>	<i>Azorhizophilus paspali</i>
<i>Acidithiobacillus albertensis</i>	<i>Actinomadura fibrosa</i>	<i>Alcaligenes fecalis</i>	<i>Akkermansia</i>	<i>Azospirillum</i>
<i>Acidithiobacillus caldus</i>	<i>Actinomadura formosensis</i>	<i>Alcantivorax</i>	<i>Akkermansia</i>	<i>Azospirillum brasilense</i>

TABLE 1-continued

Example Bacteria			Optional, the target host cells are cells of a genus or species selected from this Table and/or the production strain cells are cells of a genus or species selected from this Table
<i>Acidithiobacillus ferrooxidans</i> <i>Acidithiobacillus thiooxidans</i> <i>Acidobacterium</i> <i>Acidobacterium capsulatum</i>	<i>Actinomadura hibisca</i>	<i>Alcanivorax borkumensis</i>	<i>Azospirillum halopraeferens</i>
	<i>Actinomadura kjanitaiata</i>	<i>Alcanivorax jadensis</i>	<i>Azospirillum irakense</i>
	<i>Actinomadura latina</i>	<i>Algicola</i>	<i>Azotobacter</i>
	<i>Actinomadura livida</i>	<i>Algicola bacteriolytica</i>	<i>Azotobacter beijerinckii</i>
	<i>Actinomadura</i>	<i>Alicyclobacillus</i>	<i>Azotobacter chroococcum</i>
	<i>luteofluorescens</i>	<i>Alicyclobacillus</i>	<i>Azotobacter nigricans</i>
	<i>Actinomadura macra</i>	<i>disulfidooxidans</i>	<i>Azotobacter salinestris</i>
	<i>Actinomadura madurae</i>	<i>Alicyclobacillus</i>	<i>Azotobacter vinelandii</i>
	<i>Actinomadura oligospora</i>	<i>sendaiensis</i>	
	<i>Actinomadura pelletieri</i>	<i>Alicyclobacillus vulcanalis</i>	
<i>Bacteriovorax</i> <i>Bacteriovorax stolpii</i>	<i>Actinomadura rubrobrunea</i>	<i>Alishevianella</i>	
	<i>Actinomadura rugatobispora</i>	<i>Alishevianella fetalis</i>	
	<i>Actinomadura umbriana</i>	<i>Alkalibacillus</i>	
	<i>Actinomadura</i>	<i>Alkalibacillus</i>	
	<i>verrucospora</i>	<i>halokatiphilus</i>	
	<i>Actinomadura vinacea</i>		
	<i>Actinomadura viridilinea</i>		
	<i>Actinomadura viridis</i>		
	<i>Actinomadura ymaensis</i>		
	<i>Bacteroides</i>	<i>Bibersteinia</i>	<i>Borrelia</i>
<i>Bacteriovorax</i> <i>Bacteriovorax stolpii</i>	<i>Bacteroides cacciae</i>	<i>Bibersteinia trehalosi</i>	<i>Borrelia afzelii</i>
	<i>Bacteroides coagulans</i>	<i>Bifidobacterium</i>	<i>Borrelia americana</i>
	<i>Bacteroides eggertii</i>	<i>Bifidobacterium adolescentis</i>	<i>Borrelia burgdorferi</i>
	<i>Bacteroides fragilis</i>	<i>Bifidobacterium angulatum</i>	<i>Borrelia carolinensis</i>
	<i>Bacteroides galacturonicus</i>	<i>Bifidobacterium animalis</i>	<i>Borrelia cortacae</i>
	<i>Bacteroides helvogenes</i>	<i>Bifidobacterium asteroides</i>	<i>Borrelia garinii</i>
	<i>Bacteroides ovatus</i>	<i>Bifidobacterium bifidum</i>	<i>Borrelia japonica</i>
	<i>Bacteroides pectinophilus</i>	<i>Bifidobacterium boum</i>	<i>Bosea</i>
	<i>Bacteroides pyogenes</i>	<i>Bifidobacterium breve</i>	<i>Bosea minaitanensis</i>
	<i>Bacteroides salyersiae</i>	<i>Bifidobacterium catenulatum</i>	<i>Bosea thiooxidans</i>
<i>Bacteriovorax</i> <i>Bacteriovorax stolpii</i>	<i>Bacteroides stercoris</i>	<i>Bifidobacterium choertium</i>	<i>Brachybacterium</i>
	<i>Bacteroides suis</i>	<i>Bifidobacterium coryneforme</i>	<i>Brachybacterium</i>
	<i>Bacteroides tectus</i>	<i>Bifidobacterium cuniculi</i>	<i>alimentarium</i>
	<i>Bacteroides thetaiotaomicron</i>	<i>Bifidobacterium dentium</i>	<i>Brachybacterium facium</i>
	<i>Bacteroides uniformis</i>	<i>Bifidobacterium gallium</i>	<i>Brachybacterium</i>
	<i>Bacteroides ureolyticus</i>	<i>Bifidobacterium gallinarum</i>	<i>Brachybacterium</i>
	<i>Bacteroides vulgatus</i>	<i>Bifidobacterium indicum</i>	<i>paracongomeratum</i>
	<i>Balnearium</i>	<i>Bifidobacterium longum</i>	<i>Brachybacterium rhamnosum</i>
	<i>Balnearium lithotrophicum</i>	<i>Bifidobacterium</i>	<i>Brachybacterium</i>
	<i>Balnearix</i>	<i>magnumBifidobacterium</i>	<i>tyrofermentans</i>
<i>Bacteriovorax</i> <i>Bacteriovorax stolpii</i>	<i>Balnearix alpica</i>	<i>merycicum</i>	<i>Brachyspira</i>
	<i>Balneola</i>	<i>Bifidobacterium minimum</i>	<i>Brachyspira alvinipulli</i>
	<i>Balneola vulgaris</i>	<i>Bifidobacterium</i>	<i>Brachyspira hyodysenteriae</i>
	<i>Barnesiella</i>	<i>pseudocatenulatum</i>	<i>Brachyspira timocens</i>
	<i>Barnesiella viscericola</i>	<i>Bifidobacterium</i>	<i>Brachyspira mundochii</i>
	<i>Bartonella</i>	<i>pseudolongum</i>	<i>Brachyspira pilosicoli</i>
	<i>Bartonella alsatica</i>	<i>Bifidobacterium pullorum</i>	<i>Bradyrhizobium</i>
	<i>Bartonella bacilliformis</i>	<i>Bifidobacterium ruminantium</i>	<i>Bradyrhizobium canariense</i>
	<i>Bartonella clarridgeae</i>	<i>Bifidobacterium saeculare</i>	<i>Bradyrhizobium elkanii</i>
	<i>Bartonella doshae</i>	<i>Bifidobacterium subtile</i>	<i>Bradyrhizobium japonicum</i>

Example Bacteria

Optionally, the target host cells are cells of a genus or species selected from this Table and/or the production strain cells are cells of a genus or species selected from this Table			
<i>Bartonella elizabethae</i> <i>Bartonella grahamii</i> <i>Bartonella henselae</i> <i>Bartonella rochalimae</i> <i>Bartonella vissonii</i> <i>Bavaritococcus</i> <i>Bavaritococcus selleri</i> <i>Bdellovibrio</i> <i>Bdellovibrio bacteriovorus</i> <i>Bdellovibrio exovorus</i> <i>Beggiatoa alba</i> <i>Beijerinckia</i> <i>Beijerinckia derxii</i> <i>Beijerinckia fluminensis</i> <i>Beijerinckia indica</i> <i>Beijerinckia mobilis</i> <i>Bellella</i> <i>Bellella bullica</i> <i>Bellilinea</i> <i>Bellilinea caldifistulae</i> <i>Behavipia</i> <i>Behavipia moabensis</i> <i>Bergertella</i> <i>Bergertella denitrificans</i> <i>Beutenbergia</i> <i>Beutenbergia cavemuae</i>	<i>Bifidobacterium thermophilum</i> <i>Bilophila</i> <i>Bilophila wadsworthia</i> <i>Biostraticola</i> <i>Biostraticola tofi</i> <i>Bizonia</i> <i>Bizonia argentinensis</i> <i>Blastobacter</i> <i>Blastobacter capsulatus</i> <i>Blastobacter denitrificans</i> <i>Blastococcus</i> <i>Blastococcus aggregatus</i> <i>Blastococcus saxosidens</i> <i>Blastochloris</i> <i>Blastochloris viridis</i> <i>Blastomonas</i> <i>Blastomonas natatoria</i> <i>Blastopirellula</i> <i>Blastopirellula marina</i> <i>Blautia</i> <i>Blautia coccoides</i> <i>Blautia hansenii</i> <i>Blautia producta</i> <i>Blautia weisterae</i> <i>Bogoriella</i> <i>Bogoriella caseilytica</i> <i>Bordetella</i> <i>Bordetella avium</i> <i>Bordetella bronchiseptica</i> <i>Bordetella hinzii</i> <i>Bordetella holmesii</i> <i>Bordetella parapertussis</i> <i>Bordetella pertussis</i> <i>Bordetella pertii</i> <i>Bordetella trematum</i>	<i>Bradyrhizobium liaoningense</i> <i>Brenneria</i> <i>Brenneria albi</i> <i>Brenneria nigritfluens</i> <i>Brenneria quercina</i> <i>Brenneria quercina</i> <i>Brenneria salicis</i> <i>Brevibacillus</i> <i>Brevibacillus agri</i> <i>Brevibacillus borstelensis</i> <i>Brevibacillus brevis</i> <i>Brevibacillus centrosporus</i> <i>Brevibacillus choshinensis</i> <i>Brevibacillus invocatus</i> <i>Brevibacillus laterosporus</i> <i>Brevibacillus parabravis</i> <i>Brevibacillus reuszeri</i> <i>Brevibacterium</i> <i>Brevibacterium abidum</i> <i>Brevibacterium album</i> <i>Brevibacterium aurantiacum</i> <i>Brevibacterium celere</i> <i>Brevibacterium epidermidis</i> <i>Brevibacterium</i> <i>frigoritolerans</i> <i>Brevibacterium halotolerans</i> <i>Brevibacterium iodinum</i> <i>Brevibacterium linens</i> <i>Brevibacterium lycium</i> <i>Brevibacterium mcbrenneri</i> <i>Brevibacterium oitidis</i> <i>Brevibacterium oxydans</i> <i>Brevibacterium paucivorans</i> <i>Brevibacterium stationis</i>	<i>Burkholderia glumae</i> <i>Burkholderia graminis</i> <i>Burkholderia kururiensis</i> <i>Burkholderia multivorans</i> <i>Burkholderia phenazinium</i> <i>Burkholderia plantarii</i> <i>Burkholderia pyrocinia</i> <i>Burkholderia silvaticina</i> <i>Burkholderia stabilis</i> <i>Burkholderia thailandensis</i> <i>Burkholderia tropica</i> <i>Burkholderia umamae</i> <i>Burkholderia vietnamiensis</i> <i>Butiaxella</i> <i>Butiaxella agrestis</i> <i>Butiaxella bremeriae</i> <i>Butiaxella ferruginae</i> <i>Butiaxella javanicae</i> <i>Butiaxella izardii</i> <i>Butiaxella noackiae</i> <i>Butiaxella warmholdiae</i> <i>Buyrvibrio</i> <i>Buyrvibrio fibrisolvens</i> <i>Buyrvibrio hungatei</i> <i>Buyrvibrio proteoclasticus</i>
<i>Bacillus</i> <i>B. acidoceler</i> <i>B. acidicola</i> <i>B. acidiproducens</i> <i>B. acidocaldarius</i> <i>B. acidoterresstris</i> <i>B. acaulus</i> <i>B. aerius</i> <i>B. aerophilus</i> <i>B. agaradhaerens</i> <i>B. agri</i> <i>B. aidnensis</i> <i>B. akibai</i> <i>B. alcalophilus</i>	<i>B. aminovorans</i> <i>B. amylolyticus</i> <i>B. andresenii</i> <i>B. aneurinilyticus</i> <i>B. anthracis</i> <i>B. aquimaris</i> <i>B. arenosi</i> <i>B. arseniciselenatis</i> <i>B. arsenicus</i> <i>B. aurantiacus</i> <i>B. arvi</i> <i>B. aryabhattai</i> <i>B. asahi</i>	<i>B. taeanensis</i> <i>B. tequilensis</i> <i>B. thermantarcticus</i> <i>B. thermoaerophilus</i> <i>B. thermoamylovorans</i> <i>B. thermocatulatus</i> <i>B. thermocloacae</i> <i>B. thermocopriae</i> <i>B. thermodenitrificans</i> <i>B. halodurans</i> <i>B. halophilus</i> <i>B. halosaccharovorans</i> <i>B. hemicellulosilyticus</i> <i>B. hemicentroti</i>	<i>B. lautus</i> <i>B. lehensis</i> <i>B. lentimorbus</i> <i>B. lentus</i> <i>B. licheniformis</i> <i>B. ligniniphilus</i> <i>B. litoralis</i> <i>B. locustalis</i> <i>B. luciferensis</i> <i>B. luteolus</i> <i>B. luteus</i> <i>B. macearans</i> <i>B. macerans</i>

TABLE 1-continued

Optionally, the target host cells are cells of a genus or species selected from this Table and/or the production strain cells are cells of a genus or species selected from this Table		
Example Bacteria		
<i>B. stratosphericus</i>	<i>Campylobacter</i>	<i>Catenuloplanes</i>
<i>B. subterraneus</i>	<i>Campylobacter coli</i>	<i>Catenuloplanes atrovinosus</i>
<i>B. subtilis</i> (eg. <i>B.</i>	<i>Campylobacter concisus</i>	<i>Catenuloplanes castaneus</i>
s. subsp. <i>Inaquosorum</i> , or <i>B.</i>	<i>Campylobacter curvus</i>	<i>Catenuloplanes crispus</i>
s. subsp. <i>Subtilis</i>)	<i>Campylobacter fetus</i>	<i>Catenuloplanes indicus</i>
	<i>Campylobacter gracilis</i>	<i>Catenuloplanes japonicus</i>
	<i>Campylobacter helveticus</i>	<i>Catenuloplanes nepalensis</i>
	<i>Campylobacter hominis</i>	<i>Catenuloplanes niger</i>
	<i>Campylobacter hyoimnestinalis</i>	<i>Chryseobacterium</i>
	<i>Campylobacter jejuni</i>	<i>Chryseobacterium</i>
	<i>Campylobacter lari</i>	<i>balastium</i>
	<i>Campylobacter mucosalis</i>	<i>Citrobacter</i>
	<i>Campylobacter rectus</i>	<i>C. amalonaticus</i>
	<i>Campylobacter showae</i>	<i>C. braakii</i>
	<i>Campylobacter sputorum</i>	<i>C. diversus</i>
	<i>Campylobacter upsaliensis</i>	<i>C. farmeri</i>
	<i>Capnocytophaga</i>	<i>C. freundii</i>
	<i>Capnocytophaga canimorsus</i>	<i>C. gillenii</i>
	<i>Capnocytophaga cynodegmi</i>	<i>C. koseri</i>
	<i>Capnocytophaga gingivitis</i>	<i>C. murinae</i>
	<i>Capnocytophaga granulosa</i>	<i>C. pasteurii</i> ⁽¹⁾
	<i>Capnocytophaga haemolytica</i>	<i>C. rodentium</i>
	<i>Capnocytophaga ochracea</i>	<i>C. seelaktii</i>
	<i>Capnocytophaga sputigena</i>	<i>C. werkmanii</i>
		<i>C. youngae</i>
		<i>Clostridium</i>
		(see below)
		<i>Coccobacter</i>
		<i>Coccobacter elabens</i>
		<i>Corynebacterium</i>
		<i>Corynebacterium flavescens</i>
		<i>Corynebacterium variabile</i>
<i>Clostridium</i>		
<i>Clostridium absonum</i> , <i>Clostridium acetivulvum</i> , <i>Clostridium acidisoli</i> , <i>Clostridium aciditerans</i> , <i>Clostridium acidurici</i> , <i>Clostridium aerotolerans</i> , <i>Clostridium</i>		
<i>aestuarii</i> , <i>Clostridium abagii</i> , <i>Clostridium aldenense</i> , <i>Clostridium aldrichii</i> , <i>Clostridium algidivulvum</i> , <i>Clostridium alginiphilum</i> , <i>Clostridium alkaliphilum</i> , <i>Clostridium</i>		
<i>aminophilum</i> , <i>Clostridium aminovalericum</i> , <i>Clostridium amygdalinum</i> , <i>Clostridium amylophilum</i> , <i>Clostridium arbuti</i> , <i>Clostridium arcticum</i> , <i>Clostridium argentinense</i> , <i>Clostridium asparagiforme</i> , <i>Clostridium</i>		
<i>aurantivulvum</i> , <i>Clostridium autoethanogenum</i> , <i>Clostridium baronii</i> , <i>Clostridium bartlettii</i> , <i>Clostridium bejerinckii</i> , <i>Clostridium bifementans</i> , <i>Clostridium boltae</i> , <i>Clostridium</i>		
<i>boninense</i> , <i>Clostridium botulinum</i> , <i>Clostridium bowmanii</i> , <i>Clostridium butyricum</i> , <i>Clostridium cadaveris</i> , <i>Clostridium caenicola</i> , <i>Clostridium caninihermale</i> , <i>Clostridium carboxidivorans</i> , <i>Clostridium</i>		
<i>carnis</i> , <i>Clostridium cavendishii</i> , <i>Clostridium celatum</i> , <i>Clostridium celerecrescens</i> , <i>Clostridium cellobiosporum</i> , <i>Clostridium cellulofementans</i> , <i>Clostridium cellulolyticum</i> , <i>Clostridium</i>		
<i>cellulovorans</i> , <i>Clostridium chauvoei</i> , <i>Clostridium chromiireducens</i> , <i>Clostridium citroniae</i> , <i>Clostridium clariflavum</i> , <i>Clostridium clostridioforme</i> , <i>Clostridium coccoides</i> , <i>Clostridium</i>		
<i>cochlearium</i> , <i>Clostridium colicinis</i> , <i>Clostridium colinum</i> , <i>Clostridium collagenovorans</i> , <i>Clostridium cylindrosporum</i> , <i>Clostridium difficile</i> , <i>Clostridium diolis</i> , <i>Clostridium</i>		
<i>disporicum</i> , <i>Clostridium drakei</i> , <i>Clostridium durum</i> , <i>Clostridium estertheticum</i> , <i>Clostridium estertheticum laramiense</i> , <i>Clostridium fallax</i> , <i>Clostridium felsineum</i> , <i>Clostridium</i>		
<i>fervidum</i> , <i>Clostridium finetarium</i> , <i>Clostridium formicaceticum</i> , <i>Clostridium frigidicarnis</i> , <i>Clostridium frigoris</i> , <i>Clostridium ganghewense</i> , <i>Clostridium gasigenes</i> , <i>Clostridium ghonii</i> , <i>Clostridium glycolicum</i> , <i>Clostridium</i>		
<i>glycyrhiziniluticum</i> , <i>Clostridium grantii</i> , <i>Clostridium haemolyticum</i> , <i>Clostridium halophilum</i> , <i>Clostridium hostiforme</i> , <i>Clostridium hahewayi</i> , <i>Clostridium herbivorans</i> , <i>Clostridium hirononis</i> , <i>Clostridium</i>		

Optionally, the target host cells are cells of a genus or species selected from this Table and/or the production strain cells are cells of a genus or species selected from this Table

Clostridium histolyticum, Clostridium homopropionicum, Clostridium huckii, Clostridium hungatei, Clostridium hydroxybenzoicum, Clostridium hylemonae, Clostridium jeikeense, Clostridium indolis, Clostridium innocuum, Clostridium intestinale, Clostridium isatidis, Clostridium josui, Clostridium lactatifermentans, Clostridium lacusprofundellense, Clostridium laramiense, Clostridium lentocellum, Clostridium lenopurescens, Clostridium leptum, Clostridium limosum, Clostridium litoreale, Clostridium litusehurense, Clostridium ljungdahlii, Clostridium lortaei, Clostridium lundense, Clostridium magnum, Clostridium malenominatum, Clostridium mangenotii, Clostridium mayombei, Clostridium methoxybenzovorans, Clostridium methylpentosum, Clostridium neopropionicum, Clostridium nexile, Clostridium nitrophenolicum, Clostridium novyi, Clostridium oceanicum, Clostridium orbischdens, Clostridium oiticum, Clostridium oxalicum, Clostridium papayrosolvens, Clostridium paradoxum, Clostridium paraperfringens (Alias: C. welchii), Clostridium paraputrificum, Clostridium pasteurianum, Clostridium peptidovorans, Clostridium pereme, Clostridium perfringens, Clostridium plennigii, Clostridium phytofermentans, Clostridium piliforme, Clostridium polysaccharolyticum, Clostridium populei, Clostridium propionicum, Clostridium proteoclasticum, Clostridium proteolyticum, Clostridium psychrophilum, Clostridium puncteum, Clostridium putrificans, Clostridium putrificum, Clostridium quercetorum, Clostridium quinti, Clostridium ramosum, Clostridium rectum, Clostridium roseum, Clostridium saccharogumita, Clostridium saccharolyticum, Clostridium saccharoperbutylacetonicum, Clostridium sardintense, Clostridium sartagoforme, Clostridium scatologenes, Clostridium schimacherense, Clostridium scindens, Clostridium septicum, Clostridium sorbollii, Clostridium sphenoides, Clostridium spiriforme, Clostridium sporogenes, Clostridium sporosphaeroides, Clostridium stercorarium, Clostridium stercorarium leptospartum, Clostridium stercorarium stercoratum, Clostridium stercorarium thermolacticum, Clostridium sticklandii, Clostridium stramineosolvens, Clostridium subterminale, Clostridium sufflavum, Clostridium sulfidigenes, Clostridium symbiosum, Clostridium tagliuense, Clostridium tepidipropifidii, Clostridium terminidis, Clostridium tertium, Clostridium tetani, Clostridium tetanomorphum, Clostridium thermacetivum, Clostridium thermacetitrophicum, Clostridium thermocalcapitulum, Clostridium thermobutyricum, Clostridium thermocellum, Clostridium thermocopriace, Clostridium thermohydrosulfuricum, Clostridium thermolacticum, Clostridium thermopalmarium, Clostridium thermopapulyticum, Clostridium thermosaccharolyticum, Clostridium thermosaccharogenicus, Clostridium thermosulfurigenes, Clostridium thiosulfatireducens, Clostridium tyrobutyricum, Clostridium uliginosum, Clostridium ulunense, Clostridium ulunense, Clostridium vincentii, Clostridium viride, Clostridium xylanolyticum, Clostridium xylanovorans	Deinococcus Deinococcus aertus Deinococcus apachensis Deinococcus aquaticus Deinococcus aquatilis Deinococcus caeni Deinococcus radiodurans Deinococcus radiophilus Enterobacter kobei E. ludwigii E. mori E. nimirpressuralis E. oryzae E. pulveris E. pyrrhus E. radicicans E. taylorae E. turicensis E. sakazakii Enterobacter soli Enterococcus Enterococcus durans Enterococcus faecalis Enterococcus faecium Erwinia Erwinia hapontica Escherichia Escherichia coli Haemophilus Haemophilus aegyptius Haemophilus aphrophilus Haemophilus felis Haemophilus gallinarum Haemophilus haemolyticus Haemophilus influenzae	Delftia Delftia acidovorans Desulfovibrio Desulfovibrio desulfuricans Diplococcus Diplococcus pneumoniae Faecalibacterium Faecalibacterium prausnitzii Fangia Fangia hongkongensis Fastidiosipila Fastidiosipila sanguinis Fusobacterium Fusobacterium nucleatum	Echinicola Echinicola pacifica Echinicola vietnamensis Flavobacterium Flavobacterium antarcticum Flavobacterium aquatile Flavobacterium aquidurens Flavobacterium balustinum Flavobacterium croceum Flavobacterium cucumis Flavobacterium daejeonense Flavobacterium defluvi Flavobacterium degerlachei Flavobacterium dentrificans Flavobacterium filum Flavobacterium flevense Flavobacterium frigidarium Flavobacterium mizutaii Flavobacterium okeanoikoites Janibacter Janibacter anophelis Janibacter corallicola Janibacter limosus Janibacter melonis Janibacter terrae Janiaschia
Dactylosporangium Dactylosporangium aurantiacum Dactylosporangium fulvum Dactylosporangium matsuzakense Dactylosporangium roseum Dactylosporangium thailandense Dactylosporangium vinaceum Enterobacter E. aerogenes E. amnigenis E. agglomerans E. arachidis E. asburiae E. cancerogenus E. cloacae E. cowanii E. dissolvens E. gergoviae E. helveticus E. hormaechei E. intermedins	Gaetbulibacter Gaetbulibacter saemankumensis Gallibacterium Gallibacterium anatis Gallicola Gallicola barnesae Gartcella		

TABLE 1-continued

Optionally, the target host cells are cells of a genus or species selected from this Table and/or the production strain cells are cells of a genus or species selected from this Table		Example Bacteria	
<i>Garcella nitratireducens</i>	<i>Haemophilus paracuniculus</i>	<i>Idiomarina rumbicola</i>	<i>Jannaschia cystaurens</i>
<i>Geobacillus</i>	<i>Haemophilus parahaemolyticus</i>	<i>Idiomarina seosinensis</i>	<i>Jannaschia helgolandensis</i>
<i>Geobacillus thermoglucosidasius</i>	<i>Haemophilus parainfluenzae</i>	<i>Idiomarina zobellii</i>	<i>Jannaschia pohangensis</i>
<i>Geobacillus stearothermophilus</i>	<i>Haemophilus</i>	<i>Ignatzschineria</i>	<i>Jannaschia rubra</i>
<i>Geobacter</i>	<i>paraphrohaemolyticus</i>	<i>Ignatzschineria larvae</i>	
<i>Geobacter brennii</i>	<i>Haemophilus parvus</i>		<i>Janthinobacterium</i>
<i>Geobacter brennii</i>	<i>Haemophilus pittmaniae</i>	<i>Ignavigranum</i>	<i>Janthinobacterium</i>
<i>Geobacter chapellei</i>	<i>Hafnia</i>	<i>Ignavigranum ruoffiae</i>	<i>agardiamosum</i>
<i>Geobacter glycolae</i>	<i>Hafnia alvei</i>	<i>Ilumatobacter</i>	<i>Janthinobacterium lividum</i>
<i>Geobacter hydrogenophilus</i>	<i>Hahella</i>	<i>Ilumatobacter fluminis</i>	<i>Jejuna</i>
<i>Geobacter lovleyi</i>	<i>Hahella gonghwensis</i>	<i>Ilyobacter</i>	<i>Jejuna pallidilinea</i>
<i>Geobacter metallireducens</i>	<i>Halalkalicoccus</i>	<i>Ilyobacter delafeldii</i>	<i>Jeogalibacillus</i>
<i>Geobacter pelophilus</i>	<i>Halalkalicoccus halophilus</i>	<i>Ilyobacter insuetus</i>	<i>Jeogalibacillus</i>
<i>Geobacter pickeringii</i>	<i>Helicobacter</i>	<i>Ilyobacter polytropus</i>	<i>alimentarius</i>
<i>Geobacter sulfurreducens</i>	<i>Helicobacter pylori</i>	<i>Ilyobacter tartaricus</i>	<i>Jeogalibacillus</i>
<i>Geodermaphilus</i>			<i>Jeogalibacillus</i>
<i>Geodermaphilus obscurus</i>			<i>Jeogalibacillus</i>
<i>Gluconacetobacter</i>			<i>Jeogalibacillus</i>
<i>Gluconacetobacter xylinus</i>			<i>Jeogalibacillus</i>
<i>Gordonia</i>			<i>Jeogalibacillus</i>
<i>Gordonia rubripertincta</i>			<i>Jeogalibacillus</i>
<i>Kaistia</i>	<i>Labedella</i>	<i>Listeria ivanovii</i>	<i>Micrococcus</i>
<i>Kaistia adipata</i>	<i>Labedella gwaljiensis</i>	<i>L. marthii</i>	<i>Micrococcus luteus</i>
<i>Kaistia soli</i>	<i>Labrenzia</i>	<i>L. monocytogenes</i>	<i>Micrococcus lylae</i>
<i>Kangella</i>	<i>Labrenzia aggregata</i>	<i>L. newyorkensis</i>	<i>Moraxella</i>
<i>Kangella aquimarina</i>	<i>Labrenzia alba</i>	<i>L. riparia</i>	<i>Moraxella bovis</i>
<i>Kangella korensis</i>	<i>Labrenzia alexandrii</i>	<i>L. rocourtiae</i>	<i>Moraxella nonliquefaciens</i>
	<i>Labrenzia marina</i>	<i>L. seeligeri</i>	<i>Moraxella osloensis</i>
<i>Kersteria</i>	<i>Labrys</i>	<i>L. weihenstephanensis</i>	<i>Nakamurella</i>
<i>Kersteria gytiorum</i>	<i>Labrys methylaminiphilus</i>	<i>L. welshimeri</i>	<i>Nakamurella multipartita</i>
<i>Kilonella</i>	<i>Labrys myogitensis</i>	<i>Listonella</i>	<i>Nannocystis</i>
<i>Kilonella laminariae</i>	<i>Labrys monachus</i>	<i>Listonella anguillarum</i>	<i>Nannocystis pusilla</i>
<i>Klebstella</i>	<i>Labrys okinawensis</i>	<i>Macrocooccus</i>	<i>Natranaerobius</i>
<i>K. gramilomatis</i>	<i>Labrys portucalensis</i>	<i>Macrocooccus bovicus</i>	<i>Natranaerobius</i>
<i>K. oxytoca</i>		<i>Marinobacter</i>	<i>thermophilus</i>
<i>K. pneumoniae</i>	<i>Lactobacillus</i>	<i>Marinobacter algicola</i>	<i>Natranaerobius trueperi</i>
<i>K. terrigena</i>	[see below]	<i>Marinobacter bryozorum</i>	<i>Naxibacter</i>
<i>K. varitcola</i>	<i>Laceyella</i>	<i>Marinobacter flavinaris</i>	<i>Naxibacter alkalitolerans</i>
<i>Kluyvera</i>	<i>Laceyella putida</i>	<i>Metothermus</i>	<i>Neisseria</i>
<i>Kluyvera ascorbata</i>	<i>Lechevalieria</i>	<i>Metothermus ruber</i>	<i>Neisseria cinerea</i>
<i>Kocuria</i>	<i>Lechevalieria aerocolonigenes</i>	<i>Methylophilus</i>	<i>Neisseria denitrificans</i>
<i>Kocuria roasea</i>	<i>Legionella</i>	<i>Methylophilus methylotrophus</i>	<i>Neisseria gonorrhoeae</i>
<i>Kocuria varians</i>	[see below]	<i>Microbacterium</i>	<i>Neisseria lactamica</i>
<i>Kurthia</i>	<i>Listeria</i>	<i>Microbacterium</i>	<i>Neisseria mucosa</i>
<i>Kurthia zoefii</i>	<i>L. aquatica</i>	<i>ammoniphilum</i>	<i>Neisseria sicca</i>
	<i>L. booriae</i>	<i>Microbacterium arborescens</i>	<i>Neisseria subflava</i>
	<i>L. cornellensis</i>	<i>Microbacterium liquefaciens</i>	<i>Neptunomonas</i>
	<i>L. fleischmannii</i>	<i>Microbacterium oxydans</i>	<i>Neptunomonas japonica</i>
	<i>L. floridensis</i>		
	<i>L. grandensis</i>		

Optionally, the target host cells are cells of a genus or species selected from this Table and/or the production strain cells are cells of a genus or species selected from this Table

[illegible]

TABLE 1-continued

Optionally, the target host cells are cells of a genus or species selected from this Table and/or the production strain cells are cells of a genus or species selected from this Table		Example Bacteria	
<i>Oceanicaulis</i>	<i>Pantoea</i>	<i>Prevotella amnii</i>	<i>Quarionococcus</i>
<i>Oceanicaulis alexandrii</i>	<i>Pantoea agglomerans</i>	<i>Prevotella bergensis</i>	<i>Quarionococcus</i>
<i>Oceanicola</i>		<i>Prevotella bivia</i>	<i>australiensis</i>
<i>Oceanicola batsensis</i>	<i>Paracoccus</i>	<i>Prevotella brevis</i>	
<i>Oceanicola granulosis</i>	<i>Paracoccus alcaliphilus</i>	<i>Prevotella bryantii</i>	<i>Quinella</i>
<i>Oceanicola nankaiensis</i>	<i>Paucimonas</i>	<i>Prevotella buccae</i>	<i>Quinella ovalis</i>
<i>Oceanimonas</i>	<i>Paucimonas lemoignei</i>	<i>Prevotella buccalis</i>	
<i>Oceanimonas baumannii</i>	<i>Pectobacterium</i>	<i>Prevotella copri</i>	
<i>Oceaniserpenitilla</i>	<i>Pectobacterium aroidearum</i>	<i>Prevotella dentalis</i>	<i>Ralstonia</i>
<i>Oceaniserpenitilla halotidis</i>	<i>Pectobacterium atrosepticum</i>	<i>Prevotella denticola</i>	<i>Ralstonia eutropha</i>
<i>Oceanisphaera donghaensis</i>	<i>Pectobacterium betavascularum</i>	<i>Prevotella distens</i>	<i>Ralstonia insidiosa</i>
<i>Oceanisphaera litoralis</i>	<i>Pectobacterium cacticola</i>	<i>Prevotella histicola</i>	<i>Ralstonia mannitolitifica</i>
<i>Oceanithermus</i>	<i>Pectobacterium carnegiana</i>	<i>Prevotella intermedia</i>	<i>Ralstonia pickettii</i>
<i>Oceanithermus desulfurans</i>	<i>Pectobacterium carotovorum</i>	<i>Prevotella maculosa</i>	<i>Ralstonia</i>
<i>Oceanithermus profundus</i>	<i>Pectobacterium chrysanthemi</i>	<i>Prevotella marshii</i>	<i>pseudosolanacearum</i>
<i>Oceanobacillus</i>	<i>Pectobacterium cypripedii</i>	<i>Prevotella melanogena</i>	<i>Ralstonia syzygii</i>
<i>Oceanobacillus caeni</i>	<i>Pectobacterium rhapontici</i>	<i>Prevotella micans</i>	<i>Ralstonia solanacearum</i>
<i>Oceanospirillum</i>	<i>Pectobacterium wasabiae</i>	<i>Prevotella multiformis</i>	<i>Ramlibacter</i>
<i>Oceanospirillum litum</i>	<i>Pianococcus</i>	<i>Prevotella nigrescens</i>	<i>Ramlibacter tataouinensis</i>
	<i>Pianococcus citreus</i>	<i>Prevotella oralis</i>	
	<i>Planomicrobium</i>	<i>Prevotella oris</i>	<i>Raoultella</i>
	<i>Planomicrobium okeanokoites</i>	<i>Prevotella outorum</i>	<i>Raoultella ornithinolytica</i>
	<i>Plesiomonas</i>	<i>Prevotella pallens</i>	<i>Raoultella planticola</i>
	<i>Plesiomonas shigelloides</i>	<i>Prevotella salivae</i>	<i>Raoultella terrigena</i>
	<i>Proteus</i>	<i>Prevotella stercora</i>	<i>Rathayibacter</i>
	<i>Proteus vulgaris</i>	<i>Prevotella tanneriae</i>	<i>Rathayibacter caricis</i>
		<i>Prevotella timonensis</i>	<i>Rathayibacter festucae</i>
		<i>Prevotella veroralis</i>	<i>Rathayibacter iranica</i>
		<i>Providencia</i>	<i>Rathayibacter rathayi</i>
		<i>Providencia stuartii</i>	<i>Rathayibacter toxicus</i>
		<i>Pseudomonas</i>	<i>Rathayibacter tritici</i>
		<i>Pseudomonas aeruginosa</i>	<i>Rhodobacter</i>
		<i>Pseudomonas alcaligenes</i>	<i>Rhodobacter sphaeroides</i>
		<i>Pseudomonas anguilliseptica</i>	<i>Ruegeria</i>
		<i>Pseudomonas fluorescens</i>	<i>Ruegeria gelatinovorans</i>
		<i>Pseudoalteromonas</i>	
		<i>haloplanktis</i>	
		<i>Pseudomonas mendocina</i>	
		<i>Pseudomonas</i>	
		<i>pseudocaligenes</i>	
		<i>Pseudomonas putida</i>	
		<i>Pseudomonas tuzeri</i>	
		<i>Pseudomonas syringae</i>	
		<i>Psychrobacter</i>	
		<i>Psychrobacter faecalis</i>	
		<i>Psychrobacter</i>	
		<i>phenylpyruvicus</i>	
<i>Saccharococcus</i>	<i>Sagittula</i>	<i>Sanguibacter</i>	<i>Stenotrophomonas</i>
<i>Saccharococcus thermophilus</i>	<i>Sagittula stellata</i>	<i>Sanguibacter keddiei</i>	<i>Tatlockia maceachernii</i>
<i>Saccharomonospora</i>	<i>Salegentibacter</i>	<i>Sanguibacter suarezi</i>	<i>Tatlockia micdadei</i>

Optionally, the target host cells are cells of a genus or species selected from this Table and/or the production strain cells are cells of a genus or species selected from this Table

<i>Saccharomonospora azurea</i>	<i>Salegentibacter salegens</i>	<i>Saprospiria</i>	<i>Streptococcus</i>	<i>Tenacibaculum</i>
<i>Saccharomonospora cyanea</i>	<i>Salinicrobium</i>	<i>Saprospira grandis</i>		<i>Tenacibaculum</i>
<i>Saccharomonospora viridis</i>	<i>Salinicrobium album</i>	<i>Sarcina</i>	[also see below]	<i>amylolyticum</i>
<i>Saccharophagus</i>	<i>Salinibacter</i>	<i>Sarcina maxima</i>	<i>Streptomyces</i>	<i>Tenacibaculum discolor</i>
<i>Saccharophagus degradans</i>	<i>Salinibacter ruber</i>	<i>Sarcina ventriculi</i>	<i>Streptomyces</i>	<i>Tenacibaculum</i>
<i>Saccharopolyspora</i>	<i>Salinococcus</i>	<i>Sebadella</i>	<i>achromogenes</i>	<i>gallitatum</i>
<i>Saccharopolyspora erythraea</i>	<i>Salinococcus alkaliphilus</i>	<i>Sebadella termitidis</i>	<i>Streptomyces cesalpii</i>	<i>Tenacibaculum</i>
<i>Saccharopolyspora greggii</i>	<i>Salinococcus hispanicus</i>		<i>Streptomyces cesaepitosus</i>	<i>lutimaris</i>
<i>Saccharopolyspora hirsuta</i>	<i>Salinococcus roseus</i>	<i>Serratia</i>	<i>Streptomyces cesiastaticus</i>	<i>Tenacibaculum</i>
<i>Saccharopolyspora hordei</i>	<i>Salinispora</i>	<i>Serratia fonticola</i>	<i>Streptomyces cesexfoliatus</i>	<i>Tenacibaculum</i>
<i>Saccharopolyspora rectivirgula</i>	<i>Salinispora arenicola</i>	<i>Serratia marcescens</i>	<i>Streptomyces fimbricatus</i>	<i>skagerakense</i>
<i>Saccharopolyspora spinosa</i>	<i>Salinispora tropica</i>	<i>Sphaerotilus</i>	<i>Streptomyces fradiae</i>	<i>Teptidanaerobacter</i>
<i>Saccharopolyspora taberi</i>	<i>Salinivibrio</i>	<i>Sphaerotilus natans</i>	<i>Streptomyces fulvissimus</i>	<i>Teptidanaerobacter</i>
<i>Saccharothrix</i>	<i>Salinivibrio costicola</i>	<i>Sphingobacterium</i>	<i>Streptomyces griseoruber</i>	<i>synrophicus</i>
<i>Saccharothrix australiensis</i>	<i>Salmonella</i>	<i>Sphingobacterium multivorum</i>	<i>Streptomyces griseus</i>	<i>Teptidibacter</i>
<i>Saccharothrix coeruleofusca</i>	<i>Salmonella bongori</i>	<i>Staphylococcus</i>	<i>Streptomyces lavendulae</i>	<i>Teptidibacter</i>
<i>Saccharothrix espanaensis</i>	<i>Salmonella enterica</i>	[see below]	<i>Streptomyces</i>	<i>formicigenes</i>
<i>Saccharothrix longispora</i>	<i>Salmonella subterranea</i>		<i>phaeochromogenes</i>	<i>Teptidibacter thalassicus</i>
<i>Saccharothrix mutabilis</i>	<i>Salmonella typhi</i>		<i>Streptomyces</i>	<i>Thermus</i>
<i>Saccharothrix syringae</i>			<i>thermodiastaticus</i>	<i>Thermus aquaticus</i>
<i>Saccharothrix tangerinus</i>			<i>Streptomyces tubercidicus</i>	<i>Thermus filiformis</i>
<i>Saccharothrix texasensis</i>				<i>Thermus thermophilus</i>
<i>Staphylococcus</i>				
<i>S. arlettae</i>	<i>S. equorum</i>	<i>S. microti</i>	<i>S. schlegelii</i>	
<i>S. agnetis</i>	<i>S. felis</i>	<i>S. muscae</i>	<i>S. scituri</i>	
<i>S. aureus</i>	<i>S. fleuretti</i>	<i>S. neapalensis</i>	<i>S. simiae</i>	
<i>S. auricularis</i>	<i>S. gallinarum</i>	<i>S. pasteurii</i>	<i>S. simulans</i>	
<i>S. capitis</i>	<i>S. haemolyticus</i>	<i>S. petrasii</i>	<i>S. streptanovicii</i>	
<i>S. caprae</i>	<i>S. hominis</i>	<i>S. pettenkoferi</i>	<i>S. succinus</i>	
<i>S. carnosus</i>	<i>S. lyticus</i>	<i>S. piscifermentans</i>	<i>S. vitulinus</i>	
<i>S. cecolyticus</i>	<i>S. intermedius</i>	<i>S. pseudintermedius</i>	<i>S. warneri</i>	
<i>S. chromogenes</i>	<i>S. kloosii</i>	<i>S. pseudolugdunensis</i>	<i>S. xylosus</i>	
<i>S. cohnii</i>	<i>S. leei</i>	<i>S. pulverii</i>		
<i>S. condimenti</i>	<i>S. lentus</i>	<i>S. rostri</i>		
<i>S. delphini</i>	<i>S. lugdunensis</i>	<i>S. saccharolyticus</i>		
<i>S. devriesei</i>	<i>S. lutrae</i>	<i>S. saprophyticus</i>		
<i>S. epidermidis</i>	<i>S. lyticus</i>			
	<i>S. massiliensis</i>			
<i>Streptococcus</i>		<i>Streptococcus orisratii</i>	<i>Streptococcus thermophilus</i>	
<i>Streptococcus agalactiae</i>	<i>Streptococcus infantarius</i>	<i>Streptococcus parasanguinis</i>	<i>Streptococcus sanguinis</i>	
<i>Streptococcus anginosus</i>	<i>Streptococcus iniae</i>	<i>Streptococcus peroris</i>	<i>Streptococcus sobrinus</i>	
<i>Streptococcus bovis</i>	<i>Streptococcus intermedius</i>	<i>Streptococcus pneumoniae</i>	<i>Streptococcus suis</i>	
<i>Streptococcus canis</i>	<i>Streptococcus lactarius</i>	<i>Streptococcus</i>	<i>Streptococcus uberis</i>	
<i>Streptococcus constellatus</i>	<i>Streptococcus milleri</i>	<i>pseudopneumoniae</i>	<i>Streptococcus vestibularis</i>	
<i>Streptococcus downii</i>	<i>Streptococcus mitis</i>	<i>Streptococcus pyogenes</i>	<i>Streptococcus viridans</i>	
<i>Streptococcus dysgalactiae</i>	<i>Streptococcus mutans</i>	<i>Streptococcus ratti</i>	<i>Streptococcus</i>	
<i>Streptococcus equinus</i>	<i>Streptococcus oralis</i>	<i>Streptococcus salivarius</i>	<i>zooepidemicus</i>	
<i>Streptococcus faecalis</i>	<i>Streptococcus tigurinus</i>			
<i>Streptococcus ferus</i>				

TABLE 1-continued

Example Bacteria			Optionally, the target host cells are cells of a genus or species selected from this Table and/or the production strain cells are cells of a genus or species selected from this Table	
<i>Uliginosibacterium</i>	<i>Vagococcus</i>	<i>Vibrio</i>	<i>Virgibacillus</i>	<i>Xanthobacter</i>
<i>Uliginosibacterium gangwonense</i>	<i>Vagococcus carnisphilus</i>	<i>Vibrio aerogenes</i>	<i>Virgibacillus</i>	<i>Xanthobacter agilis</i>
<i>Ulvibacter</i>	<i>Vagococcus elongatus</i>	<i>Vibrio aestuarianus</i>	<i>Halodentitrificans</i>	<i>Xanthobacter</i>
<i>Umezawaea</i>	<i>Vagococcus fissus</i>	<i>Vibrio albensis</i>	<i>Virgibacillus</i>	<i>aminoxidans</i>
<i>Ureaplasma</i>	<i>Vagococcus lutrae</i>	<i>Vibrio alginolyticus</i>	<i>panthothenicus</i>	<i>Xanthobacter</i>
<i>Ureobacillus</i>	<i>Vagococcus salmoninarum</i>	<i>Vibrio campbellii</i>	<i>Weissella</i>	<i>autotrophicus</i>
<i>Ureobacillus composti</i>	<i>Variovorax</i>	<i>Vibrio cholerae</i>	<i>Weissella cibaria</i>	<i>Xanthobacter flavus</i>
<i>Ureobacillus suwonensis</i>	<i>Variovorax boronicumulans</i>	<i>Vibrio cincinnatiensis</i>	<i>Weissella confusa</i>	<i>Xanthobacter tagetidis</i>
<i>Ureobacillus terrenus</i>	<i>Variovorax dokdonensis</i>	<i>Vibrio corallilyticus</i>	<i>Weissella halotolerans</i>	<i>Xanthobacter viscosus</i>
<i>Ureobacillus thermophilus</i>	<i>Variovorax paradoxus</i>	<i>Vibrio cyclitrophicus</i>	<i>Weissella hellenica</i>	<i>Xanthomonas</i>
<i>Ureobacillus thermosphaericus</i>	<i>Variovorax soli</i>	<i>Vibrio diazotrophicus</i>	<i>Weissella kandleri</i>	<i>Xanthomonas</i>
	<i>Veillonella</i>	<i>Vibrio fluvialis</i>	<i>Weissella koreensis</i>	<i>albilineans</i>
	<i>Veillonella atypica</i>	<i>Vibrio furnissii</i>	<i>Weissella minor</i>	<i>Xanthomonas alfalfae</i>
	<i>Veillonella caviae</i>	<i>Vibrio gazogenes</i>	<i>Weissella</i>	<i>Xanthomonas</i>
	<i>Veillonella criceti</i>	<i>Vibrio halotitoli</i>	<i>paramesenteroides</i>	<i>arboricola</i>
	<i>Veillonella dispersa</i>	<i>Vibrio harveyi</i>	<i>Weissella soli</i>	<i>Xanthomonas</i>
	<i>Veillonella parvula</i>	<i>Vibrio ichthyoveneri</i>	<i>Weissella thailandensis</i>	<i>axonopodis</i>
	<i>Veillonella ratti</i>	<i>Vibrio mediterranei</i>	<i>Weissella viridescens</i>	<i>Xanthomonas</i>
	<i>Veillonella rodentium</i>	<i>Vibrio metschnikovii</i>	<i>Williamia</i>	<i>campestris</i>
	<i>Yersinia</i>	<i>Vibrio mytili</i>	<i>Williamia marianensis</i>	<i>Xanthomonas citri</i>
	<i>Yersinia</i>	<i>Vibrio natriegens</i>	<i>Williamia maris</i>	<i>Xanthomonas codiae</i>
	<i>Yersinia</i>	<i>Vibrio navarrensis</i>	<i>Williamia serinedens</i>	<i>Xanthomonas</i>
	<i>Yersinia</i>	<i>Vibrio nereis</i>	<i>Winogradskyella</i>	<i>cucurbitae</i>
	<i>Yersinia</i>	<i>Vibrio ordalii</i>	<i>thalassocola</i>	<i>Xanthomonas</i>
	<i>Yersinia</i>	<i>Vibrio orientalis</i>	<i>Wolbachia</i>	<i>euvasicatoria</i>
	<i>Yersinia</i>	<i>Vibrio parahaemolyticus</i>	<i>Wolbachia persica</i>	<i>Xanthomonas fragariae</i>
	<i>Yersinia</i>	<i>Vibrio pectenicida</i>	<i>Wolbachia succinigenes</i>	<i>Xanthomonas fuscans</i>
	<i>Yersinia</i>	<i>Vibrio penaeicida</i>	<i>Zobellia</i>	<i>Xanthomonas gardneri</i>
	<i>Yersinia</i>	<i>Vibrio proteolyticus</i>	<i>Zobellia galactamivorans</i>	<i>Xanthomonas horiorum</i>
	<i>Yersinia</i>	<i>Vibrio shilonii</i>	<i>Zobellia uliginosa</i>	<i>Xanthomonas hyacinthi</i>
	<i>Yersinia</i>	<i>Vibrio splendidus</i>	<i>Zoogloea</i>	<i>Xanthomonas perforans</i>
	<i>Yersinia</i>	<i>Vibrio tubiashii</i>	<i>Zoogloea ramigera</i>	<i>Xanthomonas phaseoli</i>
	<i>Yersinia</i>	<i>Vibrio vulnificus</i>	<i>Zoogloea resiniophila</i>	<i>Xanthomonas pisi</i>
	<i>Yersinia</i>			<i>Xanthomonas populi</i>
	<i>Yersinia</i>			<i>Xanthomonas theicola</i>
	<i>Yersinia</i>			<i>Xanthomonas translucens</i>
	<i>Yersinia</i>			<i>Xanthomonas</i>
	<i>Yersinia</i>			<i>vesicatoria</i>
	<i>Yersinia</i>			<i>Xylella</i>
	<i>Yersinia</i>			<i>Xylella fastidiosa</i>
	<i>Yersinia</i>			<i>Xylophilus</i>
	<i>Yersinia</i>			<i>Xylophilus ampelinus</i>
	<i>Yersinia</i>			<i>Zobellia</i>
	<i>Yersinia</i>			<i>Zobellia denitrificans</i>
	<i>Yersinia</i>			<i>Zobellia taiwanensis</i>
	<i>Yersinia</i>			<i>Zeaxanthinibacter</i>
	<i>Yersinia</i>			<i>Zeaxanthinibacter</i>
	<i>Yersinia</i>			<i>enoshimensis</i>

TABLE 1-continued

Optionally, the target host cells are cells of a genus or species selected from this Table and/or the production strain cells are cells of a genus or species selected from this Table		Example Bacteria	
<i>Xenorhabdus griffithiae</i>	<i>Yersinia</i>	<i>Yokenella regensburgei</i>	<i>Zymomonas mobilis</i>
<i>Xenorhabdus hominickii</i>	<i>Yersinia aldovae</i>	<i>Yonghaparkia</i>	<i>Zymophilus</i>
<i>Xenorhabdus koppenhoeferi</i>	<i>Yersinia hercoverti</i>	<i>Yonghaparkia alkaliphila</i>	<i>Zymophilus paucivorans</i>
<i>Xenorhabdus nematophila</i>	<i>Yersinia enterocolitica</i>	<i>Zavarzhina</i>	<i>Zymophilus raffinosivorans</i>
<i>Xenorhabdus potmarii</i>	<i>Yersinia entomophaga</i>	<i>Zavarzhina compransoris</i>	<i>Xylanibacterium</i>
<i>Xylanibacter</i>	<i>Yersinia frederiksenii</i>		<i>Xylanibacterium ulmi</i>
<i>Xylanibacter oryzae</i>	<i>Yersinia intermedia</i>		
	<i>Yersinia kristensenii</i>		

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TABLE 2

Sequences	
Nucleic acid sequences herein are written in 5' to 3' direction; amino acid sequences are written in N- to C-terminal direction.	
SEQ ID NO: 1 (P10)	
TTTCAATTAAATCATCCGGCTCGTATAATGTGTGGA	
SEQ ID NO: 2 (BCD14)	
GGGCCCAAGTTCACTTAAAAAGGAGATCAACAATGAAAGCAATTTTCGTA	
CTGAAACATCTTAATCATGCGGTGGAGGGTTTCTAATG	
SEQ ID NO: 3 (gfp)	
ATGAGCAAAGGAGAAGAACTTTTCACTGGAGTTGTC	
SEQ IDs NO: 4 & 29 (example Expression Operating Unit, EOU)	
The EOU is (in 5' to 3' direction):-	
[SEQ ID NO: 4]-[promoter]-[TIS]-[GFP-encoding nucleotide sequence]-[SEQ ID NO: 29]	
Where	
SEQ ID NO: 4 is	
GAATTCAAAGATCTTAAGTAAGTAAGAGTATACGTATATCGGCTAATAA	
CGTATGAAGGCGCTTCGGCGCCTTTTTTATGGGGGTATTTTCATCCCAA	

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TABLE 2-continued

Sequences	
5	TCCACACGTCCAACGCACAGCAAAACACCACGTCGACCCTATCAGCTGCGT
	GCTTTCCTATGAGTCGTTGCTGCATAACTTGACAATTAATCATCCGGCTCG
TATAATGTGTGGAA	
SEQ ID NO: 29 is	
GGATCCAAACTCGAGTAAGGATCTCCAGGCATCAAATAAACGAAAGGCT	
10	CAGTCGAAAGACTGGGCCTTTTCGTTTTATCTGTTGTTTGTCTGGTGAACGC
	TCTCTACTAGAGTCACACTGGCTCACCTTCGGGTGGGCCTTCTGCGTTT
ATA	
SEQ ID NO: 5 (Example Shine Dalgarno Sequence)	
AAAGAGGAGAAA	
15	SEQ ID NO: 26 (Spacer sequence)
	CTTTGCCGCGCGCTTCGTCACGTAATTCGTCGCAA
SEQ ID NO: 27 (Spacer sequence)	
GTTTGGCGATGGCGCGGTGTGGTTGTGCTTCGGCGCT	
20	SEQ ID NO: 28 (Spacer sequence)
	TGGGATGCCTACCGCAAGCAGCTTGGCCTGAA

TABLE 3

Anderson Promoter Collection				
SEQ ID NO:	Identifier	Sequence ^a	Measured Strength ^b	
6	BBa J23119	TTGACAGCTAGCTCAGTCCTAGGTATAATGCTAGC	n/a	
7	BBa J23100	TTGACGGCTAGCTCAGTCCTAGGTACAGTGCTAGC	1	
8	BBa J23101	TTTACAGCTAGCTCAGTCCTAGGTATTATGCTAGC	0.7	
9	BBa J23102	TTGACAGCTAGCTCAGTCCTAGGTACTGTGCTAGC	0.86	
10	BBa J23103	CTGATAGCTAGCTCAGTCCTAGGGATTATGCTAGC	0.01	
11	BBa J23104	TTGACAGCTAGCTCAGTCCTAGGTATTGTGCTAGC	0.72	
12	BBa J23105	TTTACGGCTAGCTCAGTCCTAGGTACTATGCTAGC	0.24	
13	BBa J23106	TTTACGGCTAGCTCAGTCCTAGGTATAGTGCTAGC	0.47	
14	BBa J23107	TTTACGGCTAGCTCAGCCCTAGGTATTATGCTAGC	0.36	
15	BBa J23108	CTGACAGCTAGCTCAGTCCTAGGTATAATGCTAGC	0.51	
16	BBa J23109	TTTACAGCTAGCTCAGTCCTAGGGACTGTGCTAGC	0.04	
17	BBa J23110	TTTACGGCTAGCTCAGTCCTAGGTACAATGCTAGC	0.33	
18	BBa J23111	TTGACGGCTAGCTCAGTCCTAGGTATAGTGCTAGC	0.58	
19	BBa J23112	CTGATAGCTAGCTCAGTCCTAGGGATTATGCTAGC	0	
20	BBa J23113	CTGATGGCTAGCTCAGTCCTAGGGATTATGCTAGC	0.01	
21	BBa J23114	TTTATGGCTAGCTCAGTCCTAGGTACAATGCTAGC	0.1	
22	BBa J23115	TTTATAGCTAGCTCAGCCCTTGGTACAATGCTAGC	0.15	
23	BBa J23116	TTGACAGCTAGCTCAGTCCTAGGGACTATGCTAGC	0.16	

TABLE 3-continued

Anderson Promoter Collection			
SEQ ID NO:	Identifier	Sequence ^a	Measured Strength ^b
24	BBa J23117	TTGACAGCTAGCTCAGTCCTAGGGATTGTGCTAGC	0.06
25	BBa J23118	TTGACGGCTAGCTCAGTCCTAGGTATTGTGCTAGC	0.56

^aalso shown in the Anderson Catalog, see parts.igem.org/Promoters/Catalog/Anderson

^bStrength is the Anderson Score (AS), e.g., a strength of 1 is a AS of 1. Reported activities of the promoters are given as the relative fluorescence of plasmids in strain TGI grown in LB media to saturation. A suitable plasmid is EX-Ptet-S-rbsRFP-P "RFP reporter" as described at parts.igem.org/Part:BBa_J61002; insertion of a promoter element between XbaI and SpeI sites results in a RFP reporter.

SEQUENCE LISTING

Sequence total quantity: 29

SEQ ID NO: 1 moltype = DNA length = 36

FEATURE Location/Qualifiers

misc_feature 1..36

note = Synthetic Construct

source 1..36

mol_type = other DNA

organism = synthetic construct

SEQUENCE: 1

tttcaattta atcatccggc tcgtataatg tgtgga 36

SEQ ID NO: 2 moltype = DNA length = 88

FEATURE Location/Qualifiers

misc_feature 1..88

note = Synthetic Construct

source 1..88

mol_type = other DNA

organism = synthetic construct

SEQUENCE: 2

gggcccaggc tcaacttaaa aggagatcaa caatgaaagc aattttcgta ctgaaacatc 60

ttaatcatgc ggtggagggt ttctaata 88

SEQ ID NO: 3 moltype = DNA length = 36

FEATURE Location/Qualifiers

misc_feature 1..36

note = Synthetic Construct

source 1..36

mol_type = other DNA

organism = synthetic construct

SEQUENCE: 3

atgagcaaag gagaagaact ttccactgga gttgtc 36

SEQ ID NO: 4 moltype = DNA length = 214

FEATURE Location/Qualifiers

misc_feature 1..214

note = Synthetic Construct

source 1..214

mol_type = other DNA

organism = synthetic construct

SEQUENCE: 4

gaattcaaaa gatcttaagt aagtaagagt atacgtatat cggtcaataa cgtattaagg 60

cgcttcggcg ccttttttta tgggggtatt ttcacccaa tccacacgac caacgcacag 120

caaacaccac gtcgacccta tcagctgcgt gctttctatg agtcgtgtgt gcataacttg 180

acaattaatc atccggctcg tataatgtgt ggaa 214

SEQ ID NO: 5 moltype = DNA length = 12

FEATURE Location/Qualifiers

misc_feature 1..12

note = Synthetic Construct

source 1..12

mol_type = other DNA

organism = synthetic construct

SEQUENCE: 5

aaagaggaga aa 12

SEQ ID NO: 6 moltype = DNA length = 35

FEATURE Location/Qualifiers

-continued

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misc_feature      1..35
                  note = Synthetic Construct
source            1..35
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 6
ttgacagcta gctcagtcct aggtataatg ctacg                      35

SEQ ID NO: 7      moltype = DNA length = 35
FEATURE           Location/Qualifiers
misc_feature      1..35
                  note = Synthetic Construct
source            1..35
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 7
ttgacgggcta gctcagtcct aggtacagtg ctacg                      35

SEQ ID NO: 8      moltype = DNA length = 35
FEATURE           Location/Qualifiers
misc_feature      1..35
                  note = Synthetic Construct
source            1..35
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 8
tttacagcta gctcagtcct aggtattatg ctacg                      35

SEQ ID NO: 9      moltype = DNA length = 35
FEATURE           Location/Qualifiers
misc_feature      1..35
                  note = Synthetic Construct
source            1..35
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 9
ttgacagcta gctcagtcct aggtactgtg ctacg                      35

SEQ ID NO: 10     moltype = DNA length = 35
FEATURE           Location/Qualifiers
misc_feature      1..35
                  note = Synthetic Construct
source            1..35
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 10
ctgatagcta gctcagtcct agggattatg ctacg                      35

SEQ ID NO: 11     moltype = DNA length = 35
FEATURE           Location/Qualifiers
misc_feature      1..35
                  note = Synthetic Construct
source            1..35
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 11
ttgacagcta gctcagtcct aggtattgtg ctacg                      35

SEQ ID NO: 12     moltype = DNA length = 35
FEATURE           Location/Qualifiers
misc_feature      1..35
                  note = Synthetic Construct
source            1..35
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 12
tttacggcta gctcagtcct aggtactatg ctacg                      35

SEQ ID NO: 13     moltype = DNA length = 35
FEATURE           Location/Qualifiers
misc_feature      1..35
                  note = Synthetic Construct
source            1..35
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 13
tttacggcta gctcagtcct aggtatagtg ctacg                      35

SEQ ID NO: 14     moltype = DNA length = 35

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FEATURE	Location/Qualifiers	
misc_feature	1..35	
	note = Synthetic Construct	
source	1..35	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 14		
tttacggcta gctcagcct	aggtattatg ctagc	35
SEQ ID NO: 15	moltype = DNA length = 35	
FEATURE	Location/Qualifiers	
misc_feature	1..35	
	note = Synthetic Construct	
source	1..35	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 15		
ctgacagcta gctcagtcct	aggtataatg ctagc	35
SEQ ID NO: 16	moltype = DNA length = 35	
FEATURE	Location/Qualifiers	
misc_feature	1..35	
	note = Synthetic Construct	
source	1..35	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 16		
tttacagcta gctcagtcct	agggactgtg ctagc	35
SEQ ID NO: 17	moltype = DNA length = 35	
FEATURE	Location/Qualifiers	
misc_feature	1..35	
	note = Synthetic Construct	
source	1..35	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 17		
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SEQ ID NO: 18	moltype = DNA length = 35	
FEATURE	Location/Qualifiers	
misc_feature	1..35	
	note = Synthetic Construct	
source	1..35	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 18		
ttgacggcta gctcagtcct	aggtatagtg ctagc	35
SEQ ID NO: 19	moltype = DNA length = 35	
FEATURE	Location/Qualifiers	
misc_feature	1..35	
	note = Synthetic Construct	
source	1..35	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 19		
ctgatatgcta gctcagtcct	agggattatg ctagc	35
SEQ ID NO: 20	moltype = DNA length = 35	
FEATURE	Location/Qualifiers	
misc_feature	1..35	
	note = Synthetic Construct	
source	1..35	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 20		
ctgatggcta gctcagtcct	agggattatg ctagc	35
SEQ ID NO: 21	moltype = DNA length = 35	
FEATURE	Location/Qualifiers	
misc_feature	1..35	
	note = Synthetic Construct	
source	1..35	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 21		
tttatggcta gctcagtcct	aggtacaatg ctagc	35

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SEQ ID NO: 22	moltype = DNA length = 35	
FEATURE	Location/Qualifiers	
misc_feature	1..35	
	note = Synthetic Construct	
source	1..35	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 22		
tttatagcta gctcagccct tggtagaatg ctacg		35
SEQ ID NO: 23	moltype = DNA length = 35	
FEATURE	Location/Qualifiers	
misc_feature	1..35	
	note = Synthetic Construct	
source	1..35	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 23		
ttgacagcta gctcagtcct agggactatg ctacg		35
SEQ ID NO: 24	moltype = DNA length = 35	
FEATURE	Location/Qualifiers	
misc_feature	1..35	
	note = Synthetic Construct	
source	1..35	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 24		
ttgacagcta gctcagtcct agggattgtg ctacg		35
SEQ ID NO: 25	moltype = DNA length = 35	
FEATURE	Location/Qualifiers	
misc_feature	1..35	
	note = Synthetic Construct	
source	1..35	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 25		
ttgacggcta gctcagtcct aggtattgtg ctacg		35
SEQ ID NO: 26	moltype = DNA length = 37	
FEATURE	Location/Qualifiers	
misc_feature	1..37	
	note = Synthetic Construct	
source	1..37	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 26		
ctttgccgcg cgcttcgtca cgtaattctc gtcgcaa		37
SEQ ID NO: 27	moltype = DNA length = 37	
FEATURE	Location/Qualifiers	
misc_feature	1..37	
	note = Synthetic Construct	
source	1..37	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 27		
gtttggcgat ggcgcggtg tggttgtgct tcggcgt		37
SEQ ID NO: 28	moltype = DNA length = 32	
FEATURE	Location/Qualifiers	
misc_feature	1..32	
	note = Synthetic Construct	
source	1..32	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 28		
tgggatgcct accgcaagca gcttggcctg aa		32
SEQ ID NO: 29	moltype = DNA length = 153	
FEATURE	Location/Qualifiers	
misc_feature	1..153	
	note = Synthetic Construct	
source	1..153	
	mol_type = other DNA	
	organism = synthetic construct	

-continued

SEQUENCE: 29

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ggatccaaac tcgagtaagg atctccaggc atcaataaaa acgaaaggct cagtcgaaag 60
actgggcctt tcgttttatt tgttggttgc cggatgaacgc totctactag agtcacactg 120
getcaccttc gggatgggct ttctgcgttt ata 153

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The invention claimed is:

1. A production strain bacterial cell comprising a nucleic acid vector for introduction into a target bacterial host cell for expression of Type I Cas3 and Cascade proteins in the target bacterial host cell, the vector comprising a first nucleotide sequence encoding a Type I Cas3 and a second nucleotide sequence encoding one or more cognate Cascade proteins, wherein the first nucleotide sequence is under the control of a promoter for controlling the expression of Type I Cas3 in the target bacterial host cell, wherein the promoter has a strength that is weaker than the Anderson Score strength of promoter BBa_J23108,

wherein the nucleic acid vector further comprises: (i) a CRISPR array for producing a crRNA in the target bacterial host cell; or (ii) a nucleotide sequence encoding a guide RNA (gRNA), wherein the crRNA or gRNA comprises a spacer sequence complementary to a target sequence of the target bacterial host cell, and wherein the production strain bacterial cell does not comprise a crRNA or gRNA operable with the Cas3 to target and cut a chromosomal sequence of the production strain cell.

2. The production strain bacterial cell of claim 1, wherein the target bacterial host cell is selected from the group consisting of *Campylobacter*, *Burkholderia*, and *Acinetobacter*.

3. The production strain bacterial cell of claim 2, wherein the promoter is operable in a target host cell selected from the group consisting of *Campylobacter jejuni*, *Campylobacter coli*, *Campylobacter lari*, *Burkholderia cepacia*, *Burkholderia cenocepacia*, *Burkholderia multivorans*, *Burkholderia vietnamiensis*, *Burkholderia stabilis*, *Burkholderia thailandensis*, *Acinetobacter baumannii*, *Acinetobacter calcoaceticus*, *Acinetobacter haemolyticus*, *Acinetobacter lwoffii*, and *Acinetobacter schindleri*.

4. The production strain bacterial cell of claim 3, wherein the Type I Cas3 and Cascade are *E. coli*, *C. difficile*, *P. aeruginosa*, *K. pneumoniae*, *P. furiosus*, or *B. halodurans* Cas and Cascade proteins.

5. The production strain bacterial cell of claim 1, wherein the nucleic acid vector comprises an operon for expression of the Type I Cas3 and Cascade proteins, and:

- (a) the first nucleotide sequence is between the promoter and the second nucleotide sequence in the operon;
- (b) the operon comprises no Cas-encoding nucleotide sequences between the promoter and the first nucleotide sequence; or
- (c) the operon comprises, in 5' to 3' direction, the promoter, the first nucleotide sequence, and the second nucleotide sequence.

6. The production strain bacterial cell of claim 1, wherein the promoter is a constitutive promoter.

7. The production strain bacterial cell of claim 1, wherein the promoter is repressible.

8. The production strain bacterial cell of claim 1, wherein the promoter has a strength that is greater than the Anderson Score strength of promoter BBa_J23114.

9. The production strain bacterial cell of claim 1, wherein the nucleic acid vector further comprises an origin of replication that is operable in the target bacterial host cell.

10. The production strain bacterial cell of claim 1, wherein the nucleic acid vector is devoid of a Cas adaption module.

11. The production strain bacterial cell of claim 1, wherein the nucleic acid vector is devoid of a nucleotide sequence encoding one or more of a Cas1, Cas2, Cas4, Cas6, Cas7, and Cas8.

12. The production strain bacterial cell of claim 1, wherein the second nucleotide sequence encodes one or more of (a)-(g):

- (a) Cas11, Cas7, and Cas8a1;
- (b) Cas8b1, Cas7, and Cas5;
- (c) Cas5, Cas8c, and Cas7;
- (d) Cas8U2, Cas7, Cas5, and Cas6;
- (e) Cas10d, Cas7, and Cas5;
- (f) Cas8e, Cas11, Cas7, Cas5, and Cas6; and
- (g) Cas8f, Cas5, Cas7, and Cas6f.

13. The production strain bacterial cell of claim 12, wherein the Type I Cas3 is a Cas3' or Cas3".

14. The production strain bacterial cell of claim 12, wherein the Type I Cas3 is a Cas3, Cas3' or Cas3", and wherein the Type I Cas3 is between the promoter and the second nucleotide sequence.

15. The production strain bacterial cell of claim 14, wherein the nucleic acid vector is devoid of a nucleotide sequence encoding a further Cas between the promoter and the Type I Cas3.

16. The production strain bacterial cell of claim 12, wherein the vector comprises the CRISPR array, the CRISPR array is cognate with the Type I Cas3, and wherein:

- (a) the CRISPR array is a Type IA array and the nucleic acid vector comprises Cas11, Cas7, and Cas8a1;
- (b) the CRISPR array is a Type IB array and the nucleic acid vector comprises Cas8b1, Cas7, and Cas5;
- (c) the CRISPR array is a Type IC array and the nucleic acid vector comprises Cas5, Cas8c, and Cas7;
- (d) the CRISPR array is a Type IU array and the nucleic acid vector comprises Cas8U2, Cas7, Cas5, and Cas6;
- (e) the CRISPR array is a Type ID array and the nucleic acid vector comprises Cas10d, Cas7, and Cas5;
- (f) the CRISPR array is a Type IE array and the nucleic acid vector comprises Cas8e, Cas11, Cas7, Cas5, and Cas6; or
- (g) the CRISPR array is a Type IF array and the nucleic acid vector comprises Cas8f, Cas5, Cas7, and Cas6f.

17. The production strain bacterial cell of claim 1, wherein the Type I Cas3 and Cascade are:

- (a) Type IA Cas and Cascade proteins;
- (b) Type IB Cas and Cascade proteins;
- (c) Type IC Cas and Cascade proteins;
- (d) Type ID Cas and Cascade proteins;
- (e) Type IE Cas and Cascade proteins;
- (f) Type IF Cas and Cascade proteins; or
- (g) Type IU Cas and Cascade proteins.

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18. The production strain bacterial cell of claim 1, wherein the Type I Cas3 and Cascade are *E. coli* Cas and Cascade proteins.

19. The production strain bacterial cell of claim 1, wherein the Type I Cas3 and Cascade are *E. coli*, *C. difficile*,
P. aeruginosa, *K. pneumoniae*, *P. furiosus*, or *B. halodurans* Cas and Cascade proteins.

20. The production strain bacterial cell of claim 1, wherein the Type I Cas3 is a Cas3 of a CRISPR/Cas locus of *E. coli*, and wherein the distance between the Cas3-encoding sequence of the locus and its cognate promoter in *E. coli* is further than the distance between the Cas3-encoding sequence and the promoter for controlling the expression of Type I Cas3 in the nucleic acid vector.

21. The production strain bacterial cell of claim 1, wherein the CRISPR array or the gRNA-encoding sequence is under the control of a second promoter that is different from the promoter that controls the expression of the Type I Cas3.

22. The production strain bacterial cell of claim 1, wherein the nucleic acid vector is a plasmid or phagemid.

23. The production strain bacterial cell of claim 1, wherein the production strain bacterial cell comprises a

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nucleotide sequence whose expression is inducible to produce phage coat proteins in the cell of the production strain, wherein the production strain bacterial cell comprises amplified copies of the nucleic acid vector,

wherein the production strain bacterial cell is capable of packaging the amplified copies of the nucleic acid vector into phage particles or non-self-replicative transduction particles for introducing the amplified copies of the nucleic acid vector into the target host cell.

24. The production strain bacterial cell of claim 23, wherein the nucleic acid vector is a plasmid or phagemid and the delivery vehicle is a non-replicative transduction particle.

25. The production strain bacterial cell of claim 1, wherein the second nucleotide sequence is under the control of the same promoter as the first nucleotide sequence.

26. The production strain bacterial cell of claim 1, wherein the target sequence of the target bacterial host cell is a chromosomal sequence of the target bacterial host cell.

27. The production strain bacterial cell of claim 1, wherein the production strain bacterial cell is an *Escherichia coli* (*E. coli*) cell.

* * * * *